

A Summary for Quant Math

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Introduction

In this book, I aim to provide an essential summary for Quantitative Mathematics, covering key areas such as Calculus, Probability Theory, Mathematical Statistics, Linear Algebra, Time Series, Regression Analysis, Stochastic Analysis. Instead of delving into nitty-gritty details of mathematical derivations, the focus will be on highlighting the most critical concepts and the fundamental knowledge you must be familiar with.

1. Calculus: Understanding the basics of differentiation and integration, the fundamental theorems, and their applications in various quantitative analysis.

2. Probability Theory: Grasping the core principles of probability, including random variables, distributions, and key theorems like the Law of Large Numbers and Central Limit Theorem.

3. Mathematical Statistics: Familiarity with statistical methods, hypothesis testing, confidence intervals, and the importance of statistical inference in data analysis.

4. Linear Algebra: Mastery of matrix operations, vector spaces, matrix decomposition, eigenvalues, and eigenvectors, which are crucial for solving systems of linear equations and for applications in various fields including machine learning.

5. Time Series: Key techniques for analyzing time series data, understanding Linear Time Series Models and Conditional heteroscedastic Time Series Models.

6. Regression Analysis: The foundation of linear regression analysis, including model fitting, assumptions, diagnostics, and interpreting results and also the basic part for non-linear regression like: Lasso, Ridge and logistics.

7. Stochastic Analysis: This part include stochastic process, random walk, martingale, brown motion, Ito's Lemma, Ito's Integral, which are the crucial part for option pricing and BSM model.

Quant interviews are hard. And for good reason, too — at the intersection of computer science, statistics, and finance lies the most lucrative job out of university, with graduate opportunities commanding up to 600,000 dollars per year in compensation from base salary, sign-on bonus, and performance bonus (and internships paying over 150 dollars per hour). Naturally, every year is more competitive than the last, and sometimes I even wonder if we'd still be able to pass the interviews at the top trading firms today. I'm uncompromising believers that you do not have to be a genius to land a role in quant and that there exists a path of persistent preparation and tenacious thoughtfulness that will lead to an offer. I carefully thought about what resources would have helped us back then that would have made our lives less stressful when preparing for interviews, so I authored this sheet in hopes that it can make your studying and interviewing processes easier. By concentrating on these pivotal topics, you will gain a solid understanding of Quantitative Mathematics that is essential for practical application in various quantitative fields.

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1 Calculus

1.1 Limits and Derivatives

1.1.1 Basics of derivatives

Let's begin with some basic definitions and equations used in limits and derivatives. Although the notations may be different, you can find these materials in any calculus textbook.

Derivative

Let $y = f(x)$, then

$$f'(x) = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The product rule

If $u = u(x)$ and $v = v(x)$ and their respective derivatives exist,

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}, \quad (uv)' = u'v + uv'$$

The quotient rule

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \left(\frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \right), \quad \left(\frac{u}{v} \right)' = \frac{u'v - uv'}{v^2}$$

The chain rule

If $y = f(u(x))$ and $u = u(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

The generalized power rule

$$\frac{d}{dx} y^n = ny^{n-1} \frac{dy}{dx} \quad \text{for } n \neq 0$$

Some useful equations

$$\begin{aligned} a^x &= e^{x \ln a}, \quad \ln(ab) = \ln a + \ln b, \quad e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, \quad \lim_{x \rightarrow 0} (1+x)^k = 1+kx \text{ for any } k \\ \lim_{x \rightarrow \infty} (\ln x / x^r) &= 0 \text{ for any } r > 0, \quad \lim_{x \rightarrow \infty} x^r e^{-x} = 0 \text{ for any } r \\ \frac{d}{dx} e^u &= e^u \frac{du}{dx}, \quad \frac{d}{dx} a^u = (a^u \ln a) \frac{du}{dx}, \quad \frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx} = \frac{u'}{u} \\ \frac{d}{dx} \sin x &= \cos x, \quad \frac{d}{dx} \cos x = -\sin x, \quad \frac{d}{dx} \tan x = \sec^2 x \end{aligned}$$

1.1.2 L'Hospital's rule

Suppose that functions $f(x)$ and $g(x)$ are differentiable at $x \rightarrow a$ and that $\lim_{x \rightarrow a} g'(x) \neq 0$. Further suppose that $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ or that $\lim_{x \rightarrow a} f(x) \rightarrow \pm\infty$ and $\lim_{x \rightarrow a} g(x) \rightarrow \pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

L'Hospital's rule converts the limit from an indeterminate form to a determinate form.

1.1.3 Partial Derivatives

Partial derivative

$$w = f(x, y) \Rightarrow \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = f_x$$

Second order partial derivatives

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

The general chain rule

Suppose that $w = f(x_1, x_2, \dots, x_m)$ and that each of variables $x_1, x_2, \dots, x_m$ is a function of the variables $t_1, t_2, \dots, t_n$. If all these functions have continuous first-order partial derivatives, then

$$\frac{\partial w}{\partial t_i} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial w}{\partial x_m} \frac{\partial x_m}{\partial t_i}$$

for each $i, 1 \leq i \leq n$.

1.2 Integration

1.2.1 Basics of integration

If we can find a function $F(x)$ with derivative $f(x)$, then we call $F(x)$ an *antiderivative* of $f(x)$.

If $f(x) = F'(x)$,

$$\int_a^b f(x) dx = \int_a^b F'(x) dx = [F(x)]_a^b = F(b) - F(a)$$

$$\frac{dF(x)}{dx} = f(x), \quad F(a) = y_a \Rightarrow F(x) = y_a + \int_a^x f(t) dt$$

The generalized power rule in reverse

$$\int u^k du = \frac{u^{k+1}}{k+1} + c \quad (k \neq -1), \quad \text{where } c \text{ is any constant.}$$

Integration by substitution

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du \quad \text{with } u = g(x), \quad du = g'(x) dx$$

Substitution in definite integrals

$$\int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Integration by parts

$$\int u dv = uv - \int v du$$

1.2.2 Multiple Integrals

Multiple integrals extend the concept of integration to functions of several variables. The most common types of multiple integrals are double and triple integrals.

General Form of Change of Variables Formula

Double Integrals

Double integrals are used to integrate over a two-dimensional region. If $f(x, y)$ is a continuous function on the region R , the double integral of f over R is denoted by:

$$\iint_R f(x, y) dA$$

where dA represents the differential area element in the xy -plane. In Cartesian coordinates, this can be written as:

$$\iint_R f(x, y) dx dy$$

or

$$\iint_R f(x, y) dy dx$$

depending on the order of integration.

For a transformation from (x, y) to (u, v) :

$$\iint_R f(x, y) dx dy = \iint_{R'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Triple Integrals

Triple integrals extend the concept of double integrals to three-dimensional regions. If $f(x, y, z)$ is a continuous function on the region V , the triple integral of f over V is denoted by:

$$\iiint_V f(x, y, z) dV$$

where dV represents the differential volume element. In Cartesian coordinates, this can be written as:

$$\iiint_V f(x, y, z) dx dy dz$$

or in different orders of integration depending on the region of integration.

For a transformation from (x, y, z) to (u, v, w) :

$$\iiint_V f(x, y, z) dx dy dz = \iiint_{V'} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

Jacobian determinant

The Jacobian determinant J is used when changing variables in multiple integrals to account for the change in the volume element.

For a transformation $(u, v) \rightarrow (x, y)$, the Jacobian determinant is given by:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

For three variables:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

This general formula applies to any coordinate transformation where the Jacobian determinant is non-zero within the region of integration.

1.2.3 Change of Variables

For both double and triple integrals, it is often useful to change variables to simplify the region of integration. Common transformations include polar coordinates for double integrals and cylindrical or spherical coordinates for triple integrals.

Polar Coordinates (Double Integrals) In polar coordinates, x and y are expressed as:

$$x = r \cos \theta, \quad y = r \sin \theta$$

- r : Radius, typically $0 \leq r \leq r_{\max}$
- θ : Angle, typically $0 \leq \theta \leq 2\pi$ (or another specified range)

and the area element dA is:

$$dA = r \, dr \, d\theta$$

Thus, a double integral in polar coordinates becomes:

$$\iint_R f(x, y) \, dA = \iint_R f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

where r and θ are within the ranges specified by the region R .

Cylindrical Coordinates (Triple Integrals) In cylindrical coordinates, x , y , and z are expressed as:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

- r : Radius, typically $0 \leq r \leq r_{\max}$
- θ : Angle, typically $0 \leq \theta \leq 2\pi$
- z : Height, typically $z_{\min} \leq z \leq z_{\max}$

and the volume element dV is:

$$dV = r \, dr \, d\theta \, dz$$

Thus, a triple integral in cylindrical coordinates becomes:

$$\iiint_V f(x, y, z) \, dV = \iiint_V f(r \cos \theta, r \sin \theta, z) r \, dr \, d\theta \, dz$$

where r , θ , and z are within the ranges specified by the region V .

Explanation

For polar coordinates:

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

For cylindrical coordinates:

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

1.3 Important Calculus Methods

1.3.1 Taylor's Series

One-dimensional Taylor's series expands function $f(x)$ as the sum of a series using the derivatives at a point $x = x_0$:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \cdots$$

If $x_0 = 0$,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

Taylor's series are often used to represent functions in power series terms. For example, Taylor's series for three common transcendental functions, e^x , $\sin x$, and $\cos x$, at $x_0 = 0$ are:

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

The Taylor's series can also be expressed as the sum of the n -th degree Taylor polynomial $T_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$ and a remainder $R_n(x)$:

$$f(x) = T_n(x) + R_n(x)$$

For some $\tilde{x}$ between x_0 and x ,

$$R_n(x) = \frac{f^{(n+1)}(\tilde{x})}{(n+1)!} |x - x_0|^{n+1}$$

Let M be the maximum of $|f^{(n+1)}(\tilde{x})|$ for all $\tilde{x}$ between x_0 and x , we get constraint

$$|R_n(x)| \leq \frac{M \times |x - x_0|^{n+1}}{(n+1)!}$$

1.3.2 Lagrange multipliers

Lagrange multipliers are a strategy for finding the local maxima and minima of a function subject to equality constraints. The method introduces new variables (Lagrange multipliers) for each constraint and forms a new function called the Lagrangian.

Problem Statement

Suppose we want to find the extrema of a function $f(x, y, \dots)$ subject to a constraint $g(x, y, \dots) = 0$.

Method**1. Form the Lagrangian:**

$$\mathcal{L}(x, y, \dots, \lambda) = f(x, y, \dots) + \lambda(g(x, y, \dots))$$

2. Set the gradient of the Lagrangian to zero:

$$\nabla \mathcal{L} = 0$$

This gives us the system of equations:

$$\begin{cases} \nabla f(x, y, \dots) + \lambda \nabla g(x, y, \dots) = 0 \\ g(x, y, \dots) = 0 \end{cases}$$

3. Solve the system of equations: Solving these equations simultaneously gives the values of $x, y, \dots, \lambda$ that provide the extrema of f under the constraint g .**Example**

Consider the function $f(x, y) = x^2 + y^2$ subject to the constraint $g(x, y) = x + y - 1 = 0$.

1. Form the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$$

2. Set the gradient to zero:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = 2x + \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = 2y + \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} = x + y - 1 = 0 \end{cases}$$

3. Solve the system of equations:

$$\begin{cases} 2x + \lambda = 0 \\ 2y + \lambda = 0 \\ x + y - 1 = 0 \end{cases}$$

From the first two equations:

$$2x = 2y \implies x = y$$

Substitute into the constraint:

$$x + x - 1 = 0 \implies 2x = 1 \implies x = \frac{1}{2}$$

Therefore:

$$y = \frac{1}{2}$$

The Lagrange multiplier is:

$$\lambda = -2x = -1$$

Hence, the function $f(x, y) = x^2 + y^2$ has a minimum at $(\frac{1}{2}, \frac{1}{2})$.

1.4 Classic Questions

1.4.1 Differentiate 1

How do you differentiate x^x with respect to x ?

solution:

Let $y = x^x$, then take logs, differentiate implicitly, use the product rule, and then use the definition of y to recover the answer:

$$\begin{aligned}\ln(y) &= x \ln(x) \\ \frac{1}{y} \frac{dy}{dx} &= \ln(x) + 1 \\ \frac{dy}{dx} &= x^x(1 + \ln(x))\end{aligned}$$

1.4.2 Differentiate 2

Define $g(x)$ by

$$g(x) = \int_0^{F(x)} h(x, y) dy.$$

What is $g'(x)$?

solution:

This is a straightforward question if you remember the Leibniz integration rule: assume $h(x, y)$ is such that $h_x(x, y)$ is continuous (where we use the subscript to denote the partial derivative), then

$$\frac{d}{dx} \left(\int_{V(x)}^{W(x)} h(x, y) dy \right) = \left(\int_{V(x)}^{W(x)} h_x(x, y) dy \right) + h(x, W(x))W'(x) - h(x, V(x))V'(x).$$

Using this rule, we then have

$$g'(x) = \int_0^{F(x)} h_x(x, y) dy + h(x, F(x))F'(x).$$

1.4.3 Differentiate 3

What is the derivative of $y = \ln x^{\ln x}$?

Solution:

Let $u = \ln y = \ln(\ln x^{\ln x}) = \ln x \times \ln(\ln x)$. Applying the chain rule and the product rule, we have

$$\frac{du}{dx} = \frac{d(\ln y)}{dx} = \frac{1}{y} \frac{dy}{dx} = \frac{d(\ln x)}{dx} \times \ln(\ln x) + \ln x \times \frac{d(\ln(\ln x))}{dx} = \frac{\ln(\ln x)}{x} + \frac{\ln x}{x \ln x}.$$

To derive $\frac{d(\ln(\ln x))}{dx}$, we again use the chain rule by setting $v = \ln x$:

$$\frac{d(\ln(\ln x))}{dx} = \frac{d(\ln v)}{dv} \frac{dv}{dx} = \frac{1}{v} \frac{1}{x} = \frac{1}{x \ln x}.$$

Therefore,

$$\frac{1}{y} \frac{dy}{dx} = \frac{\ln(\ln x)}{x} + \frac{\ln x}{x \ln x} \Rightarrow \frac{dy}{dx} = \frac{y}{x} (\ln(\ln x) + 1) = \frac{\ln x^{\ln x}}{x} (\ln(\ln x) + 1).$$

1.4.4 Integration 1

1. What is the value of $\int_{-\infty}^{\infty} e^{-x^2} dx$?
2. Show that the normal density integrates to one.

solution:

1. This is a great integration question that, to solve from scratch, involves a nice trick. First, we will consider deducing the solution by a less direct manner. If you have studied any probability theory at all, you will have seen that the density of the $N(0, 1)$ random variable is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right),$$

for $x \in \mathbb{R}$. You would also be aware that the integral of the density of a proper random variable gives 1, so with a bit of manipulation, we can use this to solve our problem:

$$1 = \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp(-y^2) dy,$$

using the substitution $y = x/\sqrt{2}$. Thus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

The alternative method of solution is as follows. Denote by I the integral in question,

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Then we have

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy.$$

By considering this as an integral over the entire plane, we change to polar coordinates setting $x = r \cos(\theta)$ and $y = r \sin(\theta)$, $r > 0$, $0 \leq \theta \leq 2\pi$. Remembering to include the Jacobian, we then have

$$I^2 = \int_0^{\infty} \int_0^{2\pi} e^{-r^2(\cos^2(\theta)+\sin^2(\theta))} r d\theta dr = \int_0^{\infty} \int_0^{2\pi} e^{-r^2} r d\theta dr = 2\pi \int_0^{\infty} e^{-r^2} r dr = \pi \int_0^{\infty} e^{-z} dz = \pi,$$

which gives $I = \sqrt{\pi}$ and confirms our previous conclusion.

2. The normal density is

$$N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

We have to show that

$$I = \int_{-\infty}^{\infty} N'(x) dx = 1.$$

The standard way to do this is to show that the square is one. First, note that the integrand is exponentially decaying and smooth, so there is no issue with the integral existing. We can write

$$I^2 = \int_{-\infty}^{\infty} N'(x) dx \int_{-\infty}^{\infty} N'(y) dy = \frac{1}{2\pi} \iint e^{-\frac{(x^2+y^2)}{2}} dx dy.$$

We now change to polar coordinates:

$$x = r \cos(\theta),$$

$$y = r \sin(\theta).$$

We then have $dx dy = r dr d\theta$. We deduce

$$I^2 = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\infty r e^{-r^2/2} dr d\theta.$$

Since

$$\frac{d}{dr} e^{-r^2/2} = -r e^{-r^2/2},$$

the integral is now straightforward, and we get

$$I^2 = \frac{1}{2\pi} 2\pi \left[-e^{-r^2/2} \right]_0^\infty = 1.$$

1.4.5 Integration 2

Compute

$$\int \frac{1}{1+x^2} dx.$$

Solution: Use the substitution $x = \tan(z)$. Then, $dx = \frac{1}{\cos^2(z)} dz$ and

$$\int \frac{1}{1+x^2} dx = \int \frac{1}{(1+\tan^2(z)) \cos^2(z)} dz = \int \frac{1}{\cos^2(z) + \sin^2(z)} dz = \int 1 dz = z + C,$$

where C is a real constant, since $\cos^2(z) + \sin^2(z) = 1$ for any z . Solving $x = \tan(z)$ for z , we obtain that $z = \arctan(x)$, and therefore

$$\int \frac{1}{1+x^2} dx = \arctan(x) + C.$$

1.4.6 What is i^i ?

Solution: The solution to this problem uses Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, which can be proven using Taylor's series. Let's look at the proof. Applying Taylor's series to $e^{i\theta}$, $\cos \theta$, and $\sin \theta$, we have

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots = 1 + i\frac{\theta}{1!} - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} + \cdots$$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots$$

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots \Rightarrow i \sin \theta = i\theta - i\frac{\theta^3}{3!} + i\frac{\theta^5}{5!} - i\frac{\theta^7}{7!} + \cdots$$

Combining these three series, it is apparent that $e^{i\theta} = \cos \theta + i \sin \theta$.

When $\theta = \pi$, the equation becomes $e^{i\pi} = \cos \pi + i \sin \pi = -1$. When $\theta = \pi/2$, the equation becomes $e^{i\pi/2} = \cos(\pi/2) + i \sin(\pi/2) = i$. So $\ln i = \ln(e^{i\pi/2}) = i\pi/2$.

Hence, $\ln(i^i) = i \ln i = i(i\pi/2) = -\pi/2 \Rightarrow i^i = e^{-\pi/2}$.

2 Linear Algebra

2.1 Basic Concepts and Notations

Linear algebra provides a concise way to represent and manipulate systems of linear equations. For example, consider the system of equations:

$$\begin{cases} 4x_1 - 5x_2 = -13 \\ -2x_1 + 3x_2 = 9 \end{cases}$$

This is a system of two equations with two variables, and as you learned in high school algebra, you can find the solutions x_1 and x_2 (excluding special cases like if the second equation is just a multiple of the first one, but in the above case, there is indeed a unique solution). In matrix notation, we can more concisely express it as:

$$A\mathbf{x} = \mathbf{b}$$

with

$$A = \begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -13 \\ 9 \end{pmatrix}$$

We can see that this matrix form of linear equations has several advantages (such as being obviously space-saving).

2.1.1 Basic Notations

We use the following notations:

- $A \in \mathbb{R}^{m \times n}$: indicates that A is a matrix with m rows and n columns consisting of real numbers.
- $\mathbf{x} \in \mathbb{R}^n$: indicates that $\mathbf{x}$ is a vector with n elements. Typically, a vector $\mathbf{x}$ is represented as a column vector: i.e., it has n rows and 1 column. If we want to explicitly represent $\mathbf{x}$ as a row vector (with 1 row and n columns), we usually write $\mathbf{x}^T$ (the transpose of $\mathbf{x}$).
- x_i denotes the i -th element of the vector $\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

- We use the notation a_{ij} (or A_{ij} , $A_{i,j}$, etc.) to represent the element in the i -th row and j -th column of A :

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

- We use $\mathbf{a}_j$ or $A_{:,j}$ to represent the j -th column of A :

$$A = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & \cdots & | \end{pmatrix}$$

- We use $\mathbf{a}_i$ or $A_{i,:}$ to represent the i -th row of A :

$$A = \begin{pmatrix} -\mathbf{a}_1^T \\ -\mathbf{a}_2^T \\ \vdots \\ -\mathbf{a}_m^T \end{pmatrix}$$

2.2 Matrix Multiplication

When two matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ are multiplied, we get:

$$C = AB \in \mathbb{R}^{m \times p}$$

where:

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Note that for the matrix product to exist, the number of columns in A must equal the number of rows in B . There are many ways to look at matrix multiplication, and we will start by examining some special cases.

2.2.1 Vector-Vector Multiplication

Given two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\mathbf{x}^T \mathbf{y}$ is usually called the dot product or inner product of the vectors, and the result is a real number:

$$\mathbf{x}^T \mathbf{y} = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \sum_{i=1}^n x_i y_i$$

Note: $\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x}$ always holds.

Given vectors $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$ (their dimensions may or may not be the same), $\mathbf{x} \mathbf{y}^T \in \mathbb{R}^{m \times n}$ is called the outer product. When $(\mathbf{x} \mathbf{y}^T)_{ij} = x_i y_j$, it is a matrix.

$$\mathbf{x} \mathbf{y}^T \in \mathbb{R}^{m \times n} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \begin{pmatrix} y_1 & y_2 & \cdots & y_n \end{pmatrix} = \begin{pmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{pmatrix}$$

An example of how to use the outer product: let $\mathbf{1} \in \mathbb{R}^n$ denote a vector of all ones of dimension n . Also, consider a matrix $A \in \mathbb{R}^{m \times n}$, where all its elements are equal to some vector $\mathbf{x} \in \mathbb{R}^m$. We can represent the matrix A using the outer product as:

$$A = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{x} & \mathbf{x} & \cdots & \mathbf{x} \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} x_1 & x_1 & \cdots & x_1 \\ x_2 & x_2 & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_m & \cdots & x_m \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} = \mathbf{x} \mathbf{1}^T$$

2.2.2 Matrix-Vector Multiplication

Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, their product is a vector $\mathbf{y} = A\mathbf{x} \in \mathbb{R}^m$. There are several ways to look at matrix-vector multiplication. We will look at each one of them in turn.

If we write A row-wise, we can express $A\mathbf{x}$ as:

$$\mathbf{y} = A\mathbf{x} = \begin{pmatrix} -\mathbf{a}_1^T - \\ -\mathbf{a}_2^T - \\ \vdots \\ -\mathbf{a}_m^T - \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{a}_1^T \mathbf{x} \\ \mathbf{a}_2^T \mathbf{x} \\ \vdots \\ \mathbf{a}_m^T \mathbf{x} \end{pmatrix}$$

In other words, the i -th component of $\mathbf{y}$ is the inner product of the i -th row of A and $\mathbf{x}$, i.e., $y_i = \mathbf{a}_i^T \mathbf{x}$. Similarly, we can write A column-wise, and the formula becomes:

$$\mathbf{y} = A\mathbf{x} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{a}^1 x_1 + \mathbf{a}^2 x_2 + \cdots + \mathbf{a}^n x_n$$

In other words, $\mathbf{y}$ is a linear combination of the columns of A , with coefficients given by the elements of $\mathbf{x}$.

So far, we have been multiplying vectors on the right-hand side using column vectors, but we can also multiply on the left-hand side using row vectors. This is written as:

$$\mathbf{y}^T = \mathbf{x}^T A \quad \text{where } A \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^m, \mathbf{y} \in \mathbb{R}^n.$$

As before, we can express $\mathbf{y}^T$ using either rows or columns of A , depending on whether we represent A by its rows or columns.

In the first case, we represent A using its columns:

$$\mathbf{y}^T = \mathbf{x}^T A = \mathbf{x}^T \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & \cdots & | \end{pmatrix} = (x_1 \ x_2 \ \cdots \ x_n) \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & \cdots & | \end{pmatrix}$$

This shows that the i -th element of $\mathbf{y}^T$ is the inner product of $\mathbf{x}$ and the i -th column of A .

Finally, based on the row representation of A , we obtain the final expression for the vector-matrix product:

$$\mathbf{y}^T = \mathbf{x}^T A = (x_1 \ x_2 \ \cdots \ x_n) \begin{pmatrix} -\mathbf{a}_1^T - \\ -\mathbf{a}_2^T - \\ \vdots \\ -\mathbf{a}_m^T - \end{pmatrix} = x_1 (-\mathbf{a}_1^T -) + x_2 (-\mathbf{a}_2^T -) + \cdots + x_n (-\mathbf{a}_m^T -)$$

So we see that $\mathbf{y}$ is a linear combination of the rows of A , where the coefficients of the linear combination are the elements of $\mathbf{x}$.

2.2.3 Matrix-Matrix Multiplication

With this knowledge, we can now look at four different (but equivalent in result) ways to multiply matrices, which is what this section defines as $C = AB$.

First, we can view matrix-matrix multiplication as a set of vector-vector multiplications. From the definition, the most obvious point is that the (i, j) element is the inner product of the i -th row of A and the j -th column of B . The formula is:

$$C = AB = \begin{pmatrix} -\mathbf{a}_1^T - \\ -\mathbf{a}_2^T - \\ \vdots \\ -\mathbf{a}_m^T - \end{pmatrix} \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}^1 & \mathbf{b}^2 & \cdots & \mathbf{b}^p \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} \mathbf{a}_1^T \mathbf{b}^1 & \mathbf{a}_1^T \mathbf{b}^2 & \cdots & \mathbf{a}_1^T \mathbf{b}^p \\ \mathbf{a}_2^T \mathbf{b}^1 & \mathbf{a}_2^T \mathbf{b}^2 & \cdots & \mathbf{a}_2^T \mathbf{b}^p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_m^T \mathbf{b}^1 & \mathbf{a}_m^T \mathbf{b}^2 & \cdots & \mathbf{a}_m^T \mathbf{b}^p \end{pmatrix}$$

Here, $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $\mathbf{a}_i \in \mathbb{R}^n$, $\mathbf{b}_j \in \mathbb{R}^p$, so their inner products can be computed. We usually use row notation for A and column notation for B .

Or, we can view it as the sum of outer products. In this case, AB is the sum of all outer products of the columns of A and the rows of B :

$$C = AB = \left(\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & & | \end{array} \right) \begin{pmatrix} -\mathbf{b}_1^T - \\ -\mathbf{b}_2^T - \\ \vdots \\ -\mathbf{b}_n^T - \end{pmatrix} = \sum_{i=1}^n \mathbf{a}^i \mathbf{b}_i^T$$

In other words, AB is the sum of all outer products of the columns of A and the rows of B . Therefore, in this situation, the outer product's dimension is $m \times p$, consistent with the resulting dimension.

Finally, we can also view matrix-matrix multiplication as a set of matrix-vector multiplications. If we represent A column-wise, we can view C as matrix-vector products given by the formula:

$$C = AB = A \begin{bmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \\ | & | & & | \end{bmatrix}$$

Here, the i -th column of the resulting matrix C is the matrix-vector product of A and the i -th column of B .

We have a similar perspective, where we use the rows of A to represent A , and the rows of C as the outer products between A and B . The formula is as follows:

$$C = AB = \begin{pmatrix} -\mathbf{a}_1^T - \\ -\mathbf{a}_2^T - \\ \vdots \\ -\mathbf{a}_m^T - \end{pmatrix} B = \begin{pmatrix} -\mathbf{a}_1^T B - \\ -\mathbf{a}_2^T B - \\ \vdots \\ -\mathbf{a}_m^T B - \end{pmatrix}$$

Here, the i -th row of C is the matrix-vector product of the i -th row of A and B : $\mathbf{c}_i^T = \mathbf{a}_i^T B$.

Additionally, understanding some basic properties of matrix multiplication is very important:

- Matrix multiplication is associative: $(AB)C = A(BC)$
- Matrix multiplication is distributive: $A(B + C) = AB + AC$
- Matrix multiplication is generally not commutative: $AB \neq BA$. (For example, suppose $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, if $m \neq p$, the product BA does not even exist!)

If you are unfamiliar with these properties, please take the time to verify them. For example, to check the associativity of matrix multiplication, suppose $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times q}$. Note that $AB \in \mathbb{R}^{m \times p}$, so $(AB)C \in \mathbb{R}^{m \times q}$. Similarly, $BC \in \mathbb{R}^{n \times q}$, so $A(BC) \in \mathbb{R}^{m \times q}$. Therefore, the dimensions of the resulting matrices are consistent. To show that matrix multiplication is associative, we check whether the (i, j) -th element of $(AB)C$ is equal to the (i, j) -th element of $A(BC)$. We can use the definition of matrix multiplication to verify this:

$$((AB)C)_{ij} = \sum_{k=1}^p (AB)_{ik} C_{kj} = \sum_{k=1}^p \left(\sum_{l=1}^n A_{il} B_{lk} \right) C_{kj} = \sum_{l=1}^n A_{il} \left(\sum_{k=1}^p B_{lk} C_{kj} \right) = \sum_{l=1}^n A_{il} (BC)_{lj} = (A(BC))_{ij}$$

2.3 Operations and Properties

2.3.1 Identity Matrix and Diagonal Matrix

The identity matrix $I \in \mathbb{R}^{n \times n}$ is a square matrix where all the diagonal elements are 1 and all other elements are 0:

$$I_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

For any matrix $A \in \mathbb{R}^{m \times n}$,

$$AI = A = IA$$

Note that, by this definition, the size of the identity matrix is not specified, but it can be inferred from the context. Usually, the context provides enough information to specify the dimension. For example, in the equation above, $AI = A$, if I is an $n \times n$ matrix, then A must be an $m \times n$ matrix.

A diagonal matrix is a square matrix where the elements outside the main diagonal are zero. The diagonal elements are usually denoted as $d_1, d_2, \dots, d_n$:

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

Specifically, the identity matrix is a diagonal matrix where all the diagonal elements are 1:

$$I = \text{diag}(1, 1, \dots, 1)$$

A is diagonalizable if and only if it has linearly independent eigenvectors. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of A , $x_1, x_2, \dots, x_n$ be the corresponding eigenvectors. and $X = [x_1 \mid x_2 \mid \dots \mid x_n]$, then

$$X^{-1}AX = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} = D \implies A = XDX^{-1} \implies A^k = XD^kX^{-1}.$$

2.3.2 Transpose

The transpose of a matrix is an important and useful operation. Given a matrix $A \in \mathbb{R}^{m \times n}$, its transpose $A^T \in \mathbb{R}^{n \times m}$ is defined as:

$$(A^T)_{ij} = A_{ji}$$

In fact, the notation A^T was previously introduced. The transpose of a matrix switches its rows and columns. Here are some properties of the transpose:

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$

2.3.3 Adjugate Matrix

The adjugate matrix (also known as the adjoint matrix) of a square matrix is the transpose of its cofactor matrix. It plays an important role in matrix theory and linear algebra, particularly in finding the inverse of a matrix. Given a square matrix A , the adjugate matrix is denoted as $\text{adj}(A)$ or A^* . The steps to compute the adjugate matrix are as follows:

1. Compute the Minor Matrix

For each element a_{ij} of the matrix A , calculate its minor. The minor of an element is the determinant of the submatrix obtained by deleting the i -th row and j -th column from A . This minor is denoted as M_{ij} .

2. Compute the Cofactor Matrix

For each minor M_{ij} , multiply it by $(-1)^{i+j}$ to obtain the cofactor C_{ij} .

3. Construct the Adjugate Matrix

Arrange all the cofactors C_{ij} into a matrix, and then transpose this matrix to obtain the adjugate matrix $\text{adj}(A)$.

The process is illustrated below:

Given a 3×3 matrix A :

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$$

Step 1: Compute the Minors

$$\begin{aligned} M_{11} &= \begin{vmatrix} 4 & 5 \\ 0 & 6 \end{vmatrix} = 4 \cdot 6 - 5 \cdot 0 = 24, \\ M_{12} &= \begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} = 0 \cdot 6 - 5 \cdot 1 = -5, \\ M_{13} &= \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix} = 0 \cdot 0 - 4 \cdot 1 = -4, \\ M_{21} &= \begin{vmatrix} 2 & 3 \\ 0 & 6 \end{vmatrix} = 2 \cdot 6 - 3 \cdot 0 = 12, \\ M_{22} &= \begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix} = 1 \cdot 6 - 3 \cdot 1 = 3, \\ M_{23} &= \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 1 \cdot 0 - 2 \cdot 1 = -2, \\ M_{31} &= \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 2 \cdot 5 - 3 \cdot 4 = 10 - 12 = -2, \\ M_{32} &= \begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = 1 \cdot 5 - 3 \cdot 0 = 5, \\ M_{33} &= \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 0 = 4. \end{aligned}$$

Step 2: Compute the Cofactors

$$\begin{aligned} C_{11} &= (-1)^{1+1} \cdot M_{11} = 1 \cdot 24 = 24, \\ C_{12} &= (-1)^{1+2} \cdot M_{12} = -1 \cdot (-5) = 5, \\ C_{13} &= (-1)^{1+3} \cdot M_{13} = 1 \cdot (-4) = -4, \\ C_{21} &= (-1)^{2+1} \cdot M_{21} = -1 \cdot 12 = -12, \\ C_{22} &= (-1)^{2+2} \cdot M_{22} = 1 \cdot 3 = 3, \\ C_{23} &= (-1)^{2+3} \cdot M_{23} = -1 \cdot (-2) = 2, \\ C_{31} &= (-1)^{3+1} \cdot M_{31} = 1 \cdot (-2) = -2, \\ C_{32} &= (-1)^{3+2} \cdot M_{32} = -1 \cdot 5 = -5, \\ C_{33} &= (-1)^{3+3} \cdot M_{33} = 1 \cdot 4 = 4. \end{aligned}$$

Thus, the cofactor matrix C is:

$$C = \begin{pmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{pmatrix}$$

Step 3: Transpose the Cofactor Matrix

The adjugate matrix $\text{adj}(A)$ is the transpose of C :

$$\text{adj}(A) = C^T = \begin{pmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{pmatrix}$$

In conclusion, the adjugate matrix of A is:

$$\text{adj}(A) = \begin{pmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{pmatrix}$$

2.3.4 Symmetric Matrix

If $A = A^T$, then matrix $A \in \mathbb{R}^{n \times n}$ is symmetric. If $A = -A^T$, then A is skew-symmetric. It can be proven that for any square matrix A , $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric. As a result, any square matrix $A \in \mathbb{R}^{n \times n}$ can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix:

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

In the above equation, the first term on the right-hand side is symmetric, and the second term is skew-symmetric. In fact, symmetric matrices are very useful in practice. They have many properties that make them much easier to deal with compared to general square matrices. Therefore, we will often encounter symmetric matrices in the context of $n \times n$ matrices.

2.3.5 Trace of a Matrix

The trace of a square matrix $A \in \mathbb{R}^{n \times n}$, denoted as $\text{tr}(A)$ (or simply tr if the matrix is clear from the context), is the sum of the diagonal elements:

$$\text{tr}(A) = \sum_{i=1}^n A_{ii}$$

The trace has the following properties (listed below):

- For any square matrix $A \in \mathbb{R}^{n \times n}$, $\text{tr}(A) = \text{tr}(A^T)$
- For any matrices $A, B \in \mathbb{R}^{n \times n}$, $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
- For any matrix $A \in \mathbb{R}^{n \times n}$, $t \in \mathbb{R}$, $\text{tr}(tA) = t\text{tr}(A)$
- For any matrices $A, B \in \mathbb{R}^{m \times n}$, $\text{tr}(AB) = \text{tr}(BA)$ (therefore, $\text{tr}(AB)$ is defined if and only if A and B are such that the product AB is square)
- For any matrices A, B, C , $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$

As an example to show the significance of these properties, we will provide a proof of one of them. Suppose $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$ (so $AB \in \mathbb{R}^{m \times m}$ is a square matrix). We want to prove that $\text{tr}(AB) = \text{tr}(BA)$. The key is to look at the definitions carefully:

$$\text{tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ji} = \sum_{j=1}^n \sum_{i=1}^m B_{ji} A_{ij} = \sum_{j=1}^n (BA)_{jj} = \text{tr}(BA)$$

2.3.6 Norm

The norm of a vector is a measure of the "length" of the vector. For example, the most commonly used norm in Euclidean space is the ℓ_2 -norm, defined as:

$$\|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

Note: $\|x\|_2 = \sqrt{x^T x}$

More formally, a norm is a function that satisfies the following four properties ($f: \mathbb{R}^n \rightarrow \mathbb{R}$):

- $f(x) \geq 0$ for all $x \in \mathbb{R}^n$ (non-negativity).
- $f(x) = 0$ if and only if $x = 0$ (definiteness).
- $f(\alpha x) = |\alpha|f(x)$ for all $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}^n$ (homogeneity).
- $f(x + y) \leq f(x) + f(y)$ for all $x, y \in \mathbb{R}^n$ (triangle inequality).

Other commonly used norms include:

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

In fact, the three norms mentioned above are all examples of ℓ_p -norms, which are defined as:

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

Many other norms exist, but they are beyond the scope of this course.

2.3.7 Linear Dependence and Rank

A set of vectors $x_1, x_2, \dots, x_n \in \mathbb{R}^n$ is linearly independent if no vector in the set can be written as a linear combination of the others. Otherwise, they are linearly dependent. Formally, the vectors are linearly independent if the equation

$$\sum_{i=1}^n \alpha_i x_i = 0$$

implies that $\alpha_i = 0$ for all i . Otherwise, they are linearly dependent.

For a set of scalars $\{a_1, \dots, a_{n-1}\} \subset \mathbb{R}$, if the vector $x_n = \sum_{i=1}^{n-1} a_i x_i$, then the vectors are linearly dependent. For example, in the system:

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

the vectors are linearly dependent because $x_2 = 2x_1$.

The column rank of a matrix $A \in \mathbb{R}^{m \times n}$ is the maximum number of linearly independent columns in the matrix. Similarly, the row rank is the maximum number of linearly independent rows. Due to the properties of matrix multiplication, **Due to the properties of matrix multiplication, the row rank and column rank of a matrix are always equal**, which is simply called the rank of the matrix.

For any matrix $A \in \mathbb{R}^{m \times n}$, the rank satisfies:

$$\text{rank}(A) \leq \min(m, n)$$

Some properties of the rank function include:

- For any $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = \text{rank}(A^T)$.
- For any matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
- For any matrices $A, B \in \mathbb{R}^{m \times n}$, $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

2.3.8 Inverse of a Matrix

For a matrix $A \in \mathbb{R}^{n \times n}$, the inverse is denoted as A^{-1} , and it is defined by the unique matrix:

$$A^{-1}A = I = AA^{-1}$$

Note that not all matrices have inverses. For example, non-square matrices do not have inverses by definition. However, some square matrices might not have inverses either. Specifically, if A^{-1} exists, we call A invertible or non-singular; otherwise, it is called non-invertible or singular. For a matrix A to have an inverse A^{-1} , it must satisfy certain conditions. Besides satisfying the basic conditions, it must also meet several other sufficient conditions. Below are properties of the inverse; suppose $A, B \in \mathbb{R}^{n \times n}$ and are non-singular matrices:

$$\begin{aligned}(A^{-1})^{-1} &= A \\ (AB)^{-1} &= B^{-1}A^{-1} \\ (A^T)^{-1} &= (A^{-1})^T\end{aligned}$$

Thus, the inverse matrix is usually denoted as A^{-1} . As an example of using inverses, consider the linear system $Ax = b$, where $A \in \mathbb{R}^{n \times n}$, $x, b \in \mathbb{R}^n$. If A is non-singular (i.e., invertible), then

$$x = A^{-1}b$$

(What if $A \in \mathbb{R}^{m \times n}$ is not a square matrix? Does this formula still work?)

2.3.9 Orthogonal Matrix

An orthogonal matrix is an important concept in linear algebra. A matrix Q is called an orthogonal matrix if it satisfies the following condition:

$$Q^T Q = Q Q^T = I$$

where Q^T is the transpose of matrix Q , and I is the identity matrix of the same dimension. This means that the inverse of an orthogonal matrix is equal to its transpose, i.e.,

$$Q^{-1} = Q^T$$

Orthogonal matrices have the following important properties:

- Preserve inner product: If Q is an orthogonal matrix, then for any vectors u and v ,

$$\langle Qu, Qv \rangle = \langle u, v \rangle$$

This means that an orthogonal transformation preserves the inner product of vectors, and thus preserves the angles and lengths of vectors.

- Preserve length: For any vector u ,

$$\|Qu\| = \|u\|$$

That is, an orthogonal matrix transformation does not change the length of a vector.

- Orthogonal column vectors: The column vectors of an orthogonal matrix are orthonormal, meaning each column vector is a unit vector and any two different column vectors are orthogonal to each other.
- Orthogonal row vectors: The row vectors of an orthogonal matrix have the same property as the column vectors, meaning each row vector is a unit vector and any two different row vectors are orthogonal to each other.

2.3.10 Quadratic Forms and Semi-Definite Matrices

Given a square matrix $A \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, the scalar value $x^T A x$ is called a quadratic form. Explicitly, we can write:

$$x^T A x = \sum_{i=1}^n x_i (A x)_i = \sum_{i=1}^n x_i \left(\sum_{j=1}^n A_{ij} x_j \right) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Note that:

$$x^T A x = (x^T A x)^T = x^T A^T x = x^T \left(\frac{1}{2} A + \frac{1}{2} A^T \right) x$$

The first equality holds because the transpose of a scalar is itself, while the second equality holds because A and A^T are generally not symmetric. Thus, we can conclude that only the symmetric part of A contributes to the quadratic form. Hence, we assume A is symmetric. We define:

- A symmetric matrix $A \in S^n$ is positive definite (PD) if $x^T A x > 0$ for all non-zero vectors $x \in \mathbb{R}^n$. This is denoted as $A > 0$, and the set of all positive definite matrices is usually denoted as $\mathbb{S}_{++}$.
- A symmetric matrix $A \in S^n$ is positive semidefinite (PSD) if $x^T A x \geq 0$ for all vectors $x \in \mathbb{R}^n$. This is denoted as $A \geq 0$, and the set of all positive semidefinite matrices is usually denoted as $\mathbb{S}_+$.
- Similarly, a symmetric matrix $A \in S^n$ is negative definite (ND) if $x^T A x < 0$ for all non-zero vectors $x \in \mathbb{R}^n$. This is denoted as $A < 0$.
- A symmetric matrix $A \in S^n$ is negative semidefinite (NSD) if $x^T A x \leq 0$ for all vectors $x \in \mathbb{R}^n$. This is denoted as $A \leq 0$.
- Finally, a symmetric matrix $A \in S^n$ is indefinite if it is neither positive semidefinite nor negative semidefinite, i.e., there exist vectors $x_1, x_2 \in \mathbb{R}^n$ such that $x_1^T A x_1 > 0$ and $x_2^T A x_2 < 0$.

Clearly, if A is positive definite, then $-A$ is negative definite, and vice versa. Similarly, if A is positive semidefinite, then $-A$ is negative semidefinite, and vice versa. If A is indefinite, then $-A$ is also indefinite.

Each of the following conditions is a necessary and sufficient condition to make a symmetric matrix A **positive semidefinite**:

1. $x^T A x \geq 0$ for any $n \times 1$ vector x .
2. All eigenvalues of A are nonnegative.
3. All the upper left (or lower right) submatrices A_K , $K = 1, \dots, n$ have nonnegative determinants.

Each of the following conditions is a necessary and sufficient condition to make a symmetric matrix A **positive definite**:

1. $x^T A x > 0$ for any nonzero $n \times 1$ vector x .
2. All eigenvalues of A are positive.
3. All the upper left (or lower right) submatrices A_K , $K = 1, \dots, n$ have positive determinants.

Example: Why is the covariance matrix semi-definite?

Solution: Covariance matrix $\mathbf{C}$ is calculated by the formula

$$\mathbf{C} \triangleq \mathbb{E}\{(\mathbf{x} - \bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})^T\}.$$

For an arbitrary real vector $\mathbf{u}$ we can write

$$\begin{aligned} \mathbf{u}^T \mathbf{C} \mathbf{u} &= \mathbf{u}^T \mathbb{E}\{(\mathbf{x} - \bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})^T\} \mathbf{u} \\ &= \mathbb{E}\{\mathbf{u}^T (\mathbf{x} - \bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{u}\} \\ &= \sigma_s^2. \end{aligned}$$

Where σ_s is the variance of the zero-mean scalar random variable S and it is a scalar real number whose value equals to.

$$\sigma_s = \mathbf{u}^T (\mathbf{x} - \bar{\mathbf{x}}) = (\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{u}.$$

Square of any real number is equal to or greater than zero. That is

$$\sigma_s^2 \geq 0.$$

Thus,

$$\mathbf{u}^T \mathbf{C} \mathbf{u} = \sigma_s^2 \geq 0.$$

Which implies that the covariance matrix of any real random vector is always semi-definite.

2.3.11 Eigenvalues and Eigenvectors

Given a square matrix $A \in \mathbb{R}^{n \times n}$, we consider the following conditions for $\lambda \in \mathbb{C}$ to be an eigenvalue of A and $\mathbf{x} \in \mathbb{C}^n$ to be the corresponding eigenvector:

$$A\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq 0$$

Intuitively, this definition means that multiplying the vector $\mathbf{x}$ by A results in a new vector that is in the same direction as $\mathbf{x}$, scaled by λ . Note that for any eigenvector $\mathbf{x} \in \mathbb{C}^n$ and scalar $c \in \mathbb{C}$, $A(c\mathbf{x}) = cA\mathbf{x} = c(\lambda\mathbf{x}) = \lambda(c\mathbf{x})$, so $c\mathbf{x}$ is also an eigenvector. Therefore, when discussing eigenvectors, we usually assume they have been normalized to length 1 (which still leaves some ambiguity since both $\mathbf{x}$ and $-\mathbf{x}$ are eigenvectors, but we must accept this).

We can rewrite the above equation to show that $(\lambda, \mathbf{x})$ is an eigenvalue-eigenvector pair:

$$(\lambda I - A)\mathbf{x} = 0, \quad \mathbf{x} \neq 0$$

This means that $\mathbf{x}$ is in the null space of $\lambda I - A$, and $\lambda I - A$ is singular. Thus, the determinant must be zero:

$$|\lambda I - A| = 0$$

Now, we can use the characteristic polynomial to expand $|\lambda I - A|$ into a polynomial in λ (of very high degree), whose degree is the dimension of A . This polynomial has eigenvalues $\lambda_1, \dots, \lambda_n$ as its roots. These are the eigenvalues of A , but some of them may be repeated. To find the eigenvector corresponding to an eigenvalue λ_i , we solve $(\lambda_i I - A)\mathbf{x} = 0$, since $\lambda_i I - A$ is singular, there is at least one non-zero solution.

Note that this method is practical for computing eigenvalues and eigenvectors numerically (remember that fully expanding the characteristic polynomial is one source of disputes).

Here are some properties of eigenvalues and eigenvectors (assuming $A \in \mathbb{R}^{n \times n}$ with eigenvalues $\lambda_1, \dots, \lambda_n$):

- The trace of A is equal to the sum of its eigenvalues:

$$\text{tr}(A) = \sum_{i=1}^n \lambda_i$$

- The determinant of A is equal to the product of its eigenvalues:

$$|A| = \prod_{i=1}^n \lambda_i$$

- The rank of A is equal to the number of non-zero eigenvalues of A .
- Suppose A is non-singular, with eigenvalue λ and eigenvector $\mathbf{x}$. Then $1/\lambda$ is the eigenvalue of A^{-1} corresponding to the eigenvector $\mathbf{x}$, i.e., $A^{-1}\mathbf{x} = (1/\lambda)\mathbf{x}$. (To prove this, take the eigenvector equation $A\mathbf{x} = \lambda\mathbf{x}$ and multiply both sides by A^{-1}).
- For a diagonal matrix with eigenvalues $d = \text{diag}(d_1, \dots, d_n)$, the eigenvalues are simply the diagonal elements $d_1, \dots, d_n$.

2.4 Matrix Decomposition

2.4.1 QR Decomposition

Definition: For a matrix $A \in \mathbb{R}^{m \times n}$, the QR decomposition is given by:

$$A = QR$$

where Q is an $m \times m$ orthogonal matrix ($Q^T Q = I$) and R is an $m \times n$ upper triangular matrix.

Process:

1. Apply the Gram-Schmidt process or Householder transformations to orthogonalize the columns of A .
2. Construct Q using the orthogonalized columns.
3. Compute $R = Q^T A$.

Applications:

- Solving linear systems
- Eigenvalue computations (QR algorithm)
- Least squares problems

2.4.2 LU Decomposition

Definition: For a matrix $A \in \mathbb{R}^{n \times n}$, the LU decomposition is:

$$A = LU$$

where A is an $n \times n$ matrix, L is a lower triangular matrix with ones on the diagonal, and U is an upper triangular matrix.

Process:

1. Apply Gaussian elimination to transform A into an upper triangular matrix U .
2. Construct L using the multipliers from the elimination process.

Applications:

- Solving linear systems
- Matrix inversion
- Determinant calculation

2.4.3 Cholesky Decomposition

Definition: For a **symmetric positive-definite matrix** $A \in \mathbb{R}^{n \times n}$, the Cholesky decomposition is:

$$A = LL^T$$

where L is a lower triangular matrix with positive diagonal elements.

Process:

1. Verify A is symmetric and positive-definite.
2. Compute the entries of L using the formula:

$$L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik}L_{jk} \right)$$

Applications:

- Solving linear systems
- Monte Carlo simulations
- Kalman filters

2.4.4 Singular Value Decomposition (SVD)

Definition: For a matrix $A \in \mathbb{R}^{m \times n}$, the singular value decomposition is:

$$A = U\Sigma V^T$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{m \times n}$ is a diagonal matrix with non-negative real numbers on the diagonal.

Process:

1. Compute the eigenvalues and eigenvectors of $A^T A$ and AA^T .
2. The columns of U are composed of the eigenvectors of AA^T .
3. The columns of V are composed of the eigenvectors of $A^T A$.
4. Construct Σ using the square roots of the non-zero eigenvalues.

Applications:

- Principal Component Analysis (PCA)
- Image compression
- Solving linear systems

2.5 Matrix Calculus

2.5.1 Matrix Gradient

Assume $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a function that takes a matrix $A \in \mathbb{R}^{m \times n}$ as input and returns a real number. The Jacobian matrix (with respect to $A \in \mathbb{R}^{m \times n}$) is defined as follows:

$$\nabla_A f(A) \in \mathbb{R}^{m \times n} = \begin{pmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \cdots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \cdots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \cdots & \frac{\partial f(A)}{\partial A_{mn}} \end{pmatrix}$$

That is, an $m \times n$ matrix:

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$$

Note that the dimensions of $\nabla_A f(A)$ are the same as the dimensions of A . In the special case where A is just a vector $\mathbf{a} \in \mathbb{R}^n$, the gradient is:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

It is important to remember that the Jacobian is defined only if the function returns a real value. For example, for $A \in \mathbb{R}^{m \times n}$ with respect to $\mathbf{x}$, we cannot take the Jacobian of $A\mathbf{x}$ since this quantity is a vector. It directly follows the equality properties of the gradient:

$$\nabla_{\mathbf{x}}(f(\mathbf{x}) + g(\mathbf{x})) = \nabla_{\mathbf{x}} f(\mathbf{x}) + \nabla_{\mathbf{x}} g(\mathbf{x})$$

For $t \in \mathbb{R}$,

$$\nabla_{\mathbf{x}}(tf(\mathbf{x})) = t\nabla_{\mathbf{x}} f(\mathbf{x})$$

In principle, the Jacobian is a natural extension of the gradient to multivariable functions. However, in practice, due to the sign notation, using the Jacobian can be very difficult. For example, suppose $A \in \mathbb{R}^{m \times n}$ is a fixed coefficient matrix and $\mathbf{z} \in \mathbb{R}^m$ is a fixed coefficient vector. Define $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ by $f(\mathbf{z}) = \mathbf{z}^T \mathbf{z}$. Therefore, $\nabla_{\mathbf{z}} f(\mathbf{z}) = 2\mathbf{z}$.

- **Gradient of a Linear Function:**

Consider the linear function $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$, where $\mathbf{a} \in \mathbb{R}^n$ is a constant vector. The gradient of f with respect to $\mathbf{x}$ is:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \nabla_{\mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}$$

- **Gradient of a Quadratic Function:**

Consider the quadratic function $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, where $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix. The gradient of f with respect to $\mathbf{x}$ is:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \nabla_{\mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = 2A\mathbf{x}$$

If A is not symmetric, we have:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = (A + A^T)\mathbf{x}$$

• **Gradient of the Quadratic Form:**

Consider the function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A \mathbf{x}$, where $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix. The gradient of f with respect to $\mathbf{x}$ is:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \nabla_{\mathbf{x}} \left(\frac{1}{2} \mathbf{x}^T A \mathbf{x} \right) = A \mathbf{x}$$

If A is not symmetric, we have:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2}(A + A^T)\mathbf{x}$$

These gradient formulas are fundamental in optimization and various fields of applied mathematics. They serve as building blocks for more complex calculations in machine learning, control theory, and other areas.

2.5.2 Derivation of Normal Equations

The normal equations are used to find the best-fitting line in the context of linear regression. Given a set of data points (x_i, y_i) for $i = 1, 2, \dots, n$, we aim to find the parameters $\mathbf{w}$ that minimize the sum of squared errors.

The linear regression model can be written as:

$$\mathbf{y} = X\mathbf{w} + \mathbf{e}$$

where:

- $\mathbf{y} \in \mathbb{R}^n$ is the vector of observed values.
- $X \in \mathbb{R}^{n \times p}$ is the design matrix, where each row represents a data point and each column represents a feature.
- $\mathbf{w} \in \mathbb{R}^p$ is the vector of parameters to be estimated.
- $\mathbf{e} \in \mathbb{R}^n$ is the vector of errors or residuals.

Objective Function

The objective is to minimize the sum of squared errors:

$$J(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|^2$$

This can be expanded as:

$$J(\mathbf{w}) = (\mathbf{y} - X\mathbf{w})^T (\mathbf{y} - X\mathbf{w})$$

Derivation

To find the minimum, we take the derivative of $J(\mathbf{w})$ with respect to $\mathbf{w}$ and set it to zero:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = \frac{\partial}{\partial \mathbf{w}} (\mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T X \mathbf{w} + \mathbf{w}^T X^T X \mathbf{w}) = 0$$

Calculating the partial derivatives:

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{y}^T \mathbf{y}) = 0$$

$$\frac{\partial}{\partial \mathbf{w}} (-2\mathbf{y}^T X \mathbf{w}) = -2X^T \mathbf{y}$$

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^T X^T X \mathbf{w}) = 2X^T X \mathbf{w}$$

Setting the derivative to zero:

$$-2X^T \mathbf{y} + 2X^T X \mathbf{w} = 0$$

Simplifying, we get the normal equation:

$$X^T X \mathbf{w} = X^T \mathbf{y}$$

Solution

The normal equation can be solved for $\mathbf{w}$ (assuming $X^T X$ is invertible):

$$\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y}$$

2.6 Determinants

2.6.1 Introduction and Properties of Determinants

A determinant is a scalar value that can be computed from the elements of a square matrix. It provides important properties of the matrix, including whether the matrix is invertible, the volume distortion during linear transformation, and the eigenvalues of the matrix.

Given an $n \times n$ matrix $A = (a_{ij})$, the determinant of A , denoted as $|A|$ or $\det(A)$, can be computed using various methods, such as expansion by minors or Laplace expansion.

1. Expansion by Minors (Cofactors)

For an $n \times n$ matrix A :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is the (i, j) -minor of A , i.e., the determinant of the $(n-1) \times (n-1)$ matrix that results from deleting the i -th row and j -th column from A .

Alternatively,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

2. Basic Properties of Determinants

- (1) **Linearity:** The determinant is a linear function of each row of the matrix.
- (2) **Multiplicative Property:** For any two $n \times n$ matrices A and B :

$$\det(AB) = \det(A) \cdot \det(B)$$

- (3) **Effect of Row (or column) Operations:**

- Swapping two rows of a matrix multiplies its determinant by -1 .
- Multiplying a row by a scalar k multiplies the determinant by k .
- Adding a scalar multiple of one row to another row does not change the determinant.

- (4) **Transpose:** The determinant of a matrix is equal to the determinant of its transpose:

$$\det(A) = \det(A^T)$$

(5) **Inverse:** If A is invertible, then:

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

(6) **Block Matrices:** For block diagonal matrices:

$$\det \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \det(A) \cdot \det(B)$$

(7) **Eigenvalues:** If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A , then:

$$\det(A) = \prod_{i=1}^n \lambda_i$$

2.6.2 Vandermonde Determinant

The Vandermonde determinant is a special type of determinant that arises in polynomial interpolation and other areas of numerical analysis.

Given numbers $x_1, x_2, \dots, x_n$, the Vandermonde matrix is:

$$V = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_n \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & x_3^{n-1} & \cdots & x_n^{n-1} \end{pmatrix}$$

The determinant of the Vandermonde matrix is given by:

$$\det(V) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

This product is zero if and only if at least two of the x_i 's are equal, which highlights the importance of distinct values in polynomial interpolation.

2.7 Classic Questions

2.7.1 Eigenvalues and Eigenvectors

If matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, what are the eigenvalues and eigenvectors of A ?

Solution:

This is a simple example of eigenvalues and eigenvectors. It can be solved using three related approaches:

Approach A: Apply the definition of eigenvalues and eigenvectors directly.

Let λ be an eigenvalue and $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be its corresponding eigenvector. By definition, we have

$$Ax = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{pmatrix} = \lambda x = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix}$$

So either $\lambda = 3$, in which case $x_1 = x_2$ (plug $\lambda = 3$ into equation $2x_1 + x_2 = \lambda x_1$) and the corresponding normalized eigenvector is $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, or $x_1 + x_2 = 0$, in which case the normalized eigenvector is $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$ and

$\lambda = 1$ (plug $x_2 = -x_1$ into equation $2x_1 + x_2 = \lambda x_1$).

Approach B: Use equation $\det(A - \lambda I) = 0$.

$$\det(A - \lambda I) = 0 \Rightarrow (2 - \lambda)(2 - \lambda) - 1 = 0.$$

Solving the equation, we have $\lambda_1 = 1$ and $\lambda_2 = 3$. Applying the eigenvalues to $Ax = \lambda x$, we can get the corresponding eigenvectors.

Approach C: Use equations $\lambda_1 \cdot \lambda_2 \cdots \lambda_n = \det(A)$ and $\sum_{i=1}^n \lambda_i = \text{trace}(A) = \sum_{i=1}^n A_{i,i}$.

$$\det(A) = 2 \times 2 - 1 \times 1 = 3 \quad \text{and} \quad \text{trace}(A) = 2 \times 2 = 4.$$

So we have

$$\begin{cases} \lambda_1 \times \lambda_2 = 3 \\ \lambda_1 + \lambda_2 = 4 \end{cases} \Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 3 \end{cases}$$

Again, applying the eigenvalues to $Ax = \lambda x$, we can get the corresponding eigenvectors.

2.7.2 Semidefiniteness Property

There are 3 random variables x, y and z . The correlation between x and y is 0.8 and the correlation between x and z is 0.8. What is the maximum and minimum correlation between y and z ?

Solution:

The problem can be solved using the positive semidefiniteness property of the correlation matrix.

Let the correlation between y and z be ρ , then the correlation matrix for x, y and z is

$$P = \begin{pmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & \rho \\ 0.8 & \rho & 1 \end{pmatrix}.$$

$$\det(P) = 1 \times \det \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} - 0.8 \times \det \begin{pmatrix} 0.8 & 0.8 \\ \rho & 1 \end{pmatrix} + 0.8 \times \det \begin{pmatrix} 0.8 & 0.8 \\ 1 & \rho \end{pmatrix}$$

$$= (1 - \rho^2) - 0.8 \times (0.8 - 0.8\rho) + 0.8 \times (0.8\rho - 0.8)$$

$$= -0.28 + 1.28\rho - \rho^2 \geq 0$$

$$\Rightarrow (\rho - 1)(\rho - 0.28) \leq 0 \Rightarrow 0.28 \leq \rho \leq 1$$

So the maximum correlation between y and z is 1, the minimum is 0.28.

2.7.3 Power to the Matrix

Let $A = \begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}$. Find $A^{10}v$, where $v = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. The answer is in the form

$$\begin{bmatrix} a \cdot b^p + c \\ d \cdot b^p - g \end{bmatrix}$$

For integers a, b, c, d, p . Find $bp + a + c + d + g$.

solution:

We first need to find the eigenvalues of A . The characteristic polynomial is $p(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$. It is easy to see that $\text{tr}(A) = 6$ and $\det(A) = 5$, so we need to solve $\lambda^2 - 6\lambda + 5 = 0$. This easily factors to $(\lambda - 5)(\lambda - 1) = 0$, so $\lambda = 1, 5$ are the eigenvalues.

Next, we want the eigenvectors corresponding to each eigenvalue. We do this by solving $(A - \lambda I_2)x = 0$. For $\lambda = 1$, this becomes

$$\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The solution set to this is $x = t \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ for $t \in \mathbb{R}$. Our eigenvector is therefore $e_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

For $\lambda = 5$, this equation becomes

$$\begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The solution set to this is $x = t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ for $t \in \mathbb{R}$. Our eigenvector is therefore $e_5 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

We now need to write v in terms of our eigenvector basis. However, it is fairly clear to see that $v = e_1 + 2e_5$. By the fact that e_1 and e_5 are eigenvalues of A , we have that

$$A^{10}v = A^{10}(e_1 + 2e_5) = A^{10}e_1 + 2A^{10}e_5 = 1^{10} \cdot e_1 + 2 \cdot 5^{10}e_5 = \begin{bmatrix} 2 \cdot 5^{10} + 1 \\ 4 \cdot 5^{10} - 2 \end{bmatrix}$$

This means our answer is $5 \cdot 10 + 1 + 1 + 4 + 2 + 2 = 59$.

2.7.4 Matrix Exponential

Find $\text{trace}(e^A)$ to 3 decimal points, where A is defined as

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}$$

The answer will be in the form $e^a + e^b$ for integers a and b . Find ab .

solution:

Recall that e^A can be rewritten as

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

In addition, note the following:

$$\begin{aligned} \text{trace}(e^A) &= \text{trace} \left(\sum_{k=0}^{\infty} \frac{1}{k!} A^k \right) \\ &= \sum_{k=0}^{\infty} \text{trace} \left(\frac{1}{k!} A^k \right) \\ &= \text{trace} \left(\begin{bmatrix} \sum_{k=0}^{\infty} \frac{3^k}{k!} & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{6^k}{k!} \end{bmatrix} \right) \\ &= e^3 + e^6 \end{aligned}$$

Therefore, our answer is $3 \cdot 6 = 18$.

2.7.5 LU and Cholesky decomposition

What is the relationship between the LU and Cholesky decompositions?

Answer:

Both the LU decomposition (without pivoting) and the Cholesky decomposition provide a computationally efficient way to solve linear systems by only using forward substitution and backward substitution, which are very fast solvers. The decompositions are similar in form, i.e., the given matrix is written as the product of a lower triangular matrix and an upper triangular matrix.

More precisely, the LU decomposition without pivoting of a nonsingular square matrix A consists of finding a lower triangular matrix L with all entries on the main diagonal equal to 1 and a nonsingular upper triangular matrix U such that

$$A = LU.$$

The matrices L and U are called the LU factors of A .

It is important to note that the LU decomposition without pivoting does not exist for every nonsingular matrix: a matrix has an LU decomposition without pivoting if and only if all the leading principal minors of the matrix are nonzero.¹² This drawback is addressed by introducing the LU decomposition with row (or column) pivoting, which exists for any nonsingular matrix.

The Cholesky decomposition of a nonsingular symmetric matrix A consists of finding a nonsingular upper triangular matrix U with positive entries on the main diagonal such that

$$A = U^T U.$$

The matrix U is called the Cholesky factor of A .

Note that not every symmetric matrix has a Cholesky decomposition: The nonsingular symmetric matrix A has a Cholesky decomposition if and only if the matrix A is symmetric positive definite, i.e., if $x^T A x > 0$ for all $x \neq 0$, or, equivalently, if all the eigenvalues of the matrix are positive.

If they exist, both decompositions are unique. In other words, if a nonsingular matrix has an LU decomposition without pivoting, the L and U factors are uniquely determined. Similarly, any symmetric positive definite matrix has a uniquely determined Cholesky factor.

2.7.6 Cholesky Decomposition and Singular Value Decomposition in Monte Carlo Simulation

How do you generate two $N(0, 1)$ (standard normal distribution) random variables with correlation ρ if you have a random number generator for standard normal distribution?

solution:

Method 1: Two $N(0, 1)$ random variables x_1, x_2 with a correlation ρ can be generated from independent $N(0, 1)$ random variables z_1, z_2 using the following equations:

$$\begin{aligned} x_1 &= z_1 \\ x_2 &= \rho z_1 + \sqrt{1 - \rho^2} z_2 \end{aligned}$$

It is easy to confirm that $\text{var}(x_1) = \text{var}(z_1) = 1$, $\text{var}(x_2) = \rho^2 \text{var}(z_1) + (1 - \rho^2) \text{var}(z_2) = 1$, and $\text{cov}(x_1, x_2) = \text{cov}(z_1, \rho z_1 + \sqrt{1 - \rho^2} z_2) = \rho$.

This approach is a basic example using Cholesky decomposition to generate correlated random numbers. To generate correlated random variables that follow a n -dimensional multivariate normal distribution $X = [X_1, X_2, \dots, X_n]^T \sim N(\mu, \Sigma)$ with mean $\mu = [\mu_1, \mu_2, \dots, \mu_n]^T$ and covariance matrix Σ (a $n \times n$ positive definite matrix), we can decompose the covariance matrix Σ into $R^T R$ and generate n independent $N(0, 1)$ random variables $z_1, z_2, \dots, z_n$. Let vector $Z = [z_1, z_2, \dots, z_n]^T$, then X can be generated as $X = \mu + R^T Z$.¹

Method 2: X can also be generated using another important matrix decomposition called **singular value decomposition (SVD)**: For any $n \times p$ matrix X , there exists a factorization of the form $X = UDV^T$, where U and V are $n \times p$ and $p \times p$ orthogonal matrices, with columns of U spanning the column space of X , and the columns of V spanning the row space; D is a $p \times p$ diagonal matrix called the singular values of X . For a positive definite covariance matrix, we have $V = U$ and $\Sigma = UDU^T$.

Furthermore, D is the diagonal matrix of eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and U is the matrix of n corresponding eigenvectors. Let $D^{1/2}$ be a diagonal matrix with diagonal elements $\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n}$, then it is clear that $D = (D^{1/2})^2 = (D^{1/2})(D^{1/2})^T$ and $\Sigma = U D^{1/2} (U D^{1/2})^T$. Again, if we generate a vector of n independent $N(0, 1)$ random variables $Z = [z_1, z_2, \dots, z_n]^T$, X can be generated as $X = \mu + (UD^{1/2})Z$.

¹This approach is a basic example using Cholesky decomposition to generate correlated random numbers.

3 Probability Theory

3.1 Fundamental of Probability

3.1.1 Basic Elements of Probability

To define the concept of probability on a set, we need some basic elements:

- **Sample space Ω :** The set of all possible outcomes of a random experiment. Here, each outcome $\omega \in \Omega$ can be considered a complete description of the state of the world at the end of the experiment.
- **Event space (event field) $\mathcal{F}$:** A collection of subsets of Ω (called events), where each subset $A \subseteq \Omega$ is considered an event.
 - Note: $\mathcal{F}$ must satisfy the following three conditions:
 1. $\emptyset \in \mathcal{F}$
 2. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$
 3. If $A_1, A_2, \dots, A_i \in \mathcal{F}$, then $(\bigcup_i A_i) \in \mathcal{F}$
- **Probability measure P :** A function P is a mapping from $\mathcal{F}$ to $\mathbb{R}$ that satisfies the following properties:
 - For each $A \in \mathcal{F}$, $P(A) \geq 0$
 - $P(\Omega) = 1$
 - If $A_1, A_2, \dots$ are mutually exclusive events (i.e., when $i \neq j$, $A_i \cap A_j = \emptyset$), then:

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$$

The above three properties are known as the axioms of probability.

Example:

Consider the experiment of rolling a six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The simplest event space is $\mathcal{F} = \{\emptyset, \Omega\}$, and another event space is the collection of all subsets of Ω . For the first event space, the only probability measure that satisfies the requirements is $P(\emptyset) = 0$, $P(\Omega) = 1$. For the second event space, a valid probability measure is to distribute the probability of each event equally, i.e., each event in this event space is assigned a probability of $1/6$; here, it is the number of elements in the event set divided by 6. For example, $P(\{1, 2, 3, 4\}) = 4/6$, $P(\{1, 2, 3\}) = 3/6$.

Properties:

- If $A \subseteq B$, then $P(A) \leq P(B)$
- $P(A \cap B) \leq \min(P(A), P(B))$
- (Boole's Inequality): $P(A \cup B) \leq P(A) + P(B)$
- $P(\Omega \setminus A) = 1 - P(A)$
- (Total Probability Law): If $A_1, \dots, A_k$ are mutually exclusive events and their union is Ω , then the sum of their probabilities is 1.

3.1.2 Conditional Probability and Independence

Suppose B is an event with $P(B) > 0$, we define the conditional probability of A given B as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

In other words, $P(A|B)$ is the probability of A occurring under the condition that B has already occurred. Two events A and B are said to be independent if $P(A \cap B) = P(A)P(B)$ (or equivalently, $P(A|B) = P(A)$). Thus, independence means that the occurrence of event B has no effect on the probability of event A .

3.1.3 Bayes Rules

When attempting to derive the conditional probability expression of one variable given another variable, a useful formula that often appears is:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Using the law of total probability, we can express $P(B)$ as

$$P(B) = \sum_i P(B|A_i) \cdot P(A_i)$$

Thus, Bayes' theorem can also be written as

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{\sum_i P(B|A_i) \cdot P(A_i)}$$

3.2 Random Variables

Consider an experiment where we flip 10 coins, and we want to know the number of heads in the outcome. Here, the sample space Ω consists of 2^{10} outcomes. For example, we may have $\omega = \{H, H, T, H, T, T, H, T, T, H\}$. However, in practical applications, we are often not concerned with the detailed reflection of all possible outcomes. Instead, we are often interested in the statistical properties of these outcomes, such as the average number of heads in 10 coin flips, or the length of a rod. In some technical fields, these statistical properties are represented by random variables.

In simple terms, a random variable is a function that maps the sample space to a real number. We usually use uppercase letters such as $X(\omega)$ to denote random variables. When we need to express a certain value of a random variable, we use lowercase letters such as x .

Example: Consider the experiment of flipping a coin, let X denote the number of heads observed in 10 flips. Suppose the outcome of the experiment is denoted by ω . Then, $X(\omega)$ can take on the values 0 to 10. This random variable X can be considered as a binomial random variable in this case.

3.2.1 Cumulative Distribution Function

To specify the distribution of a random variable, we often use functions such as the cumulative distribution function (CDF), probability density function (PDF), and probability mass function (PMF). In this section, we will focus on introducing these concepts.

The cumulative distribution function (CDF) of X is defined as:

$$F_X(x) = P(X \leq x)$$

By using this function, we can calculate the probability of X taking on values less than or equal to x .

Properties:

- $0 \leq F_X(x) \leq 1$

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow \infty} F_X(x) = 1$
- If $x \leq y$, then $F_X(x) \leq F_X(y)$

3.2.2 Probability Mass Function

PMF is for discrete random variables and gives the probability of specific values. When a random variable has a finite number of possible values, X is called a discrete random variable. One way to specify the probability distribution of a discrete random variable is to list the probabilities of each possible value it can take. Specifically, the probability mass function (PMF) is a function $p_X : \Omega \rightarrow \mathbb{R}$ defined as:

$$p_X(x) = P(X = x)$$

In the case of discrete random variables, we often use $\text{Val}(X)$ to denote the set of all possible values that X can take. For example, if $X(\omega)$ is a discrete random variable representing the number of heads in 10 coin flips, then $\text{Val}(X) = \{0, 1, 2, \dots, 10\}$.

Properties:

- $0 \leq p_X(x) \leq 1$
- $\sum_{x \in \text{Val}(X)} p_X(x) = 1$
- $\sum_{x \in A} p_X(x) = P(X \in A)$

3.2.3 Probability Density Function

PDF is for continuous random variables and gives the density of the probability over intervals. For some continuous random variables, the cumulative distribution function $F_X(x)$ is differentiable. In such cases, we define the Probability Density Function (PDF) as the derivative of the cumulative distribution function, i.e.,

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Note that the probability density function of a continuous random variable may not always exist (i.e., the derivative may not always exist). By the properties of the cumulative distribution function, for very small Δx ,

$$P(x \leq X \leq x + \Delta x) \approx f_X(x) \Delta x$$

CDF and PDF (when they exist) are used to calculate the probabilities of different events. **However, it should be noted that the value of the Probability Density Function (PDF) at a given point is not the probability of that event, i.e., $f_X(x) \neq P(X = x)$.** For example, $f_X(x)$ can be greater than 1 (but the integral of $f_X(x)$ over any subset of $\mathbb{R}$ is at most 1).

Properties:

- $f_X(x) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- $\int_A f_X(x) dx = P(X \in A)$

3.2.4 Expectation

Assume X is a discrete random variable with PMF $p_X(x)$, and $g : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary function. Under this circumstance, $g(X)$ can be considered as a random variable. We define the expectation of $g(X)$ as:

$$E[g(X)] \triangleq \sum_{x \in \text{val}(x)} g(x)p_X(x)$$

If X is a continuous random variable with PDF $f_X(x)$, then the expectation of $g(X)$ is defined as:

$$E[g(X)] \triangleq \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

Intuitively, the expectation of $g(X)$ can be considered as the weighted average of the values $g(x)$ can take, with weights given by their respective probabilities.

Properties:

- For any constant $a \in \mathbb{R}$, $E[a] = a$
- For any constant $a \in \mathbb{R}$, $E[af(X)] = aE[f(X)]$
- Linearity: $E[f(X) + g(X)] = E[f(X)] + E[g(X)]$
- For a discrete random variable X , $E[1\{X = k\}] = P(X = k)$

Example:

Suppose for some subset $A \subseteq \Omega$, we have $g(x) = 1_{\{x \in A\}}$. Calculate $E[g(X)]$?

Solution:

Under discrete conditions:

$$E[g(X)] = \sum_{x \in \text{val}(X)} 1_{\{x \in A\}} P_X(x) = \sum_{x \in A} P_X(x) = P(x \in A)$$

Under continuous conditions:

$$E[g(X)] = \int_{-\infty}^{\infty} 1_{\{x \in A\}} f_X(x)dx = \int_{x \in A} f_X(x)dx = P(x \in A)$$

3.2.5 Variance

The variance of a random variable measures the spread of its distribution around the mean. Formally, the variance of a random variable X is defined as:

$$\text{Var}(X) \triangleq E[(X - E(X))^2]$$

Using the properties above, we can derive an alternative expression for the variance:

$$E[(X - E(X))^2] = E[X^2 - 2E[X]X + E[X]^2] = E[X^2] - 2E[X]E[X] + E[X]^2 = E[X^2] - E[X]^2$$

Here, the second equation follows from the linearity of expectation, and $E[X]$ is a constant.

Properties:

- For any constant $a \in \mathbb{R}$, $\text{Var}[a] = 0$

- For any constant $a \in \mathbb{R}$, $\text{Var}[af(X)] = a^2 \text{Var}[f(X)]$

Example:

- Calculate the expectation and variance of a uniformly distributed random variable on $[0, 1]$, with PDF $p_X(x) = 1$, elsewhere $p_X(x) = 0$.

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x dx = \frac{1}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$$

3.2.6 Moments

In probability theory, moments are quantitative measures related to the shape of a distribution's probability density function. Here are the primary moments used in probability theory:

1. First Moment (Mean or Expected Value) The first moment about the origin is the mean (or expected value) of a random variable X . It is denoted by μ or $E[X]$ and is calculated as:

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (\text{for continuous random variables})$$

$$E[X] = \sum_{x \in \text{range of } X} x P(X = x) \quad (\text{for discrete random variables})$$

2. Second Moment (Variance) The second moment about the mean is the variance, which measures the spread or dispersion of the random variable from the mean. It is denoted by σ^2 or $\text{Var}(X)$ and is calculated as:

$$\text{Var}(X) = E[(X - E[X])^2]$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

3. Third Moment (Skewness) The third moment about the mean measures the asymmetry of the distribution. It is calculated as:

$$\text{Skewness} = E \left[\left(\frac{X - E[X]}{\sigma} \right)^3 \right]$$

A positive skewness indicates a distribution with a tail on the right side, and a negative skewness indicates a distribution with a tail on the left side.

4. Fourth Moment (Kurtosis) The fourth moment about the mean measures the "tailedness" of the distribution. It is calculated as:

$$\text{Kurtosis} = E \left[\left(\frac{X - E[X]}{\sigma} \right)^4 \right]$$

Higher kurtosis indicates a distribution with heavier tails, while lower kurtosis indicates a distribution with

lighter tails.

5. Central Moments The n -th central moment of a random variable X is the expected value of the n -th power of the deviations from the mean. It is given by:

$$\mu_n = E[(X - E[X])^n]$$

For example:

- The first central moment is zero.
- The second central moment is the variance.
- The third and fourth central moments are related to skewness and kurtosis, respectively.

3.3 Common Univariate Random Variables

3.3.1 Discrete Random Variables

- **Bernoulli Distribution:** The probability of flipping heads is p (where $0 \leq p \leq 1$). If heads occurs, it is 1, otherwise it is 0.

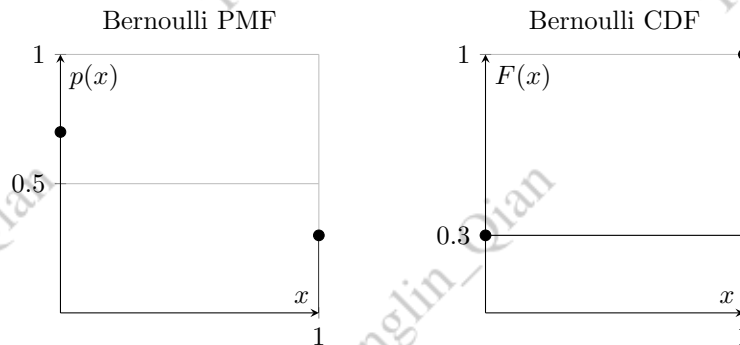
– **PMF:**

$$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

– **CDF:**

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

– **PMF and CDF Plots:**



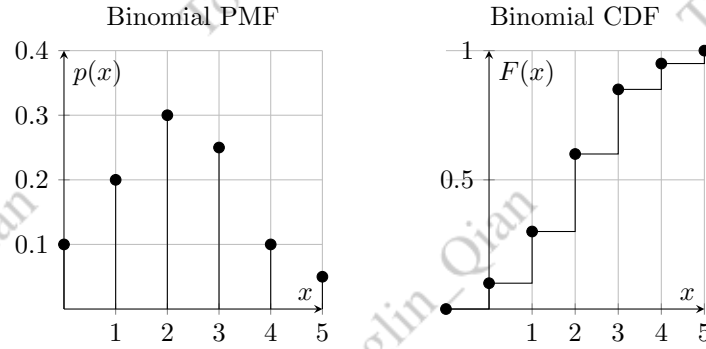
- **Binomial Distribution:** The number of heads in n independent flips of a coin with probability p of heads (where $0 \leq p \leq 1$).

– **PMF:**

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

– **CDF:**

$$F(x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1 - p)^{n-k}$$



– **PMF and CDF Plots:**

- **Geometric Distribution:** The number of trials needed to get the first head when flipping a coin with probability p of heads (where $p > 0$).

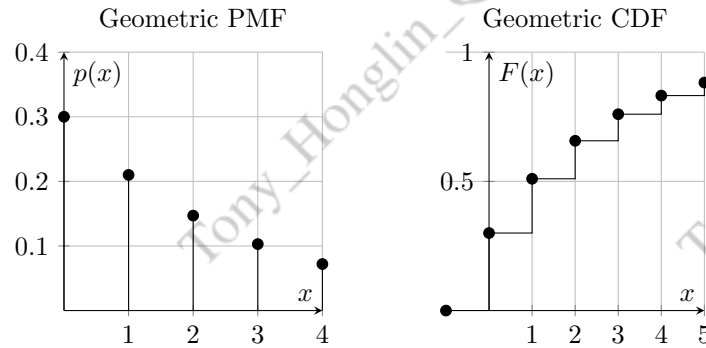
– **PMF:**

$$p(x) = (1-p)^x p, \quad x = 0, 1, 2, \dots$$

– **CDF:**

$$F(x) = 1 - (1-p)^{x+1}, \quad x = 0, 1, 2, \dots$$

– **PMF and CDF Plots:**



- **Poisson Distribution:** The Poisson distribution is a commonly used discrete probability distribution that describes the number of events occurring within a fixed interval of time or space. (where $\lambda > 0$).

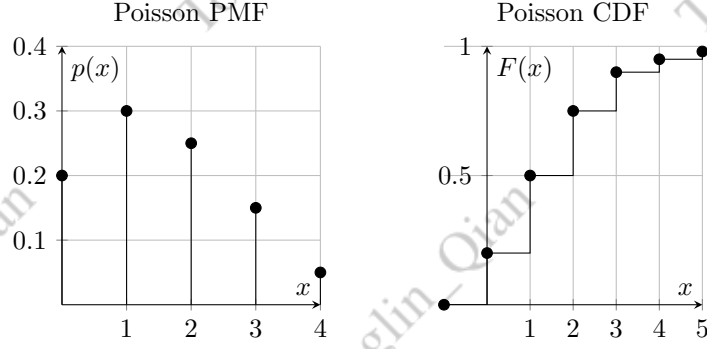
– **PMF:**

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

– **CDF:**

$$F(x) = e^{-\lambda} \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k}{k!}$$

– **PMF and CDF Plots:**



3.3.2 Continuous Random Variables

- **Uniform Distribution:**

- **Description:** The uniform distribution is used to model situations where all outcomes in a given interval are equally likely. It is commonly used in simulations and for generating random numbers.
- **PDF:**

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- **CDF:**

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b \end{cases}$$

- **Exponential Distribution:**

- **Description:** The exponential distribution is used to model the time between events in a Poisson process. It is often used in reliability analysis, survival analysis, and queuing theory.
- **PDF:**

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- **CDF:**

$$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

Memoryless Property The memoryless property can be formally stated as:

$$P(X > s + t \mid X > t) = P(X > s)$$

for all $s, t \geq 0$.

In words, this means that the probability that the event will take more than $s + t$ time given that it has already taken more than t time is the same as the probability that it will take more than s time from the beginning.

- **Normal Distribution:**

- **Description:** The normal distribution is used to model many natural phenomena and measurement errors due to its property of the central limit theorem. It is widely used in statistics, finance, and natural sciences.
- **PDF:**

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- CDF:

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma\sqrt{2}} \right) \right]$$

- **Log-normal Distribution:**

- **Description:** The log-normal distribution is used to model data that are positively skewed, such as incomes, stock prices, and certain types of biological data.
- **PDF:**

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}, \quad x > 0$$

- CDF:

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{\ln x - \mu}{\sigma\sqrt{2}} \right) \right], \quad x > 0$$

- **Chi-Square Distribution:**

- **Description:** The chi-square distribution is used in hypothesis testing, particularly in tests of independence and goodness-of-fit. It is also used in estimating variances.
- **Random Variable Definition:** If $Z_1, Z_2, \dots, Z_k$ are independent standard normal random variables, then the chi-square distribution with k degrees of freedom is defined as:

$$X = \sum_{i=1}^k Z_i^2$$

- **PDF:**

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$$

- **CDF:**

$$F(x) = \frac{\gamma\left(\frac{k}{2}, \frac{x}{2}\right)}{\Gamma\left(\frac{k}{2}\right)}$$

- **t-Distribution:**

- **Description:** The t-distribution is used in statistics for hypothesis testing and constructing confidence intervals when the sample size is small and the population standard deviation is unknown.
- **Random Variable Definition:** If Z is a standard normal random variable and V is a chi-square random variable with ν degrees of freedom, and Z and V are independent, then the t-distribution can be defined as:

$$T = \frac{Z}{\sqrt{V/\nu}}$$

- **PDF:**

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

- **CDF:**

$$F(x) = I_{\frac{x^2}{\nu+x^2}}\left(\frac{1}{2}, \frac{\nu}{2}\right)$$

- **F-Distribution:**

- **Description:** The F-distribution is used in analysis of variance (ANOVA) and in comparing two variances. It is commonly used to test the hypothesis that two populations have the same variance.

- **Random Variable Definition:** If U and V are independent chi-square random variables with ν_1 and ν_2 degrees of freedom respectively, then the F-distribution can be defined as:

$$F = \frac{(U/\nu_1)}{(V/\nu_2)}$$

- **PDF:**

$$f(x) = \frac{\sqrt{\frac{(\nu_1 x)^{\nu_1} \nu_2^{\nu_2}}{(\nu_1 x + \nu_2)^{\nu_1 + \nu_2}}}}{x B\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)}$$

- **CDF:**

$$F(x) = I_{\frac{\nu_1 x}{\nu_1 x + \nu_2}}\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)$$

3.4 Box-Muller Transform

The Box-Muller transform is a method for generating normally distributed random numbers from uniformly distributed random numbers. It has two common forms: the basic form and the polar form.

3.4.1 Basic Form

The basic form of the Box-Muller transform uses two independent uniformly distributed random variables U_1 and U_2 , which are in the range $(0, 1]$, to generate two independent standard normal random variables Z_0 and Z_1 .

The basic formulas are:

$$Z_0 = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$$

$$Z_1 = \sqrt{-2 \ln U_1} \sin(2\pi U_2)$$

3.4.2 Polar Form

The polar form of the Box-Muller transform can be derived by considering the transformation in polar coordinates. Given two independent uniformly distributed random variables U_1 and U_2 , with values in $(0, 1]$, define R and Θ as follows:

$$R = \sqrt{-2 \ln U_1}$$

$$\Theta = 2\pi U_2$$

Then, the two standard normal random variables Z_0 and Z_1 can be expressed as:

$$Z_0 = R \cos(\Theta)$$

$$Z_1 = R \sin(\Theta)$$

Distributions Summary Table

Distribution	PDF/PMF	Mean & Variance
Bernoulli	$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$	$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p)$
Binomial	$p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$	$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p)$
Geometric	$p(x) = (1 - p)^x p$	$\mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}$
Poisson	$p(x) = \frac{\lambda^k e^{-\lambda}}{k!}$	$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda$
Uniform	$f(x) = \frac{1}{b - a}, \quad a \leq x \leq b$	$\mathbb{E}[X] = \frac{a + b}{2}, \quad \text{Var}(X) = \frac{(b - a)^2}{12}$
Exponential	$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$	$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$
Normal	$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2$
Log-normal	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}, \quad x > 0$	$\mathbb{E}[X] = e^{\mu + \frac{\sigma^2}{2}}, \quad \text{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$
t-Distribution	$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$\mathbb{E}[X] = 0 (\nu > 1), \quad \text{Var}(X) = \frac{\nu}{\nu - 2} (\nu > 2)$
Chi-Square	$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$	$\mathbb{E}[X] = k, \quad \text{Var}(X) = 2k$
F-Distribution	$f(x) = \frac{\sqrt{\frac{(\nu_1 x)^{\nu_1} \nu_2^{\nu_2}}{(\nu_1 x + \nu_2)^{\nu_1 + \nu_2}}}}{xB\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)}$	$\mathbb{E}[X] = \frac{\nu_2}{\nu_2 - 2} (\nu_2 > 2), \quad \text{Var}(X) = \frac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)} (\nu_2 > 4)$

3.5 Multivariate Random Variables

Multivariate random variables involve two or more random variables and their joint behavior. Understanding multivariate random variables is crucial in various fields such as statistics, economics, engineering, and the natural sciences because real-world phenomena often depend on multiple interrelated factors.

3.5.1 Discrete Multivariate Random Variables

A discrete multivariate random variable is a vector of discrete random variables. For instance, consider two discrete random variables X and Y which can take values in a finite or countable set. Their joint probability distribution is given by the joint probability mass function (pmf):

$$P(X = x_i, Y = y_j) = p_{ij}, \quad i, j = 1, 2, \dots$$

Expectation and Variance

- **Expectation (Mean):** The expected value of a function $g(X, Y)$ of the random variables X and Y is defined as:

$$E[g(X, Y)] = \sum_i \sum_j g(x_i, y_j) P(X = x_i, Y = y_j)$$

- **Variance:** The variance of a discrete random variable X is given by:

$$\text{Var}(X) = E[(X - E(X))^2] = \sum_i (x_i - E(X))^2 P(X = x_i)$$

For the covariance between X and Y , we have:

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = \sum_i \sum_j (x_i - E(X))(y_j - E(Y)) P(X = x_i, Y = y_j)$$

3.5.2 Continuous Multivariate Random Variables

A continuous multivariate random variable is a vector of continuous random variables. For instance, consider two continuous random variables X and Y . Their joint probability distribution is described by the joint probability density function (pdf):

$$f(x, y), \quad -\infty < x, y < +\infty$$

Expectation and Variance

- **Expectation:** The expected value of a function $g(X, Y)$ of the random variables X and Y is defined as:

$$E[g(X, Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f(x, y) dx dy$$

- **Variance:** The variance of a continuous random variable X is given by:

$$\text{Var}(X) = E[(X - E(X))^2] = \int_{-\infty}^{+\infty} (x - E(X))^2 f(x) dx$$

For the covariance between X and Y , we have:

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - E(X))(y - E(Y)) f(x, y) dx dy$$

3.5.3 Cumulative Distribution Function (CDF) and Probability Density Function (PDF)

For multivariate random variables, the cumulative distribution function (CDF) and probability density function (PDF) describe the probability distribution of the variables.

CDF

- **Discrete Case:** For discrete random variables X and Y , the joint cumulative distribution function $F_{X,Y}(x, y)$ is defined as:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \sum_{x_i \leq x} \sum_{y_j \leq y} P(X = x_i, Y = y_j)$$

- **Continuous Case:** For continuous random variables X and Y , the joint cumulative distribution function $F_{X,Y}(x, y)$ is defined as:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) du dv$$

PDF and its Relation to CDF

- **Discrete Case:** The joint probability mass function $p_{X,Y}(x, y)$ is related to the joint cumulative distribution function $F_{X,Y}(x, y)$ by:

$$p_{X,Y}(x, y) = P(X = x, Y = y) = F_{X,Y}(x, y) - F_{X,Y}(x-1, y) - F_{X,Y}(x, y-1) + F_{X,Y}(x-1, y-1)$$

- **Continuous Case:** The joint probability density function $f_{X,Y}(x, y)$ is the partial derivative of the joint cumulative distribution function $F_{X,Y}(x, y)$:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y)$$

3.5.4 Quantile Function

The Quantile Function $Q(p)$ is the inverse of the CDF. It describes the value that a random variable X takes given a specific probability p . It is defined as:

$$Q(p) = F^{-1}(p)$$

Relationship

1. From CDF to Quantile Function:

Given the CDF $F(x)$, the quantile function $Q(p)$ can be obtained by finding its inverse. Specifically, if $F(x)$ is strictly monotonically increasing, then:

$$Q(p) = \inf\{x \mid F(x) \geq p\}$$

where $\inf$ denotes the infimum.

2. From Quantile Function to CDF:

Given the quantile function $Q(p)$, the CDF $F(x)$ can be constructed. If $Q(p)$ is strictly monotonically increasing, then:

$$F(x) = P(X \leq x) = p$$

where p satisfies $Q(p) = x$.

Example

Suppose a random variable X follows a standard normal distribution. Its CDF $F(x)$ and quantile function $Q(p)$ are as follows:

- The CDF of the standard normal distribution, $F(x)$, does not have a simple closed-form expression but can be obtained through numerical integration.
- The quantile function of the standard normal distribution, $Q(p)$, also does not have a simple closed-form expression but can be obtained through lookup tables or numerical computation.

For example:

- If $p = 0.5$ (the median), then $Q(0.5) = 0$ because the standard normal distribution is symmetric.
- If $x = 0$, then $F(0) = 0.5$.

3.5.5 Conditional Expectation

Conditional expectation is a crucial concept in understanding the dependence structure between random variables. Given two random variables X and Y , the conditional expectation of Y given $X = x$ is the expected value of Y when X is known to be x .

Discrete Case

For discrete random variables, the conditional expectation of Y given $X = x_i$ is defined as:

$$E(Y | X = x_i) = \sum_j y_j P(Y = y_j | X = x_i)$$

Continuous Case

For continuous random variables, the conditional expectation of Y given $X = x$ is defined as:

$$E(Y | X = x) = \int_{-\infty}^{+\infty} y f_{Y|X}(y | x) dy$$

where $f_{Y|X}(y | x)$ is the conditional density of Y given $X = x$.

Properties**1. Iterated conditioning:**

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]] = \int_{-\infty}^{\infty} \mathbb{E}[Y|X = x] f_X(x) dx$$

For example, if X and Y are discrete,

$$\begin{aligned} \sum_i E[Y | X = x_i] P(X = x_i) &= \sum_i \sum_j y_j P(Y = y_j | X = x_i) P(X = x_i) \\ &= \sum_i \sum_j y_j P(X = x_i, Y = y_j) \end{aligned}$$

2. **Take out what is known:** if $Z = f(X)$,

$$E[ZY | X] = ZE[Y | X]$$

3. **If X and Y are independent,** then

$$E[Y | X] = E[Y].$$

3.5.6 Properties of Multivariate Random Variables

1. **Linearity of Expectation:** For any constants a and b ,

$$E(aX + bY) = aE(X) + bE(Y)$$

2. **Covariance and Correlation:**

- Covariance measures the joint variability of two random variables:

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

- The correlation coefficient is a normalized version of the covariance that measures the strength and direction of the linear relationship between two random variables:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

3. **Independence:** Two random variables X and Y are independent if and only if their joint pdf/pmf/cdf is the product of their marginal pdfs/pmfs/cdfs:

$$f(x, y) = f_X(x)f_Y(y) \quad \text{or} \quad P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j) \quad \text{or} \quad F(x, y) = F_X(x)F_Y(y)$$

3.5.7 Examples of Multivariate Distributions

1. **Multivariate Normal Distribution**

A random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^\top$ follows a multivariate normal distribution if it has the probability density function:

$$f(\mathbf{X}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \right),$$

where $\boldsymbol{\mu}$ is the mean vector and $\boldsymbol{\Sigma}$ is the covariance matrix.

2. **Multinomial Distribution**

A random vector $\mathbf{X} = (X_1, X_2, \dots, X_k)$ follows a multinomial distribution if it represents the counts of k different outcomes in n independent trials, each with probability p_i of outcome i , and has the probability mass function:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k},$$

where $x_1 + x_2 + \dots + x_k = n$ and $\sum_{i=1}^k p_i = 1$.

3.6 Sample Moments

Sample moments are statistical measures that describe the characteristics of a sample distribution. These include sample mean and sample variance. Here is an introduction to sample mean, sample variance, and related theorems:

3.6.1 Sample Mean

The sample mean is the arithmetic average of all observations in the sample, representing the central tendency of the sample data.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

where X_i is the i -th observation, and n is the sample size.

3.6.2 Sample Variance

The sample variance measures the dispersion of the sample data from the mean. It is the average of the squared differences between each observation and the sample mean.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

where $\bar{X}$ is the sample mean, and n is the sample size. Note that we use $n-1$ instead of n to correct for bias in estimating the population variance.

Why the Difference Between Dividing by n and $n-1$

The reason for dividing by n in the sample mean and by $n-1$ in the sample variance lies in their different properties when estimating population parameters.

The formula for the sample mean is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

In calculating the sample mean, we directly sum all sample values and divide by the sample size n . This is because the sample mean is an unbiased estimator of the population mean, ensuring accuracy by using n .

The formula for the sample variance is:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We use $n-1$ instead of n for the following reasons:

1. **Unbiased Estimation:** Using $n-1$ instead of n ensures that the sample variance is an unbiased estimator of the population variance. If we used n , the sample variance would underestimate the population variance because the sample mean $\bar{X}$ is itself a random variable computed from the sample data, reducing the degrees of freedom by 1. Thus, using $n-1$ adjusts for this bias, making the sample variance a more accurate estimate of the population variance.

2. **Degrees of Freedom:** While there are n observations in the sample, one degree of freedom is lost because the sample mean $\bar{X}$ is used in the calculation. Therefore, $n-1$ represents the remaining degrees of freedom, ensuring a more accurate estimate.

3.6.3 Law of Large Numbers (LLN)

The Law of Large Numbers states that as the sample size becomes large, the sample mean will almost surely converge to the population mean. In other words, with an increasing sample size, the sample mean will approach the population mean.

Formula and Principle:

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables from the same population with mean μ . For any $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| < \epsilon \right) = 1$$

This indicates that the sample mean $\bar{X}$ will converge to the population mean μ with probability 1 as the sample size n increases.

3.6.4 Central Limit Theorem (CLT)

The Central Limit Theorem states that as the sample size becomes large, the distribution of the sample mean will approach a normal distribution, regardless of the population's distribution. The CLT is one of the most important results in probability theory, addressing when the distribution of a sum of random variables is approximately normal.

Main Forms of the CLT

1. Lindeberg-Levy Central Limit Theorem

Let $\{X_n\}$ be a sequence of independent and identically distributed random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2 > 0$. Define:

$$Y_n^* = \frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

Then for any real number y :

$$\lim_{n \rightarrow \infty} P(Y_n^* \leq y) = \Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{t^2}{2}} dt$$

2. De Moivre-Laplace Central Limit Theorem

Consider n independent Bernoulli trials, where the probability of event A occurring in each trial is p (with $0 < p < 1$). Let S_n be the number of times event A occurs in n trials. Define:

$$Y_n^* = \frac{S_n - np}{\sqrt{npq}}$$

where $q = 1 - p$. Then for any real number y :

$$\lim_{n \rightarrow \infty} P(Y_n^* \leq y) = \Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{t^2}{2}} dt$$

In summary, sample mean and sample variance are crucial statistical measures that describe the characteristics of sample data, while the Law of Large Numbers and the Central Limit Theorem provide insights into the behavior and distribution of these measures in large samples. These concepts and theorems form the foundational knowledge in statistics, widely applied in various practical data analysis and statistical inference tasks.

3.7 Probability Formula Summary

(1) Random events and Probability

1. Relations and Operations of Events

1. Sub-event: $A \subseteq B$, if A occurs, then B occurs.
2. Equivalent event: $A = B$, i.e., $A \subseteq B$, $B \subseteq A$.
3. Union event: $A \cup B$ (or $A + B$), at least one of A or B occurs.
4. Difference event: $A - B$, A occurs but B does not occur.
5. Intersection event: $A \cap B$ (or AB), A and B occur simultaneously.
6. Mutually exclusive events: $A \cap B = \emptyset$.
7. Complementary events: $A \cap B = \emptyset$, $A \cup B = \Omega$, $A = \bar{B}$.

2. Algebra of Sets

1. Commutative Law: $A \cup B = B \cup A, A \cap B = B \cap A$
2. Associative Law: $(A \cup B) \cup C = A \cup (B \cup C); (A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Law: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

3. De Morgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

4. Complete Event Group

$$A_1, A_2, \dots, A_n \text{ are mutually exclusive, and } \bigcup_{i=1}^n A_i = \Omega$$

5. Basic Concepts of Probability

1. Probability: A measure of the likelihood of an event occurring, strictly defined as follows: The probability $P(g)$ is a function defined on the event set that satisfies the following three conditions:
 - (a) For any event A , $P(A) \geq 0$
 - (b) For the certain event Ω , $P(\Omega) = 1$
 - (c) For $A_1, A_2, \dots, A_n$, if $A_i \cap A_j = \emptyset$ ($i \neq j$), then $P(\bigcup A_i) = \sum P(A_i)$
2. Basic Properties of Probability
 - (a) $P(\bar{A}) = 1 - P(A)$
 - (b) $P(A - B) = P(A) - P(AB)$
 - (c) $P(A \cup B) = P(A) + P(B) - P(AB)$, especially, when $B \subseteq A$, $P(A - B) = P(A) - P(B)$ and $P(B) \leq P(A)$; $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$
 - (d) If $A_1, A_2, \dots, A_n$ are mutually exclusive, then $P(\bigcup A_i) = \sum P(A_i)$
3. Classical Probability: If all outcomes of an experiment are equally likely, then the probability is calculated by the formula: $P(A) = \frac{\text{number of outcomes in event } A}{\text{total number of outcomes}}$
4. Geometric Probability: If the sample space Ω is a subset of a distance space, and each sample point is equally likely to occur, then the probability is calculated by the formula: $P(A) = \frac{\text{measure of event } A}{\text{measure of sample space } \Omega}$

6. Fundamental Formulas of Probability

1. Conditional Probability: $P(B|A) = \frac{P(AB)}{P(A)}$ is the probability of event B occurring under the condition that event A occurs.
2. Total Probability Formula: $P(A) = \sum P(A|B_i)P(B_i)$, where B_i are mutually exclusive and $\bigcup B_i = \Omega$
3. Bayes' Theorem:

$$P(B|A) = \frac{P(A|B)P(B)}{\sum P(A|B_j)P(B_j)}, \quad j = 1, 2, \dots, n$$

4. Multiplication Rule:

$$P(A_1 A_2) = P(A_1)P(A_2|A_1) = P(A_2)P(A_1|A_2)$$

$$P(A_1 A_2 \cdots A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 A_2) \cdots P(A_n|A_1 A_2 \cdots A_{n-1})$$

7. Independence of Events

1. A and B are independent if and only if $P(AB) = P(A)P(B)$
2. If A, B, C are pairwise independent, then $P(AB) = P(A)P(B)$, $P(BC) = P(B)P(C)$, $P(AC) = P(A)P(C)$
3. If A, B, C are mutually independent, then $P(ABC) = P(A)P(B)P(C)$

8. Independent Repetition of Experiments

Repeating an experiment independently n times, with the probability of event A occurring each time being p , the probability of A occurring k times in n trials is:

$$P(X = k) = C_n^k p^k (1 - p)^{n-k}$$

9. Important Formulas and Conclusions

1. $P(\bar{A}) = 1 - P(A)$
2. $P(A \cup B) = P(A) + P(B) - P(AB)$
3. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$
4. General Addition Rule: $P(\bigcup_{i=1}^n A_i) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k})$
5. The total probability formula: $P(A) = \sum P(A|B_i)P(B_i)$
6. Bayes' theorem: $P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum P(A|B_j)P(B_j)}$
7. Conditional Probability $P(\cdot|B)$ satisfies all properties of probability. For example:

$$P(\bar{A}_1|B) = 1 - P(A_1|B)$$

$$P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1 A_2|B)$$

$$P(A_1 A_2|B) = P(A_1|B)P(A_2|A_1 B)$$

8. If $A_1, A_2, \dots, A_m$ and $B_1, B_2, \dots, B_n$ are mutually independent, then $f(A_1, A_2, \dots, A_m)$ and $g(B_1, B_2, \dots, B_n)$ are also mutually independent, where $f(\cdot)$ and $g(\cdot)$ represent events obtained by any operations on the corresponding events. Additionally, an event with probability 1 (or 0) is independent of any other event.

(2) Random Variables and Their Probability Distributions

1. Random Variables and Probability Distributions

Selecting a random variable and defining its probability distribution strictly within the sample space, and taking any real-valued function as a random variable, the probability distribution function is called the probability distribution of the random variable.

2. Concepts and Properties of Distribution Functions

Definition: $F(x) = P(X \leq x)$, $-\infty < x < \infty$

Properties:

1. $0 \leq F(x) \leq 1$
2. $F(x)$ is non-decreasing
3. $F(x)$ is right-continuous: $\lim_{x \rightarrow \alpha^+} F(x) = F(\alpha)$
4. $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$

3. Probability Distribution of Discrete Random Variables

$$P(X = x_i) = p_i, \quad i = 1, 2, \dots, n, \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1$$

4. Probability Density of Continuous Random Variables

Probability density $f(x)$: non-negative and satisfies:

$$1. \int_{-\infty}^{\infty} f(x)dx = 1$$

$$2. P(a \leq X \leq b) = \int_a^b f(x)dx$$

5. Common Distributions

$$1. \text{Bernoulli distribution: } P(X = k) = p^k(1-p)^{1-k}, \quad k = 0, 1$$

$$2. \text{Binomial distribution: } B(n, p) : P(X = k) = C_n^k p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$3. \text{Poisson distribution: } P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \lambda > 0, k = 0, 1, 2, \dots$$

$$4. \text{Uniform distribution: } U(a, b) : f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

$$5. \text{Normal Distribution: } N(\mu, \sigma^2):$$

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \sigma > 0, -\infty < x < +\infty$$

$$6. \text{Exponential Distribution: } E(\lambda):$$

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$7. \text{Geometric Distribution: } G(p):$$

$$P(X = k) = (1-p)^{k-1}p, \quad 0 < p < 1, k = 1, 2, \dots$$

$$8. \text{Hypergeometric Distribution: } H(N, M, n):$$

$$P(X = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \dots, \min(n, M)$$

6. Probability Distributions of Random Variables

1. Discrete:

$$P(X = x_i) = p_i, \quad Y = g(X)$$

Then:

$$P(Y = y_j) = \sum_{g(x_i)=y_j} P(X = x_i)$$

2. Continuous:

$$X \sim f_X(x), \quad Y = g(X)$$

Then:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \int_{g(x) \leq y} f_X(x)dx, \quad f_Y(y) = F'_Y(y)$$

7. Important Formulas and Conclusions

1. $X \sim N(0, 1) \Rightarrow \Phi(0) = \frac{1}{2}, \Phi(-a) = P(X \leq -a) = 1 - \Phi(a)$
2. $X \sim N(\mu, \sigma^2) \Rightarrow \frac{X-\mu}{\sigma} \sim N(0, 1), P(X \leq a) = \Phi\left(\frac{a-\mu}{\sigma}\right)$
3. $X \sim E(\lambda) \Rightarrow P(X \leq s+t|X > s) = P(X > t)$
4. $X \sim G(p) \Rightarrow P(X = m+k|X > m) = P(X = k)$
5. The distribution function of a discrete random variable is a step function; the distribution function of a continuous random variable is a continuous function, but not necessarily differentiable.
6. Both non-discrete and non-continuous random variables can exist.

(3) Multivariate Random Variables and Their Distributions**1. Joint Distribution of Bivariate Random Variables**

For the random vector (X, Y) composed of two random variables, the joint distribution is $F(x, y) = P(X \leq x, Y \leq y)$.

2. Distribution of Bivariate Discrete Random Variables

1. Joint probability distribution $P\{X = x_i, Y = y_j\} = p_{ij}, i, j = 1, 2, \dots$
2. Marginal distributions:

$$p_{i.} = \sum_{j=1}^{\infty} p_{ij}, i = 1, 2, \dots, \quad p_{.j} = \sum_{i=1}^{\infty} p_{ij}, j = 1, 2, \dots$$

3. Conditional probability distribution:

$$P\{X = x_i | Y = y_j\} = \frac{p_{ij}}{p_{.j}} \quad P\{Y = y_j | X = x_i\} = \frac{p_{ij}}{p_{i.}}$$

3. Density of Bivariate Continuous Random Variables

1. Joint probability density $f(x, y)$:

- (a) $f(x, y) \geq 0$
- (b) $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$

2. Distribution function:

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv$$

3. Marginal probability density:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

4. Conditional probability density:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \quad f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

4. Common Joint Distributions of Bivariate Random Variables

1. Bivariate uniform distribution: $(x, y) \sim U(D)$, $f(x, y) = \begin{cases} \frac{1}{S(D)}, & (x, y) \in D \\ 0, & \text{otherwise} \end{cases}$

2. Bivariate normal distribution: $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$:

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho \frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right]\right)$$

5. Independence and Correlation of Random Variables

- X and Y are independent $\Leftrightarrow F(x, y) = F_X(x)F_Y(y)$:

- $\Leftrightarrow p_{ij} = p_{i\cdot} \cdot p_{\cdot j}$ (discrete)

- $\Leftrightarrow f(x, y) = f_X(x)f_Y(y)$ (continuous)

- Correlation of X and Y :

Correlation coefficient $\rho_{XY} = 0$, X and Y are uncorrelated, otherwise X and Y are correlated.

6. Probability Distributions of Simple Functions of Two Random Variables

1. Discrete:

$$P(X = x_i, Y = y_j) = p_{ij}, Z = g(X, Y) \Rightarrow \\ P(Z = z_k) = P(g(X, Y) = z_k) = \sum_{g(x_i, y_j) = z_k} P(X = x_i, Y = y_j)$$

2. Continuous:

$$(X, Y) \sim f(x, y), Z = g(X, Y) \Rightarrow \\ F_Z(z) = P(g(X, Y) \leq z) = \iint_{g(x, y) \leq z} f(x, y) dx dy, \quad f_Z(z) = F'_Z(z)$$

7. Important Formulas and Conclusions

1. Marginal density formula:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

2. $P\{(X, Y) \in D\} = \iint_D f(x, y) dx dy$

3. If (X, Y) follows a bivariate normal distribution $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, then:

(a) $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$

(b) X and Y are independent $\Leftrightarrow \rho = 0$, i.e., X and Y are uncorrelated.

(c) $C_1X + C_2Y \sim N(C_1\mu_1 + C_2\mu_2, C_1^2\sigma_1^2 + C_2^2\sigma_2^2 + 2C_1C_2\sigma_1\sigma_2\rho)$

(d) The conditional distribution of X given $Y = y$ is:

$$N\left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2}(y - \mu_2), \sigma_1^2(1 - \rho^2)\right)$$

(e) The conditional distribution of Y given $X = x$ is:

$$N\left(\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1), \sigma_2^2(1 - \rho^2)\right)$$

4. If X and Y are independent and follow $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively, then:

$$(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, 0), \quad C_1X + C_2Y \sim N(C_1\mu_1 + C_2\mu_2, C_1^2\sigma_1^2 + C_2^2\sigma_2^2)$$

5. If X and Y are independent, and $f(x)$ and $g(x)$ are continuous functions, then $f(X)$ and $g(Y)$ are also independent.

(4) Numerical Characteristics of Random Variables

1. Mathematical Expectation

- Discrete case: $P\{X = x_i\} = p_i$

$$E(X) = \sum_i x_i p_i$$

- Continuous case: $X \sim f(x)$

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

Properties:

1. $E(C) = C, E[E(X)] = E(X)$
2. $E(C_1X + C_2Y) = C_1E(X) + C_2E(Y)$
3. If X and Y are independent, then $E(XY) = E(X)E(Y)$
4. $[E(XY)]^2 \leq E(X^2)E(Y^2)$

2. Variance

$$D(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2$$

3. Standard Deviation

$$\sqrt{D(X)}$$

4. Discrete case

$$D(X) = \sum_i [x_i - E(X)]^2 p_i$$

5. Continuous case

$$D(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f(x) dx$$

Properties:

1. $D(C) = 0, D[E(X)] = 0, D[D(X)] = 0$
2. If X and Y are independent, then $D(X \pm Y) = D(X) + D(Y)$
3. $D(C_1X + C_2) = C_1^2 D(X)$
4. $D(X) < E(X - C)^2, C \neq E(X)$
5. $D(X) = 0 \Leftrightarrow P\{X = C\} = 1$

6. Mathematical Expectation of Functions of Random Variables

- For function $Y = g(X)$:

– Discrete case: $P\{X = x_i\} = p_i$

$$E(Y) = \sum_i g(x_i)p_i$$

– Continuous case: $X \sim f(x)$

$$E(Y) = \int_{-\infty}^{+\infty} g(x)f(x)dx$$

7. Covariance

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

8. Correlation Coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)D(Y)}}$$

k-th order raw moment $E(X^k)$; k-th order central moment $E[(X - E(X))^k]$

Properties:

1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
2. $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$
3. $\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$
4. $|\rho(X, Y)| \leq 1$
5. $\rho(X, Y) = 1 \Leftrightarrow P(Y = aX + b) = 1$, where $a > 0$
6. $\rho(X, Y) = -1 \Leftrightarrow P(Y = aX + b) = 1$, where $a < 0$

9. General rules of variance and covariance:

$$\text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$$

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

10. LLN and CLT

Law of Large Numbers (LLN) The Law of Large Numbers states that the sample average converges to the expected value as the sample size grows. Formally, if $X_1, X_2, \dots, X_n$ are i.i.d. random variables with expected value $\mu = \mathbb{E}[X_i]$, then:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty$$

where $\xrightarrow{P}$ denotes convergence in probability.

Central Limit Theorem (CLT) The Central Limit Theorem states that the distribution of the sample mean approximates a normal distribution as the sample size becomes large. Formally, if $X_1, X_2, \dots, X_n$ are i.i.d. random variables with mean $\mu = \mathbb{E}[X_i]$ and variance $\sigma^2 = \text{Var}(X_i)$, then the standardized sample mean:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty$$

where $\xrightarrow{D}$ denotes convergence in distribution and $\mathcal{N}(0, 1)$ is the standard normal distribution.

3.8 Classic Questions

3.8.1 lognormal

If X and Y are lognormal random variables, is their product XY lognormally distributed?

solution:

First, note that, if X and Y are independent lognormal random variables, then XY is lognormally distributed, since $\ln(XY) = \ln(X) + \ln(Y)$ is the sum of two independent normal random variables, and therefore it is normally distributed.

In a more general setting, if $\ln(X)$ and $\ln(Y)$ have joint normal distribution, then $\ln(X) + \ln(Y)$ is normally distributed and therefore XY is a lognormal random variable.

Otherwise, $\ln(X) + \ln(Y)$ may not be normally distributed even if $\ln(X)$ and $\ln(Y)$ are normally distributed, in which case XY is not lognormally distributed.

3.8.2 LLN

What is the law of large numbers?

solution:

There is a strong law of large numbers and there is a weak law of large numbers. The strong law of large numbers states that the average of a large number of independent identically distributed integrable random variables converges almost surely to their common mean; in the case of the weak law of large numbers, the convergence is only in probability.

More precisely, let $X_1, X_2, \dots$ be a sequence of independent identically distributed random variables with finite expected value $\mu = \mathbb{E}[X_i]$, and let $S_n = X_1 + \dots + X_n$.

The strong law of large numbers states that $\frac{S_n}{n} \rightarrow \mu$ almost surely, i.e.,

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1. \quad (3.118)$$

The weak law of large numbers states that $\frac{S_n}{n} \rightarrow \mu$ in probability, i.e.,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) = 0, \quad \forall \epsilon > 0. \quad (3.119)$$

Note that, if a sequence of random variables converges almost surely, then it also converges in probability, and therefore, if (3.118) holds true, then (3.119) also holds true. This is the sense in which the strong law of large numbers is "stronger" than the weak law of large numbers.

3.8.3 CLT

What is the central limit theorem?

solution:

The central limit theorem states that the limiting distribution of the centered and scaled sum of an independent identically distributed sequence of random variables is a normal distribution if the common distribution of the random variables has finite variance.

More precisely, let $X_1, X_2, \dots$ be a sequence of independent identically distributed random variables with finite expected value $\mu = E[X_i]$ and finite variance $\sigma^2 = \text{var}[X_i]$. Let $S_n = X_1 + \dots + X_n$. Then,

$$\lim_{n \rightarrow \infty} \frac{S_n - n\mu}{\sigma\sqrt{n}} = Z,$$

where Z is the standard normal distribution, and the convergence is in distribution, i.e.,

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq t\right) = P(Z \leq t).$$

Putting together the Law of Large Numbers and the Central Limit Theorem, the following approximation as n goes to infinity holds:

$$\frac{S_n}{n} \approx \mu + \frac{\sigma}{\sqrt{n}}Z.$$

3.8.4 Sum Exceedance I

Let $X_1, X_2, \dots$ be IID $\text{Unif}(0, 1)$ random variables and let $N = \min\{n : X_1 + \dots + X_n > \ln(2)\}$. Find $E[N]$.

solution:

$$E(N) = \sum_{k=0}^{\infty} P(N > k) = 1 + \sum_{k=1}^{\infty} P(N > k)$$

$$P(N \geq k) = P\left(\sum_{i=1}^k X_i \leq \ln 2\right)$$

Let $S_n = X_1 + X_2 + \dots + X_n$,

Prove: $P(S_n \leq t) = \frac{t^n}{n!}$

$$P(S_1 \leq t) = t$$

$$P(S_2 \leq t) = \int_0^t f(X_2) \cdot P(S_1 \leq t - X_2) dX_2 = \int_0^t (t - X_2) dX_2 = \frac{t^2}{2}$$

...

$$P(S_{n+1} \leq t) = \int_0^t f(X_{n+1}) P(S_n \leq t - X_{n+1}) dX_{n+1} = \int_0^t \frac{(t - X_{n+1})^n}{n!} dX_{n+1} = \frac{1}{n!} \left[-\frac{(t - X_{n+1})^{n+1}}{n+1} \right]_0^t = \frac{t^{n+1}}{(n+1)!}$$

$$\text{So, } E(N) = 1 + \sum_{k=1}^{\infty} \frac{(\ln 2)^k}{k!} = e^{\ln 2} = 2$$

3.8.5 First Ace

On average, how many cards in a normal deck of 52 playing cards do you need to flip over to observe your first ace?

solution:

The four aces are dispersed throughout the deck and cut the 48 remaining cards into 5 distinct sections, each of some random length $X_i \in [0, 48]$ where $1 \leq i \leq 5$; that is, $\sum_{i=1}^5 X_i = 48$. Furthermore, by symmetry, $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \cdots = \mathbb{E}[X_5]$, as in the absence of any additional information, none of the sections is expected to be any larger or smaller than any other. Thus, by linearity of expectation:

$$\mathbb{E}\left[\sum_{i=1}^5 X_i\right] = 5 \times \mathbb{E}[X_1] = 48 \implies \mathbb{E}[X_1] = \frac{48}{5}$$

We have found that the expected number of cards in the first section is $\frac{48}{5}$, so we will observe the first ace on the next card. Hence, we expect to flip $\frac{48}{5} + 1 = 10.6$ cards to observe the first ace.

3.8.6 Coupon collection

There are N distinct types of coupons in cereal boxes and each type, independent of prior selections, is equally likely to be in a box.

A. If a child wants to collect a complete set of coupons with at least one of each type, how many coupons (boxes) on average are needed to make such a complete set?

B. If the child has collected n coupons, what is the expected number of distinct coupon types?³¹

Solution:

For part **A**, let X_i , $i = 1, 2, \dots, N$, be the number of additional coupons needed to obtain the i -th type after $(i-1)$ distinct types have been collected. So the total number of coupons needed is $X = X_1 + X_2 + \cdots + X_N = \sum_{i=1}^N X_i$.

For any i , $i-1$ distinct types of coupons have already been collected. It follows that a new coupon will be of a different type with probability $1 - (i-1)/N = (N-i+1)/N$. Essentially to obtain the i -th distinct type, the random variable X_i follows a geometric distribution with $p = (N-i+1)/N$ and $\mathbb{E}[X_i] = N/(N-i+1)$. For example, if $i = 1$, we simply have $X_i = \mathbb{E}[X_i] = 1$.

$$\therefore \mathbb{E}[X] = \sum_{i=1}^N \mathbb{E}[X_i] = \sum_{i=1}^N \frac{N}{N-i+1} = N \left(\frac{1}{N} + \frac{1}{N-1} + \cdots + \frac{1}{1} \right).$$

For part **B**, let Y be the number of distinct types of coupons in the set of n coupons. We introduce indicator random variables I_i , $i = 1, 2, \dots, N$, where

$$I_i = \begin{cases} 1, & \text{if at least one coupon of the } i\text{-th type is in the set of } n \text{ coupons} \\ 0, & \text{otherwise} \end{cases}$$

So we have $Y = I_1 + I_2 + \cdots + I_N = \sum_{i=1}^N I_i$.

For each collected coupon, the probability that it is not the i -th coupon type is $\frac{N-1}{N}$.

Since all n coupons are independent, the probability that none of the n coupons is the i -th coupon type is $P(I_i = 0) = \left(\frac{N-1}{N}\right)^n$ and we have $\mathbb{E}[I_i] = P(I_i = 1) = 1 - \left(\frac{N-1}{N}\right)^n$.

$$\therefore \mathbb{E}[Y] = \sum_{i=1}^N \mathbb{E}[I_i] = N - N \left(\frac{N-1}{N} \right)^n.$$

3.8.7 Hit Or Miss

A particle starts at $(4, 4)$. Each turn, it moves either 1 unit in the $-x$ direction, 1 unit in the $-y$ direction, or 1 unit in each of the $-x$ and $-y$ directions. During a turn, the particle decides how it will move randomly such that the probability of each option is $\frac{1}{3}$. The particle repeatedly takes turns until it hits the x or y axes for the first time. Compute the probability that the particle hits the origin.

solution:

In order to hit the origin, the particle must arrive at $(1, 1)$. To do so, the particle can (1) move 3 left and 3 down, (2) move 2 left, 2 down, and 1 diagonal, (3) move 1 left, 1 down, and 2 diagonal, or (4) move down 3 consecutive diagonals. The probability that the particle arrives at $(1, 1)$ is then

$$\frac{1}{3^6} \cdot \binom{6}{3} + \frac{1}{3^5} \cdot \binom{5}{2, 2, 1} + \frac{1}{3^4} \cdot \binom{4}{1, 1, 2} + \frac{1}{3^3} = \frac{245}{3^6}$$

Finally, there is a $\frac{1}{3}$ chance that the particle reaches the origin from $(1, 1)$, as it has to move diagonally from $(1, 1)$, so our answer is

$$\frac{245}{3^6} \cdot \frac{1}{3} = \frac{245}{2187}$$

3.8.8 Correlation and Variance

Suppose we have that $\sigma_X = 4$, $\sigma_Y = 7$, and $\sigma_{X+Y} = 11$. What is $\rho(X, Y)$?

Solution:

We are going to generalize this to when $\sigma_X = a$, $\sigma_Y = b$, and $\sigma_{X+Y} = a + b$. We know that

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \implies \text{Cov}(X, Y) = \frac{1}{2}((a + b)^2 - a^2 - b^2) = ab$$

After, we know that

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{ab}{ab} = 1$$

3.8.9 Drunk passenger

A line of 100 airline passengers are waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n -th passenger in line has a ticket for the seat number n . Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their proper seats unless it is already occupied; In that case, they will randomly choose a free seat. You're person number 100. What is the probability that you end up in your seat (i.e., seat 100) ?

Solution:

Let's consider seats 1 and 100. There are two possible outcomes:

Let E_1 : Seat #1 is taken before #100;

Let E_2 : Seat #100 is taken before #1.

If any passenger takes seat #100 before #1 is taken, surely you will not end up in your own seat. But if any passenger takes #1 before #100 is taken, you will definitely end up in your own seat. By symmetry, either outcome has a probability of 0.5. So the probability that you end up in your seat is 50%.

In case this over-simplified version of reasoning is not clear to you, consider the following detailed explanation: If the drunk passenger takes #1 by chance, then it's clear all the rest of the passengers will have the correct seats. If he takes #100, then you will not get your seat. The probabilities that he takes #1 or #100 are equal. Otherwise, assume that he takes the n -th seat, where n is a number between 2 and 99. Everyone between 2 and $(n - 1)$ will get his own seat. That means the n -th passenger essentially becomes the new "drunk" guy with designated seat #1. If he chooses #1, all the rest of the passengers will have the

correct seats. If he takes #100, then you will not get your seat. (The probabilities that he takes #1 or #100 are again equal.) Otherwise, he will just make another passenger down the line the new "drunk" guy with designated seat #1, and each new "drunk" guy has an equal probability of taking #1 or #100. Since at all jump points there's an equal probability for the "drunk" guy to choose seat #1 or #100, by symmetry, the probability that you, as the 100th passenger, will sit in #100 is 0.5.

3.8.10 Chess tournament

A chess tournament has 2^n players with skills $1 > 2 > \dots > 2^n$. It is organized as a knockout tournament, so that after each round only the winner proceeds to the next round. Except for the final, opponents in each round are drawn at random. Let's also assume that when two players meet in a game, the player with better skills always wins. What's the probability that players 1 and 2 will meet in the final?

Solution:

There are at least two approaches to solve the problem. The standard approach applies the multiplication rule based on conditional probability, while a counting approach is far more efficient. (We will cover conditional probability in detail in the next section.)

Let's begin with the conditional probability approach, which is easier to grasp. Since there are 2^n players, the tournament will have n rounds (including the final). For round 1, players $2, 3, \dots, 2^n$ each have $\frac{1}{2^n-1}$ probability to be 1's rival, so the probability that 1 and 2 do not meet in round 1 is

$$\frac{2^n - 2}{2^n - 1} = \frac{2 \times (2^{n-1} - 1)}{2^n - 1}.$$

Condition on that 1 and 2 do not meet in round 1, 2^{n-1} players proceed to the 2nd round and the conditional probability that 1 and 2 will not meet in round 2 is

$$\frac{2^{n-1} - 2}{2^{n-1} - 1} = \frac{2 \times (2^{n-2} - 1)}{2^{n-1} - 1}.$$

We can repeat the same process until the $(n-1)$ th round, in which there are $2^2 (= 2^n / 2^{n-2})$ players left and the conditional probability that 1 and 2 will not meet in round $(n-1)$ is

$$\frac{2^2 - 2}{2^2 - 1} = \frac{2 \times (2^2 - 1)}{2^2 - 1}.$$

Let E_1 be the event that 1 and 2 do not meet in round 1;
 E_2 be the event that 1 and 2 do not meet in rounds 1 and 2;
 $\vdots$
 E_{n-1} be the event that 1 and 2 do not meet in rounds 1, 2, $\dots$, $n-1$.

Apply the multiplication rule, we have

$$\begin{aligned} P(1 \text{ and } 2 \text{ meet in the } n\text{th game}) &= P(E_1) \times P(E_2 | E_1) \times \dots \times P(E_{n-1} | E_1 E_2 \dots E_{n-2}) \\ &= \frac{2 \times (2^{n-1} - 1)}{2^n - 1} \times \frac{2 \times (2^{n-2} - 1)}{2^{n-1} - 1} \times \dots \times \frac{2 \times (2^2 - 1)}{2^2 - 1} = \frac{2^{n-1}}{2^n - 1}. \end{aligned}$$

3.8.11 Application Letter

You're sending job applications to 5 firms: Morgan Stanley, Lehman Brothers, UBS, Goldman Sachs, and Merrill Lynch. You have 5 envelopes on the table neatly typed with names and addresses of people at these 5 firms. You even have 5 cover letters personalized to each of these firms. Your 3-year-old tried to be helpful and stuffed each cover letter into each of the envelopes for you. Unfortunately she randomly put letters into

envelopes without realizing that the letters are personalized. What is the probability that all 5 cover letters are mailed to the wrong firms?

Solution:

Let A_i be the event that the i -th envelope went to the right.

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} P(A_1 \cdots A_k)$$

Thus,

$$1 - P\left(\bigcup_{i=1}^n A_i\right) = 1 - \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} P(A_1 \cdots A_k)$$

Simplifying,

$$\begin{aligned} &= 1 - \left(1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \frac{1}{5!}\right) \\ &= 1 - \frac{19}{30} = \frac{11}{30} \end{aligned}$$

3.8.12 Unfair coin

You are given 1000 coins. Among them, 1 coin has heads on both sides. The other 999 coins are fair coins. You randomly choose a coin and toss it 10 times. Each time, the coin turns up heads. What is the probability that the coin you choose is the unfair one?

Solution:

This is a classic conditional probability question that uses Bayes' theorem. Let A be the event that the chosen coin is the unfair one, then A^c is the event that the chosen coin is a fair one. Let B be the event that all ten tosses turn up heads. Apply Bayes' theorem, we have

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)}.$$

The priors are $P(A) = \frac{1}{1000}$ and $P(A^c) = \frac{999}{1000}$. If the coin is unfair, it always turns up heads, so $P(B | A) = 1$. If the coin is fair, each time it has $\frac{1}{2}$ probability of turning up heads. So

$$P(B | A^c) = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}.$$

Plug in all the available information and we have the answer:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)} = \frac{\frac{1}{1000} \times 1}{\frac{1}{1000} \times 1 + \frac{999}{1000} \times \frac{1}{1024}} \approx 0.5.$$

3.8.13 Monty Hall problem

Monty Hall problem is a probability puzzle based on an old American show Let's Make a Deal. The problem is named after the show's host. Suppose you're on the show now, and you're given the choice of 3 doors. Behind one door is a car; behind the other two, goats. You don't know ahead of time what is behind each of the doors. You pick one of the doors and announce it. As soon as you pick the door, Monty opens one of the other two doors that he knows has a goat behind it. Then he gives you the option to either keep your original choice or switch to the third door. Should you switch? What is the probability of winning a car if you switch?

Solution:

Using a switching strategy, you win the car if and only if you originally pick a door with a goat, which has a probability of $2/3$ (You pick a door with a goat, Monty shows a door with another goat, so the one you switch to must have a car behind it. If you originally picked the door with the car, which has a probability of $1/3$, you will lose by switching. So your probability of winning by switching is actually $2/3$).

Bayes method:

- A: guest chooses door 1, car is in door 1
 B: guest chooses door 1, car is in door 2
 C: guest chooses door 1, car is in door 3
 D: host opens door 2

$$P(D|A) = \frac{1}{2}, \quad P(D|B) = 0, \quad P(D|C) = 1$$

$$\therefore P(D) = P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C)$$

$$= \frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$\therefore P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} (\text{don't Switch})$$

$$P(C|D) = \frac{P(D|C)P(C)}{P(D)} = \frac{1 \times \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3} (\text{Switch})$$

3.8.14 Amoeba Population

There is a one amoeba in a pond. After every minute the amoeba may die, stay the same, split into two or split into three with equal probability. All its offspring, if it has any, will behave the same (and independent of other amoebas. What is the probability the amoeba population will die out?

Solution:

This is just another standard conditional probability problem once you realize we need to derive the probability conditioned on what happens to the amoeba one minute later. Let $P(E)$ be the probability that the amoeba population will die out, and apply the law of total probability conditioned on what happens to the amoeba one minute later:

$$P(E) = P(E | F_1)P(F_1) + P(E | F_2)P(F_2) + \cdots + P(E | F_n)P(F_n).$$

For the original amoeba, as stated in the question, there are four possible mutually exclusive events each with probability $1/4$. Let's denote F_1 as the event the amoeba dies; F_2 as the event that it stays the same; F_3 as the event that it splits into two; F_4 as the event that it splits into three. For event F_1 , $P(E | F_1) = 1$ since no amoeba is left.

$P(E | F_2) = P(E)$ since the state is the same as the beginning. For F_3 , there are two amoebas; either behaves the same as the original one. The amoeba population will die only if both amoebas die out. Since they are independent, the probability that they both will die out is $P(E)^2$. Similarly, we have $P(F_4) = P(E)^3$. Plugging in all the numbers, the equation becomes

$$P(E) = \frac{1}{4} \times 1 + \frac{1}{4} \times P(E) + \frac{1}{4} \times P(E)^2 + \frac{1}{4} \times P(E)^3.$$

Solve this equation with the restriction $0 < P(E) < 1$, and we will get $P(E) = \sqrt{2} - 1 \approx 0.414$. (The other two roots of the equation are 1 and $-\sqrt{2} - 1$.)

3.8.15 Probability of triangle

A stick is cut twice randomly (each cut point follows a uniform distribution on the stick), what is the probability that the 3 segments can form a triangle?

Solution:

Without loss of generality, let's assume that the length of the stick is 1. Let's also label the point of the first cut as x and the second cut as y .

If $x < y$, then the three segments are x , $y - x$, and $1 - y$. The conditions to form a triangle are:

$$\begin{aligned}x + (y - x) &> 1 - y \implies y > \frac{1}{2} \\x + (1 - y) &> y - x \implies y < \frac{1}{2} + x \\(y - x) + (1 - y) &> x \implies x < \frac{1}{2}\end{aligned}$$

The total shadowed area represents the region where 3 segments can form a triangle, which is $1/4$ of the square. So the probability is $1/4$.

3.8.16 Russian roulette series

Let's play a traditional version of Russian roulette. A single bullet is put into a 6-chamber revolver. The barrel is randomly spun so that each chamber is equally likely to be under the hammer. Two players take turns to pull the trigger—with the gun unfortunately pointing at one's own head without further spinning until the gun goes off and the person who gets killed loses. If you, one of the players, can choose to go first or second, how will you choose? And what is your probability of loss?

Solution:

Many people have the wrong impression that the first person has a higher probability of loss. After all, the first player has a $\frac{1}{6}$ chance of getting killed in the first round before the second player starts. Unfortunately, this is one of the few times that intuition is wrong. Once the barrel is spun, the position of the bullet is fixed. If you go first, you lose if and only if the bullet is in chambers 1, 3, and 5. So the probability that you lose is the same as the second player, $\frac{1}{2}$. In that sense, whether to go first or second does not matter.

Now, let's change the rule slightly. We will spin the barrel again after every trigger pull. Will you choose to be the first or the second player? And what is your probability of loss?

Solution:

The difference is that each run now becomes independent. Assume that the first player's probability of losing is p , then the second player's probability of losing is $1 - p$. Let's condition the probability on the first person's first trigger pull. He has $\frac{1}{6}$ probability of losing in this run. Otherwise, he essentially becomes the second player in the game with new (conditional) probability of losing $1 - p$. That happens with probability $\frac{5}{6}$. That gives us

$$p = 1 \times \frac{1}{6} + (1 - p) \times \frac{5}{6} \implies p = \frac{6}{11}.$$

So you should choose to be the second player and have $\frac{5}{11}$ probability of losing.

If instead of one bullet, two bullets are randomly put in the chamber. Your opponent played the first and he was alive after the first trigger pull. You are given the option whether to spin the barrel. Should you spin the barrel?

Solution:

If you spin the barrel, the probability that you will lose in this round is $\frac{2}{6}$. If you don't spin the barrel, there are only 5 chambers left and your probability of losing in this round (conditioned on that your opponent

survived) is $\frac{2}{5}$. So you should spin the barrel.

What if the two bullets are randomly put in two consecutive positions? If your opponent survived his first round, should you spin the barrel?

Solution:

Now we have to condition our probability on the fact that the positions of the two bullets are consecutive. Let's label the empty chambers as 1, 2, 3, and 4; label the ones with bullets 5 and 6. Since your opponent survived the first round, the possible position he encountered is 1, 2, 3, or 4 with equal probability. With $\frac{1}{4}$ chance, the next one is a bullet (the position was 4). So if you don't spin, the chance of survival is $\frac{3}{4}$. If you spin the barrel, each position has equal probability of being chosen, and your chance of survival is only $\frac{2}{3}$. So you should not spin the barrel.

3.8.17 Gambler's ruin problem

A gambler starts with an initial fortune of i dollars. On each successive game, the gambler wins \$1 with probability p , $0 < p < 1$, or loses \$1 with probability $q = 1 - p$. He will stop if he either accumulates N dollars or loses all his money. What is the probability that he will end up with N dollars?

Solution:

From any initial state i (the dollars the gambler has), $0 \leq i \leq N$, let P_i be the probability that the gambler's fortune will reach N instead of 0. The next state is either $i + 1$ with probability p or $i - 1$ with probability q . So we have

$$P_i = pP_{i+1} + qP_{i-1} \implies P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}) = \left(\frac{q}{p}\right)^2 (P_{i-1} - P_{i-2}) = \cdots = \left(\frac{q}{p}\right)^i (P_1 - P_0).$$

We also have the boundary probabilities $P_0 = 0$ and $P_N = 1$.

So starting from P_2 , we can successively evaluate P_i as an expression of P_1 :

$$P_1 = pP_2 + qP_0 \implies P_2 = \frac{1}{p}P_1 = \left[1 + \frac{q}{p}\right] P_1,$$

$$P_3 = \left[1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2\right] P_1,$$

$\vdots$

$$P_i = \left[1 + \frac{q}{p} + \cdots + \left(\frac{q}{p}\right)^{i-1}\right] P_1.$$

Extending this expression to P_N , we have:

$$P_N = 1 = \left[1 + \frac{q}{p} + \cdots + \left(\frac{q}{p}\right)^{N-1}\right] P_1 = \begin{cases} \frac{1-(q/p)^N}{1-q/p} P_1, & \text{if } q/p \neq 1, \\ NP_1, & \text{if } q/p = 1 \end{cases}$$

$$\implies P_1 = \begin{cases} \frac{1-q/p}{1-(q/p)^N}, & \text{if } q/p \neq 1 \\ \frac{1}{N}, & \text{if } q/p = 1 \end{cases}$$

$$\implies P_i = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)^N} P_1, & \text{if } p \neq 1/2 \\ \frac{i}{N}, & \text{if } p = 1/2 \end{cases}$$

3.8.18 Conditional Expectation I

Consider the following joint pdf:

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} c(1 - x_2) & 0 \leq x_1 \leq x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

where c is a constant such that f_{X_1, X_2} is a valid joint pdf. Compute $\mathbb{E}[X_1]$.

Solution:

We first need to compute c .

$$\iint_{\mathbb{R}^2} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$$

$$\int_0^1 \int_0^{x_2} c(1 - x_2) dx_1 dx_2 = 1$$

$$\int_0^1 cx_2(1 - x_2) dx_2 = 1$$

$$c \int_0^1 \left(\frac{x_2^2}{2} - \frac{x_2^3}{3} \right) dx_2 = 1$$

$$c \left[\frac{x_2^2}{2} - \frac{x_2^3}{3} \right]_0^1 = 1$$

$$c = 6$$

Next, we need to determine the conditional pdf $f_{X_1|X_2}(x_1|x_2)$. Recall

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_2}(x_2)}$$

We must compute the marginal pdf of X_2 .

$$\int_0^{x_2} 6(1 - x_1) dx_1 = 6x_2(1 - x_2)$$

$$f_{X_2}(x_2) = \begin{cases} 6x_2(1 - x_2) & 0 \leq x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Plugging in, we find

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_2}(x_2)} = \begin{cases} \frac{1}{x_2} & 0 \leq x_1 \leq x_2 \\ 0 & \text{otherwise} \end{cases}$$

The last step is to apply the law of total expectation.

$$\begin{aligned} \mathbb{E}[X_1] &= \mathbb{E}[\mathbb{E}[X_1|X_2 = x_2]] \\ &= \int_0^1 \mathbb{E}[X_1|X_2 = x_2] f_{X_2}(x_2) dx_2 \\ &= \int_0^1 \left(\int_0^{x_2} \frac{x_1}{x_2} dx_1 \right) 6x_2(1 - x_2) dx_2 \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \left(\frac{x_2}{2}\right) 6x_2(1-x_2) dx_2 = \int_0^1 3x_2^2(1-x_2) dx_2 = 3 \int_0^1 x_2^2 - x_2^3 dx_2 \\
&= 3 \left[\frac{x_2^3}{3} - \frac{x_2^4}{4} \right]_0^1 = \frac{3}{12} = \frac{1}{4}
\end{aligned}$$

3.8.19 Conditional Expectation II

Suppose $X \sim \text{Pois}(\lambda)$, where $\lambda \sim \text{Exp}(1)$. Compute $\mathbb{E}[X]$.

Solution:

Let's compute $\mathbb{E}[X]$ using the law of total expectation.

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid \lambda]]$$

If the value of λ is known, then X follows a set Poisson distribution, meaning we can describe its expectation.

$$\mathbb{E}[X \mid \lambda] = \lambda$$

Recall

$$f_\lambda(t) = \begin{cases} e^{-t} & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

So, it follows from the law of total expectation,

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}[\mathbb{E}[X \mid \lambda]] \\
&= \int_0^\infty f_\lambda(t) \mathbb{E}[X \mid t] dt \\
&= \int_0^\infty t e^{-t} dt \\
&= [-\lambda e^{-\lambda} - e^{-\lambda}]_0^\infty \\
&= 1
\end{aligned}$$

3.8.20 Expected value of max and min

Let $X_1, X_2, \dots, X_n$ be IID random variables with uniform distribution between 0 and 1. What are the cumulative distribution function, the probability density function and expected value of $Z_n = \max(X_1, X_2, \dots, X_n)$? What are the cumulative distribution function, the probability density function and expected value of $Y_n = \min(X_1, X_2, \dots, X_n)$?

Solution:

This is a direct test of textbook knowledge. For uniform distribution on $[0, 1]$, $F_X(x) = x$ and $f_X(x) = 1$. Applying $F_X(x)$ and $f_X(x)$ to $Z_n = \max(X_1, X_2, \dots, X_n)$ we have

$$\begin{aligned}
P(Z_n \leq x) &= (P(X \leq x))^n \Rightarrow F_{Z_n}(x) = (F_X(x))^n = x^n \\
&\Rightarrow f_{Z_n}(x) = n f_X(x) (F_X(x))^{n-1} = n x^{n-1}
\end{aligned}$$

and

$$E[Z_n] = \int_0^1 x f_{Z_n}(x) dx = \int_0^1 n x^n dx = \frac{n}{n+1} [x^{n+1}]_0^1 = \frac{n}{n+1}.$$

Applying $F_X(x)$ and $f_X(x)$ to $Y_n = \min(X_1, X_2, \dots, X_n)$ we have

$$P(Y_n \geq x) = (P(X \geq x))^n \Rightarrow F_{Y_n}(x) = 1 - (1 - F_X(x))^n = 1 - (1 - x)^n \\ \Rightarrow f_{Y_n}(x) = n f_X(x)(1 - F_X(x))^{n-1} = n(1 - x)^{n-1}$$

and

$$E[Y_n] = \int_0^1 x f_{Y_n}(x) dx = \int_0^1 nx(1 - x)^{n-1} dx = \int_0^1 n(1 - y)y^{n-1} dy = [y^n]_0^1 - \frac{n}{n+1} [y^{n+1}]_0^1 = \frac{1}{n+1}.$$

3.8.21 Correlation of max and min

Let X_1 and X_2 be IID random variables with uniform distribution between 0 and 1, $Y = \min(X_1, X_2)$ and $Z = \max(X_1, X_2)$. What is the probability of $Y \geq y$ given that $Z \leq z$ for any $y, z \in [0, 1]$? What is the correlation of Y and Z ?

Solution:

This problem is another demonstration that a figure is worth a thousand words. As shown in Figure, the probability that $Z \leq z$ is simply the square with side length z . So $P(Z \leq z) = z^2$. Since $Z = \max(X_1, X_2)$ and $Y = \min(X_1, X_2)$, we must have $Y \leq Z$ for any pair of X_1 and X_2 . So if $y > z$, $P(Y \geq y | Z \leq z) = 0$. For $y \leq z$, that X_1 and X_2 satisfies $Y \geq y$ and $Z \leq z$ is the square with vertices (y, y) , (z, y) , (z, z) , and (y, z) , which has an area $(z - y)^2$. So $P(Y \geq y | Z \leq z) = (z - y)^2$. Hence

$$P(Y \geq y | Z \leq z) = \begin{cases} \frac{(z-y)^2}{z^2}, & \text{if } 0 \leq z \leq 1 \text{ and } 0 \leq y \leq z \\ 0, & \text{otherwise.} \end{cases}$$

Now let's move on to calculate the correlation of Y and Z .

$$\text{corr}(Y, Z) = \frac{\text{cov}(Y, Z)}{\text{std}(Y) \times \text{std}(Z)} = \frac{E[YZ] - E[Y]E[Z]}{\sqrt{E[Y^2] - E[Y]^2} \times \sqrt{E[Z^2] - E[Z]^2}}$$

Using previous problem's conclusions, we have $E[Y] = \frac{1}{2+1} = \frac{1}{3}$, $E[Z] = \frac{2}{2+1} = \frac{2}{3}$.

From the pdfs of Y and Z , $f_Y(x) = n(1 - x)^{n-1} = 2(1 - x)$ and $f_Z(z) = nz^{n-1} = 2z$, we can also get $E[Y^2] = \int_0^1 2(1 - y)y^2 dy = \frac{2}{3} - \frac{2}{4} = \frac{1}{6}$ and $E[Z^2] = \int_0^1 2z^3 dz = \frac{2}{4} = \frac{1}{2}$, which give us the variances: $\text{var}(Y) = E[Y^2] - E[Y]^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$ and $\text{var}(Z) = \frac{2}{4} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$.

To calculate $E[YZ]$, we can use $E[YZ] = \int_0^1 \int_0^z yzf(y, z) dy dz$. To solve this equation, we need $f(y, z)$. We can see that when $0 \leq z \leq 1$ and $0 \leq y \leq z$, $F(y, z)$ is the shadowed area with probability

$$F(y, z) = P(Y \leq y \cap Z \leq z) = P(Z \leq z) - P(Y \geq y \cap Z \leq z) = z^2 - (z - y)^2 = 2zy - y^2$$

$$\therefore f(y, z) = \frac{\partial}{\partial y \partial z} F(y, z) = 2$$

$$\text{and } E[YZ] = \int_0^1 \int_0^z 2yz dy dz = \int_0^1 z [y^2]_0^z dz = \int_0^1 z^3 dz = \frac{1}{4}.$$

An alternative and simpler approach to calculate $E[YZ]$ is again to take advantage of symmetry. Notice that no matter $x_1 \leq x_2$ or $x_1 > x_2$, we always have $yz = x_1 x_2$ ($z = \max(x_1, x_2)$ and $y = \min(x_1, x_2)$).

$$\therefore E[YZ] = \int_0^1 \int_0^1 x_1 x_2 dx_1 dx_2 = E[X_1]E[X_2] = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Hence

$$\text{cov}(Y, Z) = E[YZ] - E[Y]E[Z] = \frac{1}{4} - \frac{1}{3} \times \frac{2}{3} = \frac{1}{36}$$

and

$$\text{corr}(Y, Z) = \frac{\text{cov}(Y, Z)}{\sqrt{\text{var}(Y)} \times \sqrt{\text{var}(Z)}} = \frac{\frac{1}{36}}{\sqrt{\frac{1}{18}} \times \sqrt{\frac{1}{18}}} = \frac{1}{2}.$$

Sanity check: That Y and Z have positive autocorrelation makes sense since when Y becomes large, Z tends to become large as well ($Z \geq Y$).

4 Mathematical Statistics

4.1 Parameter Estimation

In statistics, parameter estimation is the process of using sample data to estimate the parameters of a population. Common methods of parameter estimation include point estimation, method of moments estimation, and maximum likelihood estimation. Below are descriptions of these methods.

4.1.1 Point Estimation

Point estimation is a method of estimating the value of an unknown parameter using a single value or point from the sample data. The objective of point estimation is to find an estimator that is as close as possible to the unknown population parameter. Examples of point estimators include:

- **Sample Mean** ($\bar{X}$): The average of all the sample observations, used to estimate the population mean.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- **Sample Variance** (S^2): An estimator for the population variance, calculated as follows:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

4.1.2 Method of Moments Estimation

The method of moments estimation involves equating sample moments to population moments to solve for the parameters. A moment is a specific quantitative measure of the shape of a probability distribution. The k -th sample moment about the origin is defined as:

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k$$

For example, the first sample moment about the origin is the sample mean:

$$m_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

The second sample moment about the origin is:

$$m_2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

To estimate the parameters using the method of moments, we set the sample moments equal to the corresponding theoretical moments of the population distribution. For instance, consider a distribution with parameters μ (mean) and σ^2 (variance). The first two theoretical moments about the origin are:

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \mu^2 + \sigma^2$$

By equating the sample moments to the theoretical moments, we get the following system of equations:

$$\begin{aligned} \bar{X} &= \mu \\ m_2 &= \mu^2 + \sigma^2 \end{aligned}$$

Solving these equations simultaneously allows us to find the method of moments estimators for μ and σ^2 . For example, solving for σ^2 gives:

$$\sigma^2 = m_2 - \mu^2$$

Thus, the method of moments estimators for the mean μ and variance σ^2 are:

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = m_2 - \bar{X}^2 = m_2 - \hat{\mu}^2$$

By solving these equations, we can find the estimates for μ and σ^2 .

4.1.3 Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE) is a method of estimating the parameters of a statistical model by maximizing the likelihood function. The likelihood function measures the probability of observing the given sample data as a function of the parameters.

Given a sample $X_1, X_2, \dots, X_n$ from a population with a probability density function (pdf) $f(X; \theta)$, the likelihood function $L(\theta)$ is defined as:

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

The MLE for the parameter θ is the value that maximizes $L(\theta)$. Often, it is easier to work with the natural logarithm of the likelihood function, known as the log-likelihood function:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

The maximum likelihood estimator $\hat{\theta}$ is obtained by solving the equation:

$$\frac{\partial \ell(\theta)}{\partial \theta} = 0$$

These methods provide different approaches to parameter estimation, each with its own advantages and applications depending on the nature of the data and the statistical model being used.

4.1.4 Fisher Information

For a probability distribution $p(x|\theta)$, where θ is an unknown parameter, Fisher Information $I(\theta)$ is defined as:

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln p(X|\theta) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln p(X|\theta) \right]$$

Where:

- $p(X|\theta)$ is the probability density function of the observed data X , given the parameter θ .
- $\ln p(X|\theta)$ is the log-likelihood function.
- $\frac{\partial}{\partial \theta} \ln p(X|\theta)$ is the derivative of the log-likelihood function with respect to θ , known as the score function.

4.1.5 Cramér-Rao Bound

The Cramér-Rao bound gives a lower limit on the variance of any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

This inequality shows that the variance of an unbiased estimator cannot be smaller than the inverse of the Fisher Information. A higher Fisher Information value implies a lower bound on the estimator's variance, indicating more precise estimates.

4.2 Degrees of Freedom

In mathematical statistics, the term **degrees of freedom** (DOF) is crucial for various statistical analyses. It typically refers to the number of independent values or quantities that can vary in the analysis without violating any constraints. Here are some common contexts where degrees of freedom are used:

1. Sample Mean Degrees of Freedom

When calculating the sample mean, the degrees of freedom are usually the sample size minus one. For example, if you have a sample of size n , you lose one degree of freedom because one parameter (the sample mean) is estimated from the data. Thus, the degrees of freedom for the sample variance calculation would be $n - 1$.

$$\text{Degrees of Freedom} = n - 1$$

2. t-Distribution Degrees of Freedom

In the context of t-tests, the degrees of freedom are related to the sample size. For a one-sample t-test, the degrees of freedom are $n - 1$, where n is the sample size. The larger the degrees of freedom, the closer the t-distribution is to the standard normal distribution.

$$\text{Degrees of Freedom} = n - 1$$

3. Degrees of Freedom in Regression Analysis

In regression analysis, the degrees of freedom depend on the number of parameters in the model. Typically, the total degrees of freedom are the number of observations minus the number of estimated parameters (including the intercept). For instance, for a linear regression model with n observations and p predictors, the degrees of freedom for the residuals would be $n - p - 1$.

$$\text{Degrees of Freedom} = n - p - 1$$

4. Chi-Square Distribution Degrees of Freedom

In chi-square tests, the degrees of freedom are often associated with the number of categories. For example, if you have a categorical variable with k categories, the degrees of freedom for the chi-square test are $k - 1$.

$$\text{Degrees of Freedom} = k - 1$$

In summary, degrees of freedom refer to the number of independent pieces of information available to estimate another piece of information. Correctly calculating and understanding degrees of freedom is essential in statistical analyses as they directly impact the results and interpretations of statistical tests.

4.3 Hypothesis Testing

In parametric hypothesis testing, the distribution type of the population is known, and the purpose of hypothesis testing is to test the population parameters. For this reason, the researcher needs to put forward a hypothesis in advance to judge whether the hypothesis is true based on the sample statistics. In parametric hypothesis testing, the "hypothesis" is a statement about the specific numerical value of the population parameter. In order to make the sample statistics used as evidence necessarily support and only support one hypothesis, a pair of logically complete and mutually exclusive hypotheses for the population parameters must be established, that is, the null hypothesis (null hypothesis, recorded as H_0), alternative hypothesis (alternative hypothesis, recorded as H_1). Usually, the hypothesis that the researcher hopes to collect evidence to reject is regarded as the null hypothesis, and the hypothesis that the researcher hopes to collect evidence to support is regarded as the alternative hypothesis.

A **confidence interval** refers to the probability that a population parameter will fall between a set of values for a certain proportion of times. If a point estimate is generated from a statistical model of 10.00 with a 95 percent confidence interval of 9.50 - 10.50, it can be inferred that there is a 95

4.3.1 Z-test

The Z-test is used primarily **when the sample size is large** ($n > 30$) **and the population variance is known**. It is appropriate for testing the mean of a population versus a standard or comparing the means of two different populations under the assumption of normal distribution.

Formulas

- Single sample:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

where $\bar{X}$ is the sample mean, μ_0 is the population mean, σ is the population standard deviation, and n is the sample size.

- Two-sample:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\bar{X}_1$ and $\bar{X}_2$ are the sample means of two populations, σ_1 and σ_2 are the standard deviations of the two populations, and n_1 and n_2 are the sample sizes of the two populations.

Hypotheses

- Single sample:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0 \text{ or } \mu > \mu_0 \text{ or } \mu < \mu_0$$

- Two-sample:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2 \text{ or } \mu_1 > \mu_2 \text{ or } \mu_1 < \mu_2$$

Confidence Intervals

- Single sample:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Two-sample:

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

4.3.2 t-test

The t-test is used **when the population variance is unknown and the sample size is small** ($n < 30$). It is used to determine if there are significant differences between the mean of a sample and a population mean (single-sample t-test), between the means of two independent samples (independent t-test), or between paired samples (paired t-test).

Formulas

- Single-sample t-test:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

where $\bar{X}$ is the sample mean, μ_0 is the population mean, s is the sample standard deviation, and n is the sample size.

- Independent samples (equal variances):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

where $\bar{X}_1$ and $\bar{X}_2$ are the sample means of two independent samples, s_p is the pooled standard deviation, n_1 and n_2 are the sample sizes of the two samples.

- Paired samples:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n - 1}}$$

where $\bar{d}$ is the mean of the differences between paired observations, d_i are the differences for each pair of observations, s_d is the standard deviation of the differences, and n is the number of pairs.

Hypotheses

- Single-sample t-test:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0 \text{ or } \mu > \mu_0 \text{ or } \mu < \mu_0$$

- Independent samples (equal variances):

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2 \text{ or } \mu_1 > \mu_2 \text{ or } \mu_1 < \mu_2$$

- Paired samples:

$$H_0 : \mu_d = 0, \quad H_1 : \mu_d \neq 0 \text{ or } \mu_d > 0 \text{ or } \mu_d < 0$$

Confidence Intervals

- Single-sample t-test:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2, n-1}$ is the critical value from the t-distribution with $n - 1$ degrees of freedom corresponding to the desired confidence level α .

- Independent samples (equal variances):

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $t_{\alpha/2, n_1+n_2-2}$ is the critical value from the t-distribution with $n_1 + n_2 - 2$ degrees of freedom corresponding to the desired confidence level α .

- Paired samples:

$$\bar{d} \pm t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}$$

where $t_{\alpha/2, n-1}$ is the critical value from the t-distribution with $n - 1$ degrees of freedom corresponding to the desired confidence level α .

4.3.3 Chi-square test

The Chi-square test is used for categorical data to test hypotheses about the frequency distribution of categorical variables. It is applicable in situations like goodness-of-fit tests to see if sample data matches a population distributed a certain way, or tests of independence to explore the relationship between two variables.

Formula

•

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed frequency, E_i is the expected frequency under the null hypothesis.

Hypotheses

- H_0 states no significant difference between observed and expected frequencies
- H_1 indicates a significant difference

Confidence Intervals

- Chi-square tests are not typically used to calculate confidence intervals, as they assess the fit or independence of categorical data.

4.3.4 F-test

The F-test is used to compare the variances of two populations and is often used in ANOVA (Analysis of Variance) to test the equality of means across different groups when the dependent variable is continuous and the independent variables are categorical.

Formula

•

$$F = \frac{s_1^2}{s_2^2}$$

where s_1^2 and s_2^2 are the sample variances of the two populations being compared.

Hypotheses

- $H_0 : \sigma_1^2 = \sigma_2^2$
- $H_1 : \sigma_1^2 \neq \sigma_2^2$

Confidence Intervals

- The F-test is primarily used for testing the equality of variances between two datasets, and it does not directly provide a confidence interval.

4.4 Rejection Regions for Different Statistical Tests**Z-test Rejection Region**

- **One-tailed test:** The rejection region is $\{Z \geq z_\alpha\}$ or $\{Z \leq z_\alpha\}$, where z_α is the $1 - \alpha$ percentile of the standard normal distribution.
- **Two-tailed test:** The rejection region is $\{Z \leq z_{\alpha/2}\} \cup \{Z \geq z_{1-\alpha/2}\}$, where $z_{\alpha/2}$ and $z_{1-\alpha/2}$ are the $\alpha/2$ and $1 - \alpha/2$ percentiles of the standard normal distribution, respectively.

t-test Rejection Region

- **One-tailed test:** The rejection region is $\{t \geq t_{\alpha, df}\}$ or $\{t \leq t_{\alpha, df}\}$, where $t_{\alpha, df}$ is the $1 - \alpha$ percentile of the t-distribution with df degrees of freedom.
- **Two-tailed test:** The rejection region is $\{t \leq t_{\alpha/2, df}\} \cup \{t \geq t_{1-\alpha/2, df}\}$, where $t_{\alpha/2, df}$ and $t_{1-\alpha/2, df}$ are the $\alpha/2$ and $1 - \alpha/2$ percentiles of the t-distribution with df degrees of freedom, respectively.

Chi-square Test Rejection Region

- **Rejection Region:** Typically a right-tailed test, the rejection region is $\{\chi^2 \geq \chi_{\alpha, df}^2\}$, where $\chi_{\alpha, df}^2$ is the $1 - \alpha$ percentile of the chi-square distribution with df degrees of freedom.

F-test Rejection Region

- **Rejection Region:** Usually a right-tailed test, the rejection region is $\{F \geq F_{\alpha, df1, df2}\}$, where $F_{\alpha, df1, df2}$ is the $1 - \alpha$ percentile of the F-distribution with $df1$ and $df2$ degrees of freedom.

4.5 Type I Error and Type II Error**4.5.1 Type I Error**

- **Definition:** Type I error occurs when the null hypothesis (H_0) is true but is incorrectly rejected.
- **Symbol:** The probability of a Type I error is denoted by α , also known as the significance level.
- **Impact:** This means we believe there is an effect or difference when in reality, there isn't one.
- **Example:** In a drug trial, if the null hypothesis (H_0) is "the drug is ineffective" and the alternative hypothesis (H_1) is "the drug is effective," a Type I error would occur if we incorrectly reject H_0 when the drug is actually ineffective.

4.5.2 Type II Error

- **Definition:** Type II error occurs when the null hypothesis (H_0) is false but is incorrectly failed to be rejected.
- **Symbol:** The probability of a Type II error is denoted by β .
- **Impact:** This means we fail to detect an effect or difference that actually exists.
- **Example:** In the same drug trial, a Type II error would occur if we fail to reject H_0 when the drug is actually effective.

Table of Type I and Type II Errors

	Fail to Reject H_0	Reject H_0
H_0 is True	Correct Decision	Type I Error (α)
H_0 is False	Type II Error (β)	Correct Decision

Table 1: Type I and Type II Errors

4.6 P-value

The p-value is a crucial concept in statistics, used in hypothesis testing to help decide whether to reject the null hypothesis. The p-value measures the probability of observing the data, or more extreme data, under the assumption that the null hypothesis is true.

Process of Using p-values

1. **Set the Null Hypothesis:** This is the baseline assumption you wish to test, typically representing no effect or no difference.
2. **Collect Data and Compute Statistics:** Conduct an experiment or gather data, then calculate an appropriate statistic, such as a mean difference or ratio.

3. **Calculate the p-value:** The p-value is the probability of obtaining the observed statistic, or more extreme, given that the null hypothesis is true. The smaller this value, the less likely it is to observe such an outcome under the null hypothesis.
4. **Set the Significance Level (α):** This is usually 0.05 or 0.01, indicating the risk level of incorrectly rejecting the null hypothesis you are willing to accept.
5. **Make a Decision:**
 - If the p-value is less than or equal to α , reject the null hypothesis, considering the results statistically significant.
 - If the p-value is greater than α , do not reject the null hypothesis.

For example, if a drug test results in a p-value of 0.03 and your significance level is set at 0.05, you would reject the null hypothesis (that the drug is ineffective) and conclude that the drug is effective, supported by statistical evidence.

Note: It is important to remember that the p-value is not evidence of the null hypothesis being correct or incorrect, but rather the probability of observing the current or more extreme outcomes if the null hypothesis were true. It represents a conditional probability, not a direct probability.

4.7 Classic Questions

4.7.1 Statistical Test Review I

We want to test whether or not a coin is balanced based on the number of heads, X , that appear after 36 tosses of the coin. Let's say we use the rejection region $|x - 18| \geq 4$. What is the value of α to the nearest thousandth?

solution:

Let us define

$$H_0 : p = 0.5$$

$$H_a : p \neq 0.5$$

Our rejection region is $|x - 18| \geq 4$. Recall, by definition, that α is the probability that the test statistic, X , is within the rejection region when H_0 is true. Hence,

$$\begin{aligned} \alpha &= \sum_{x=0}^{14} \binom{36}{x} (0.5)^x (0.5)^{36-x} + \sum_{x=22}^{36} \binom{36}{x} (0.5)^x (0.5)^{36-x} \\ &\approx 0.243 \end{aligned}$$

4.7.2 Statistical Test Review II

We want to test whether or not a coin is balanced based on the number of heads, X , that appear after 36 tosses of the coin. Let's say we use the rejection region $|x - 18| \geq 4$. If $p = 0.7$, what is the value of β to the nearest ten thousandth?

solution:

Let us define

$$H_0 : p = 0.5$$

$$H_a : p \neq 0.5$$

Our rejection region is $|x - 18| \geq 4$. Recall, by definition, that β is the probability that the test statistic, X , is not within the rejection region when H_0 is rejected, or in this case, $p = 0.7$. Let $Y \sim \text{Binom}(0.7)$. Then,

$$\begin{aligned} \beta &= \mathbb{P}(Y \leq 21) - \mathbb{P}(Y \leq 14) \\ &\approx 0.0916 \end{aligned}$$

4.7.3 Statistical Test Review III

Trader wages have mean 132 and standard deviation 25. QuantHomies, a proprietary trading firm, employs 40 traders and pays them a wage of 122. Using an $\alpha = 0.01$ level test (note that $z_{0.01} = -2.326$), determine whether QuantHomies can be accused of paying wages that are below the industry standard. Respond 1 if yes, 0 if no.

solution:

We define

$$H_0 : \mu = 132$$

$$H_a : \mu < 132$$

Here, $n = 40$, which is large enough for us to let $\mu_{\text{sample}} = 122$ be a point estimator of μ that is approximately normally distributed. We can then define our test statistic as

$$Z = \frac{\mu_{\text{sample}} - \mu}{\sigma/\sqrt{40}}$$

Under the null hypothesis, we expect $Z \sim \mathcal{N}(0, 1)$. We reject H_0 if $Z < z_{0.01} = -2.326$.

$$Z = \frac{\sqrt{40}(122 - 132)}{25} \approx -2.53 < -2.326$$

We may reject the null hypothesis, which means we can accuse QuantHomies of paying wages that are below the industry standard. We should respond with 1.

4.7.4 Statistical Test Review V

Rishab wants to use a Z test to test a hypothesis involving the population mean. Which of the following conditions must be met in order for Rishab to go through with the Z test?

- (1) Population variance must be known,
- (2) sample must be a random sample of the population,
- (3) distribution of sample mean must be approximately normal,
- (4) the sample size is at least 30.

Please respond with the conditions concatenated in increasing order. For example, if 1 and 3 must be met, respond with 13.

solution:

The population variance must be known, though one could make an argument that it can be estimated accurately by the sample variance when $n \geq 30$. The sample must of course be a random sample of the population. The distribution of the sample mean must be approximately normal, which suggests $n \geq 30$ by the Central Limit Theorem. However, if it is given that the population distribution is normal, then it is not necessarily the case that $n \geq 30$ must be satisfied to proceed with the Z test. Therefore, our answer is 123.

4.7.5 Statistical Test Review IV

QuantGuide's competitor, QuantChaperone, offers three plans: A , B , C . 400 of the first 1000 plans sold are of plan A . Can one conclude that customers have a preference for plan A ? Respond 1 if yes, 0 if no.

solution:

Let p denote the probability that a customer chooses plan A . We define

$$H_0 : p = \frac{1}{3}$$

$$H_a : p > \frac{1}{3}$$

For a Bernoulli random variable X with parameter p , recall that $\mu = p$ and $\sigma^2 = p(1-p)$. Since $n = 1000$, we have a sufficiently large sample size and may assume standard normally distributed $\hat{p}$, which we define as our sample proportion: $\hat{p} = 0.4$. Our test statistic is

$$\begin{aligned} Z &= \frac{\hat{p} - \mu}{\sigma/\sqrt{n}} \\ &= \frac{\sqrt{1000}(0.4 - \frac{1}{3})}{\sqrt{\frac{1}{3} \cdot \frac{2}{3}}} \\ &\approx 4.47 \\ \Rightarrow \mathbb{P}(Z > 4.47) &\approx 3.4 \times 10^{-6} \end{aligned}$$

This p-value is much smaller than typical significance levels for α such as $\alpha = 0.05$ or $\alpha = 0.01$. Since the p-value is less than α , we may reject H_0 , which tells us that there is enough evidence to suggest that customers have a preference for plan A. We should answer with 1.

4.7.6 Statistical Test Review VI

Andy claims that 20% of the public loves QuantGuide. Aaron dislikes Andy and wants to prove him wrong—specifically, he thinks Andy's estimate is too large. He takes a random sample of 100 people to test his claim. With $\alpha = 0.05$ (note that $z_{0.05} = -1.645$), what is the maximum number of people in the sample that may love QuantGuide such that Aaron can refute Andy's claim?

solution:

Let p denote the probability that a person loves QuantGuide. We define

$$H_0 : p = 0.2$$

$$H_a : p < 0.2$$

For a Bernoulli random variable X with parameter p , recall that $\mu = p$ and $\sigma^2 = p(1-p)$. Let $\hat{p}$ denote the sample proportion. Since $n = 100$, our sample size is sufficiently large to assume that $\hat{p}$ is normally distributed with mean p and variance $p(1-p)$. Our test statistic is

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{\hat{p} - 0.2}{0.04}$$

In order to reject the null hypothesis, we need our test statistic Z to be less than $z_{0.05} = -1.645$. With some simple algebra we determine $\hat{p} < 0.1342$. The maximum integer number of people that may love QuantGuide such that Andy's claim can be refuted is therefore 13.

4.7.7 Statistical Test Review VII

Andy claims that 20% of the public loves QuantGuide. Aaron dislikes Andy and wants to prove him wrong—specifically, he thinks Andy's estimate is too large. He takes a random sample of 100 people to test his claim. It turns out that 15 of the 100 people love QuantGuide, so Aaron argues that 15% of the public loves QuantGuide. Compute, to the nearest thousandth, the value of β for Aaron's alternative hypothesis under a 0.05-level test (note that $z_{0.05} = -1.645$).

solution:

Let p denote the probability that a person loves QuantGuide. We define

$$H_0 : p = 0.2$$

$$H_a : p = 0.15$$

Our test statistic is

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{\hat{p} - 0.2}{0.04}$$

Z should be a standard normal random variable under the null hypothesis. Under the alternative hypothesis,

$$Z_a = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \approx \frac{\hat{p} - 0.15}{0.357}$$

Recall, by definition, that β is the probability that the test statistic, X , is not within the rejection region when H_0 is rejected. So,

$$\begin{aligned}\beta &= \mathbb{P}\left(\frac{10(\hat{p} - 1.5)}{0.357} > \frac{0.4 \cdot (-1.645 + 5) - 1.5}{0.357}\right) \\ &= \mathbb{P}\left(\frac{10(\hat{p} - 1.5)}{0.357} \leq 0.04426\right) \approx 0.671\end{aligned}$$

4.7.8 Statistical Test Review VIII

Consider a random sample of size 25 from the distribution $\mathcal{N}(160, 900)$. To the nearest ten thousandth, what is the probability that the sample mean is 165 or greater?

solution:

Let $\hat{\mu}$ denote the sample mean. Our test statistic is defined as follows:

$$\begin{aligned}z &= \frac{\hat{\mu} - \mu}{\sigma/\sqrt{n}} \\ &= \frac{\sqrt{25}(165 - 160)}{30} \\ &= \frac{5}{6}\end{aligned}$$

The desired probability is simply $\mathbb{P}(Z \geq \frac{5}{6}) = 1 - \mathbb{P}(Z \leq \frac{5}{6})$ for $Z \sim \mathcal{N}(0, 1)$. With a calculator, we find this value to be 0.2023.

5 Time Series

5.1 Stationarity

Stationarity is the foundation of time series analysis. A time series can be denoted by $Y = \{Y_1, \dots, Y_t, \dots\}$. Stationarity is divided into strict stationarity and weak stationarity.

5.1.1 Strictly Stationary

A time series Y is strictly stationary if for any time t and any time intervals $s_1, \dots, s_n$,

$$(Y_{s_1}, \dots, Y_{s_n}) \sim (Y_{t+s_1}, \dots, Y_{t+s_n})$$

where $\sim$ means identically distributed.

5.1.2 Weakly Stationary

A time series Y is weakly stationary if it satisfies the following two conditions:

- $\mathbb{E}[Y_t] = \mu$, i.e., the mean μ does not depend on time t
- The covariance $\text{Cov}(Y_t, Y_{t-h}) = \gamma(h)$ is finite and depends only on the lag h :

$$\text{Var}(Y_t) = \gamma(0), \quad \forall t$$

$$\gamma(h) = \gamma(-h) = \text{Cov}(Y_t, Y_{t+h}), \quad \forall t$$

A strictly stationary process with finite covariances is weakly stationary, but the converse is not necessarily true.

5.2 Autocovariance, Autocorrelation, ACF, and PACF

In time series analysis, autocovariance and autocorrelation are key concepts used to understand the relationships between different time points in a series.

5.2.1 Autocovariance

The autocovariance function measures the covariance of the time series with a lagged version of itself. For a time series $\{X_t\}$, the autocovariance at lag h is defined as:

$$\gamma(h) = \text{Cov}(X_t, X_{t-h}) = \mathbb{E}[(X_t - \mu)(X_{t-h} - \mu)]$$

where μ is the mean of the time series X_t .

5.2.2 Autocorrelation

The autocorrelation function standardizes the autocovariance function to produce a dimensionless measure. It is defined as the autocovariance at lag h divided by the variance:

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

where $\gamma(0)$ is the variance of the time series.

5.2.3 Autocorrelation Function (ACF)

The autocorrelation function (ACF) is a sequence of autocorrelations at different lags. For a time series $\{X_t\}$, the ACF at lag h is:

$$\text{ACF}(h) = \rho(h)$$

It provides a measure of how the observations in a time series are related to each other at different lags.

5.2.4 Partial Autocorrelation Function (PACF)

The partial autocorrelation function (PACF) measures the correlation between the time series observations at different lags, after controlling for the correlations at all shorter lags. For a time series $\{X_t\}$, the PACF at lag h is defined as the partial correlation of X_t and X_{t-h} , controlling for the intermediate values $X_{t-1}, \dots, X_{t-h+1}$.

5.2.5 Ljung-Box Test

The Ljung-Box test is a statistical test used to check for the presence of autocorrelation in a time series. Specifically, it tests whether the values in a time series are correlated with previous values (lags). Autocorrelation means that past values influence future values, violating the assumption of randomness in time series data.

The Ljung-Box test checks whether the autocorrelations for multiple lags (up to m) are jointly zero, meaning that the time series data is independent of its own past values. If significant autocorrelation is detected, it may suggest that a model (such as an ARIMA model) could be useful to capture these dependencies. The test is widely used in the analysis of residuals from time series models to verify if any autocorrelation remains unexplained by the model.

1. State the Hypotheses

The hypotheses for the Ljung-Box test can be written mathematically as:

- Null hypothesis H_0 : The autocorrelations for all lags up to m are zero.

$$H_0 : \rho_1 = \rho_2 = \dots = \rho_m = 0$$

- Alternative hypothesis H_1 : At least one of the autocorrelations up to lag m is non-zero.

$$H_1 : \rho_k \neq 0 \quad \text{for at least one } k \in \{1, 2, \dots, m\}$$

2. Compute Autocorrelation Function (ACF)

For each lag k , calculate the autocorrelation $\hat{\rho}_k$ in the time series using the following formula:

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^n (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^n (y_t - \bar{y})^2}$$

where:

- y_t is the value of the time series at time t ,
- $\bar{y}$ is the mean of the time series,
- n is the total number of observations.

3. Select the Number of Lags m

Decide the number of lags m to test. Common practice is to select m based on sample size n , such as $m = \sqrt{n}$, or another appropriate value based on domain-specific knowledge.

4. Calculate the Ljung-Box Statistic

The Ljung-Box statistic Q is calculated using the following formula:

$$Q = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{n-k}$$

where:

- n is the sample size,
- m is the number of lags,
- $\hat{\rho}_k$ is the autocorrelation at lag k .

5. Determine the Critical Value

The Ljung-Box statistic Q follows a chi-squared (χ^2) distribution with m degrees of freedom. The critical value is determined from the chi-squared distribution table using:

- Degrees of freedom $df = m$,
- Significance level α , typically 0.05 or 0.01.

6. Perform the Hypothesis Test

Compare the calculated Q statistic with the chi-squared critical value $\chi_{\alpha,m}^2$:

- If $Q > \chi_{\alpha,m}^2$, reject the null hypothesis H_0 and conclude that the series has significant autocorrelation.
- If $Q \leq \chi_{\alpha,m}^2$, do not reject the null hypothesis, indicating no significant autocorrelation.

7. Interpret the Results

- If the null hypothesis H_0 is rejected, the time series shows significant autocorrelation at one or more lags.
- If the null hypothesis is not rejected, there is no evidence of autocorrelation, and the time series might be considered random.

Key Takeaways

The Ljung-Box test is useful for:

- Testing whether residuals from a time series model are white noise.
- Identifying if a model captures the time series' dependency structure adequately.
- Validating assumptions in time series forecasting and analysis.

5.3 White Noise

A weakly stationary process $\{\epsilon_t\}$ is a white noise if it satisfies the following conditions:

1. $\mathbb{E}[\epsilon_t] = 0$
2. $\text{Var}(\epsilon_t) = \sigma_\epsilon^2$
3. $\text{Cov}(\epsilon_s, \epsilon_t) = 0$ for $s \neq t$

This can be denoted as:

$$\epsilon_t \sim WN(0, \sigma_\epsilon^2)$$

Gaussian White Noise

It is a Gaussian white noise if $\epsilon_t \sim N(0, \sigma_\epsilon^2)$ and the ϵ_t are i.i.d. (independent and identically distributed).

5.4 Linear Time Series Models**5.4.1 Autoregressive (AR) Model with a Constant Term****General Expression:**

$$X_t = \delta + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t$$

where:

- X_t is the value of the time series at time t ,
- δ is the constant term, representing the mean level of the series when other terms are zero,
- $\phi_1, \phi_2, \dots, \phi_p$ are the parameters of the model, indicating the impact of past values on the current value,
- p is the order of the AR model, specifying the number of lagged terms of the series that are used as predictors,
- ϵ_t is the white noise error term at time t .

Expectation:

$$E(X_t) = \frac{\delta}{1 - \sum_{i=1}^p \phi_i}$$

Variance:

$$\text{Var}(X_t) = \frac{\sigma^2}{1 - \sum_{i=1}^p \phi_i^2}$$

Autocorrelation Function (ACF):

$$\rho(k) = \sum_{i=1}^p \phi_i \rho(k-i)$$

Yule-Walker Equations: The Yule-Walker equations for an AR(p) model provide a method for estimating the model parameters ϕ_i using the autocorrelations of the observed time series. The equations are given by:

$$\begin{aligned} \rho(1) &= \phi_1 + \phi_2 \rho(1) + \phi_3 \rho(2) + \cdots + \phi_p \rho(p-1) \\ \rho(2) &= \phi_1 \rho(1) + \phi_2 + \phi_3 \rho(1) + \cdots + \phi_p \rho(p-2) \\ &\vdots \\ \rho(p) &= \phi_1 \rho(p-1) + \phi_2 \rho(p-2) + \cdots + \phi_p \end{aligned}$$

These equations can be solved to find the ϕ_i values that best fit the observed autocorrelations.

Matrix Form of Yule-Walker Equations: The matrix form of the Yule-Walker equations for an AR(p) model is:

$$\begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) & \rho(2) & \cdots & \rho(p-1) \\ \rho(1) & 1 & \rho(1) & \cdots & \rho(p-2) \\ \rho(2) & \rho(1) & 1 & \cdots & \rho(p-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \rho(p-3) & \cdots & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix}$$

This matrix equation can be solved to obtain the ϕ_i values.

Partial Autocorrelation Function (PACF): The PACF for an AR(p) model is zero at lag k for $k > p$, and significant at lag p .

5.4.2 Moving Average (MA) Model with a Constant Term

General Expression:

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

where:

- X_t is the value of the time series at time t ,
- μ is the constant term or the mean level of the series,
- ϵ_t are the white noise error terms at time t ,
- $\theta_1, \theta_2, \dots, \theta_q$ are the parameters of the model,
- q is the order of the MA model, indicating the number of lagged error terms included.

Expectation:

$$E(X_t) = \mu$$

Variance:

$$\text{Var}(X_t) = \sigma^2 \left(1 + \sum_{i=1}^q \theta_i^2 \right)$$

Autocorrelation Function (ACF): For lag k greater than q , the autocorrelation is zero:

$$\rho(k) = 0 \quad \text{for } k > q$$

Partial Autocorrelation Function (PACF): The PACF for an MA(q) model decays gradually for all lags.

5.4.3 ARMA Model with a Constant Term

An ARMA model combines both Autoregressive (AR) and Moving Average (MA) components. The general form of an ARMA(p, q) model with a constant term is given by:

General Expression:

$$X_t = \delta + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

Expectation:

$$E(X_t) = \frac{\delta}{1 - \sum_{i=1}^p \phi_i}$$

Variance: The variance of an ARMA(p, q) model is generally more complex to derive directly due to the interaction between the AR and MA components. However, for simpler cases like ARMA(1,1), we can express the variance as:

$$\frac{1 + \theta_1^2 + 2\phi_1\theta_1}{1 - \phi_1^2} \sigma^2$$

For general ARMA(p, q) models, the variance can be computed using advanced techniques or numerical methods.

Autocorrelation Function (ACF): The ACF of an ARMA(p, q) model exhibits a mixture of the behavior of AR and MA models. It decays in a more complex pattern, typically combining exponential decay and cut-offs.

Partial Autocorrelation Function (PACF): Similarly, the PACF of an ARMA(p, q) model combines the characteristics of AR and MA models. It shows significant values up to lag p and then decays gradually.

Conclusion

The ARMA model provides a versatile framework for modeling time series data by combining the strengths of both AR and MA models. This flexibility allows for a better fit to complex data patterns that neither AR nor MA models alone can adequately capture. When applying ARMA models, careful parameter estimation and model validation are essential to ensure accurate and reliable results.

5.4.4 Calculation Examples

Assume we have an AR(1) model and an MA(1) model with constant terms, parameters ϕ_1 and θ_1 , and white noise variance σ^2 .

AR(1) Model with a Constant Term

$$X_t = \delta + \phi_1 X_{t-1} + \epsilon_t$$

Expectation:

$$E(X_t) = \frac{\delta}{1 - \phi_1}$$

Variance:

$$\text{Var}(X_t) = \frac{\sigma^2}{1 - \phi_1^2}$$

ACF:

$$\rho(k) = \phi_1^k$$

PACF: At lag 1, PACF is ϕ_1 ; at other lags, it is 0.

MA(1) Model with a Constant Term

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Expectation:

$$E(X_t) = \mu$$

Variance:

$$\text{Var}(X_t) = \sigma^2(1 + \theta_1^2)$$

ACF:

$$\rho(0) = 1, \quad \rho(1) = \frac{\theta_1}{1 + \theta_1^2}, \quad \rho(k) = 0 \quad \text{for } k > 1$$

PACF: The PACF decays gradually for all lags.

To illustrate ARMA models, let's consider ARMA(1,1) and ARMA(2,1) models.

ARMA(1,1) Model with a Constant Term

$$X_t = \delta + \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Expectation:

$$E(X_t) = \frac{\delta}{1 - \phi_1}$$

Variance:

$$\frac{1 + \theta_1^2 + 2\phi_1\theta_1}{1 - \phi_1^2} \sigma^2$$

ACF:

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{(\phi_1 + \theta_1)(1 + \phi_1\theta_1)}{1 + \theta_1^2 + 2\phi_1\theta_1}$$

$$\rho(h) = \phi_1^{h-1} \rho(1), \quad h \geq 2$$

PACF: The PACF exhibits significant value at lag 1 and then decays gradually for subsequent lags.

5.4.5 ARIMA Model

The ARIMA (Autoregressive Integrated Moving Average) model is a popular statistical method used for time series analysis and forecasting. It combines three components: autoregression (AR), differencing (I), and moving average (MA).

Autoregressive (AR) Part The autoregressive part involves regressing the variable on its own lagged values. The AR(p) model can be written as:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t \quad (1)$$

where:

- X_t is the value of the time series at time t
- $\phi_1, \phi_2, \dots, \phi_p$ are the parameters of the model
- ϵ_t is the error term (white noise)
- p is the order of the autoregressive part

Differencing (I) Part Differencing is used to make the time series stationary by removing trends and seasonality. The differenced series Y_t is given by:

$$Y_t = \Delta^d X_t = (1 - L)^d X_t \quad (2)$$

where:

- Y_t is the differenced time series
- L is the lag operator
- d is the order of differencing

Moving Average (MA) Part The moving average part involves modeling the error term as a linear combination of past error terms. The MA(q) model is written as:

$$X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} \quad (3)$$

where:

- $\theta_1, \theta_2, \dots, \theta_q$ are the parameters of the model
- q is the order of the moving average part

ARIMA Model Representation Combining the AR, I, and MA parts, the ARIMA model is denoted as ARIMA(p, d, q), where:

- p is the number of lag observations included in the model (order of AR)
- d is the number of times that the raw observations are differenced (order of I)
- q is the size of the moving average window (order of MA)

5.4.6 Unit-Root Test

The **ADF Test** (Augmented Dickey-Fuller Test) is a statistical test used in time series analysis to determine whether a time series has a **unit root**, meaning it is non-stationary. A time series with a unit root is unpredictable and thus harder to model for forecasting.

The key equation for the ADF test is an extension of the autoregressive model:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t$$

where:

- $\Delta y_t = y_t - y_{t-1}$ is the **first difference** of the time series.
- α is a **constant term** (intercept).
- βt represents the **time trend**.
- γ is the coefficient being tested for the existence of a unit root.
- δ_i are the **lag terms** of the differenced series, added to address higher-order correlation.
- ϵ_t is the **white noise error term**.

Null and Alternative Hypotheses

- **Null Hypothesis** (H_0): The time series has a unit root (i.e., it is non-stationary). This implies $\gamma = 0$.
- **Alternative Hypothesis** (H_1): The time series is stationary. This implies $\gamma < 0$.

ADF Test Statistic

The ADF test statistic is calculated by:

$$\text{ADF Statistic} = \frac{\hat{\gamma}}{\text{Standard Error of } \hat{\gamma}}$$

where:

- $\hat{\gamma}$ is the estimated value of γ from the regression.
- Standard Error of $\hat{\gamma}$ is the standard error of the estimate.

Decision Criteria

The ADF statistic is compared to critical values at various significance levels (1%, 5%, 10%):

- If the ADF statistic is less than the critical value (in absolute terms), reject the null hypothesis and conclude that the series is stationary.
- If the ADF statistic is greater than the critical value (in absolute terms), fail to reject the null hypothesis, meaning the series has a unit root (non-stationary).

Models in ADF Test

Depending on the assumptions, the ADF test uses different models:

1. **Without constant and trend:**

$$\Delta y_t = \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t$$

2. **With constant but without trend:**

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t$$

3. **With constant and trend:**

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t$$

Critical Values and p-value

The test results will often provide:

- The **ADF statistic** (test statistic calculated).
- **p-value** to indicate the significance of the result. If $p\text{-value} < 0.05$, reject the null hypothesis.
- **Critical values** at 1%, 5%, and 10% levels, compared against the ADF statistic.

5.5 Conditional heteroscedastic Time Series Models

5.5.1 ARCH Model

The Autoregressive Conditional Heteroskedasticity (ARCH) model, introduced by Robert Engle in 1982, is used to model and forecast the volatility of time series data. The ARCH model captures the time-varying nature of volatility, which is often observed in financial time series data.

ARCH($\bar{p}$) Process: The ARCH($\bar{p}$) model expresses the current conditional variance as a function of past squared observations. The mathematical formulation of the ARCH($\bar{p}$) model is given by:

$$a_t = \sigma_t \epsilon_t \quad (4)$$

$$\sigma_t = \sqrt{w + \alpha_1 a_{t-1}^2 + \cdots + \alpha_{\bar{p}} a_{t-\bar{p}}^2} \quad (5)$$

where:

- a_t is the time series value at time t
- σ_t is the conditional standard deviation at time t
- $\epsilon_t \sim N(0, 1)$ is a Gaussian white noise
- $w > 0$ is a constant
- $\alpha_i \geq 0$ are the parameters of the model
- $\alpha_1 + \cdots + \alpha_{\bar{p}} < 1$

Zero conditional and unconditional means:

- $\mathbb{E}[a_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\epsilon_t | \mathcal{F}_{t-1}] = 0$
- $\mathbb{E}[a_t] = \mathbb{E}[\mathbb{E}[a_t | \mathcal{F}_{t-1}]] = 0$

Conditional Variance and Unconditional Variance

- **Conditional variance**

$$\begin{aligned} \text{var}(a_t | \mathcal{F}_{t-1}) &= \mathbb{E}[a_t^2 | \mathcal{F}_{t-1}] \\ &= \sigma_t^2 \mathbb{E}[\epsilon_t^2 | \mathcal{F}_{t-1}] \\ &= \sigma_t^2 = w + \alpha_1 a_{t-1}^2 + \cdots + \alpha_{\bar{p}} a_{t-\bar{p}}^2 \end{aligned}$$

- **Unconditional variance**

$$\begin{aligned} \gamma_a(0) &= \text{var}(a_t) = \mathbb{E}[a_t^2] \\ &= \mathbb{E}[\mathbb{E}[a_t^2 | \mathcal{F}_{t-1}]] \\ &= \mathbb{E}[w + \alpha_1 a_{t-1}^2 + \cdots + \alpha_{\bar{p}} a_{t-\bar{p}}^2] \\ &= w + \alpha_1 \mathbb{E}[a_{t-1}^2] + \cdots + \alpha_{\bar{p}} \mathbb{E}[a_{t-\bar{p}}^2] \\ &= w + \alpha_1 \gamma_a(0) + \cdots + \alpha_{\bar{p}} \gamma_a(0) \\ \gamma_a(0) &= \frac{w}{1 - (\alpha_1 + \cdots + \alpha_{\bar{p}})} \end{aligned}$$

5.5.2 GARCH Model

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, developed by Tim Bollerslev in 1986, extends the ARCH model by allowing past conditional variances to be included in the current conditional variance equation. This makes the model more flexible and able to capture more complex volatility patterns. Financial time series often exhibit volatility clustering (extreme values followed by extreme values, large variance followed by large variance)

GARCH($\bar{p}, \bar{q}$) Process: The GARCH($\bar{p}, \bar{q}$) model expresses the current conditional variance as a function of past squared observations and past conditional variances. The mathematical formulation of the GARCH($\bar{p}, \bar{q}$) model is given by:

$$a_t = \sigma_t \epsilon_t \quad (6)$$

$$\sigma_t = \sqrt{w + \alpha_1 a_{t-1}^2 + \cdots + \alpha_{\bar{p}} a_{t-\bar{p}}^2 + \beta_1 \sigma_{t-1}^2 + \cdots + \beta_{\bar{q}} \sigma_{t-\bar{q}}^2} \quad (7)$$

where:

- a_t is the time series value at time t
- σ_t is the conditional standard deviation at time t
- $\epsilon_t \sim N(0, 1)$ is a Gaussian white noise
- $w > 0$ is a constant
- $\alpha_i \geq 0$ are the parameters of the ARCH part of the model
- $\beta_i \geq 0$ are the parameters of the GARCH part of the model
- $\alpha_1 + \cdots + \alpha_{\bar{p}} + \beta_1 + \cdots + \beta_{\bar{q}} < 1$

Applications ARCH and GARCH models are widely used in finance for modeling and forecasting the volatility of asset returns. They are particularly useful for risk management, option pricing, and financial market analysis.

Example

- Consider a GARCH(1,1) model

$$a_t = \sigma_t \epsilon_t$$

$$\sigma_t = \sqrt{w + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2}$$

where $\epsilon \sim N(0, 1)$ is a Gaussian white noise, $w > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, $\alpha_1 + \beta_1 < 1$

- a_t has zero conditional and unconditional means;
- conditional variance of a_t :

$$\text{var}(a_t | \mathcal{F}_{t-1}) = \sigma_t^2$$

- Unconditional variance of GARCH(1,1)

$$\begin{aligned} \text{var}(a_t) &= \mathbb{E}[a_t^2] \\ &= \mathbb{E}[\mathbb{E}[a_t^2 | \mathcal{F}_{t-1}]] \\ &= \mathbb{E}[\sigma_t^2] \\ &= \mathbb{E}[w + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2] \\ &= w + \alpha_1 \text{var}(a_{t-1}) + \beta_1 \text{var}(a_{t-1}) \\ \gamma_a(0) &= \frac{w}{1 - (\alpha_1 + \beta_1)} \end{aligned}$$

5.5.3 TGARCH Model

(Threshold GARCH): Based on the GARCH model, it considers that positive and negative shocks may have different impacts on volatility by introducing a threshold to model the effects of different types of shocks.

(1) General Form of the TGARCH(p, q) Model

The conditional variance equation models the dynamics of volatility, incorporating asymmetric effects:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^q \gamma_i \epsilon_{t-i}^2 I(\epsilon_{t-i} < 0) + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Alternatively, it can be written as:

$$\sigma_t^2 = \omega + \sum_{i=1}^q (\alpha_i + \gamma_i I(\epsilon_{t-i} < 0)) \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

where:

- σ_t^2 : The conditional variance (volatility) at time t .
- ω : The constant term, representing the long-term average variance.
- α_i : ARCH term coefficients, measuring the impact of past squared residuals.
- γ_i : Asymmetric effect coefficients, capturing the additional impact of negative shocks.
- β_j : GARCH term coefficients, representing the persistence of volatility.
- $I(\epsilon_{t-i} < 0)$: Indicator function, equal to 1 if $\epsilon_{t-i} < 0$, and 0 otherwise.

Explanation of Each Parameter

1. ω (Omega - Constant Term)

- **Meaning:** Represents the base level of variance or the long-term average variance.
- **Role:** When there are no new shocks, the conditional variance tends toward ω .
- **Requirement:** $\omega > 0$ to ensure that variance is positive.

2. α_i (Alpha - ARCH Term Coefficients)

- **Meaning:** Measures the impact of past squared residuals on current variance.
- **Role:** Captures the effect of new market information or shocks on volatility.
- **Requirement:** $\alpha_i \geq 0$ to maintain non-negativity of variance.

3. γ_i (Gamma - Asymmetric Effect Coefficients)

- **Meaning:** Measures the additional impact of negative shocks on current variance.
- **Role:** When $\epsilon_{t-i} < 0$ (negative shock), $I(\epsilon_{t-i} < 0) = 1$, so $\gamma_i \epsilon_{t-i}^2$ adds to the variance. When $\epsilon_{t-i} \geq 0$, $I(\epsilon_{t-i} < 0) = 0$, so there is no additional impact.
- **Interpretation:** If $\gamma_i > 0$, negative shocks increase volatility more than positive shocks of the same magnitude (leverage effect). If $\gamma_i < 0$, negative shocks decrease volatility (less common).
- **Requirement:** Typically $\gamma_i \geq 0$ to reflect the leverage effect in financial markets.

4. β_j (Beta - GARCH Term Coefficients)

- **Meaning:** Measures the persistence of volatility, capturing the influence of past variances.
- **Role:** Reflects the long-term impact or persistence of shocks on volatility.
- **Requirement:** $\beta_j \geq 0$ to ensure non-negativity of variance.

5. $I(\epsilon_{t-i} < 0)$ (Indicator Function)

- **Meaning:** Identifies past negative shocks.
- **Values:** $I(\epsilon_{t-i} < 0) = 1$ if $\epsilon_{t-i} < 0$, and $I(\epsilon_{t-i} < 0) = 0$ if $\epsilon_{t-i} \geq 0$.
- **Role:** Allows the model to distinguish between the effects of positive and negative shocks on volatility.

6. ϵ_t (Epsilon - Residual Term)

- **Meaning:** The difference between the observed value and the conditional mean.
- **Relationship:** $\epsilon_t = \sigma_t z_t$, where σ_t is the conditional standard deviation and z_t is a standardized random variable.
- **Role:** Contains unpredictable random shocks, forming the basis for modeling volatility.

7. σ_t^2 (Sigma Squared - Conditional Variance)

- **Meaning:** The variance of the time series at time t conditional on past information.
- **Role:** Reflects dynamic changes in volatility influenced by past shocks and variances.

8. z_t (Standardized Random Variable)

- **Meaning:** A random variable with mean zero and variance one.
- **Assumption:** Typically $z_t \sim N(0,1)$, but other distributions (e.g., t-distribution) can be used to capture heavy tails.
- **Role:** Introduces randomness, generating residuals with specific distributional properties.

Summary

The TGARCH model extends the traditional GARCH model by introducing asymmetric terms to more accurately describe the behavior of volatility in financial time series. It is particularly useful for capturing the leverage effect, where negative shocks have a more pronounced impact on volatility than positive shocks. Understanding each parameter's meaning and role is crucial for correctly specifying and interpreting the model.

(2) Differences Between Bootstrap and Simulation in TARCH Models

TARCH (Threshold Autoregressive Conditional Heteroskedasticity) models are econometric models used for time series data to capture asymmetric effects in volatility. When analyzing TARCH models, bootstrap and simulation are two commonly used methods. They differ in their application purposes and procedures. Below is a detailed comparison:

Bootstrap in TARCH Models

Bootstrap in TARCH models is typically used to estimate the distribution of model parameters, their standard errors, and to construct confidence intervals. The steps are as follows:

1. **Fit the Model:** Fit the TARCH model to the original data to obtain parameter estimates.
2. **Extract Residuals:** Extract the residuals from the fitted TARCH model.
3. **Resample Residuals:** Perform resampling with replacement on the residuals to create multiple new residual series.
4. **Generate New Samples:** Use the resampled residuals and the fitted model parameters to generate multiple new time series samples.
5. **Refit the Model:** Refit the TARCH model to each new time series sample and record the parameter estimates.
6. **Construct Distribution:** Use the multiple sets of parameter estimates to construct their empirical distribution, estimate standard errors, and build confidence intervals.

The advantage of the bootstrap method is that it allows for the estimation of parameter precision without making strong distributional assumptions, and it is particularly useful for small sample sizes.

Simulation in TARCH Models

Simulation in TARCH models is used to generate time series data to evaluate the model's predictive performance and stability. The steps are as follows:

1. **Set Initial Conditions:** Define initial conditions such as initial values and model parameters.
2. **Generate Random Errors:** Generate a sequence of random errors based on a predefined distribution (e.g., normal distribution).
3. **Run Simulation:** Use the recursive formula of the TARCH model and the generated random errors to produce simulated time series data.
4. **Analyze Results:** Analyze the simulated data to calculate statistics such as mean, variance, and other metrics to assess model performance.

The advantage of simulation is that it allows researchers to generate large amounts of synthetic data to study the TARCH model's behavior under different scenarios and test its robustness in extreme conditions.

Key Differences

- **Purpose:**
 - **Bootstrap:** Mainly used to estimate the distribution, standard errors, and confidence intervals of model parameters.
 - **Simulation:** Mainly used to generate time series data to evaluate predictive performance and model stability.
- **Method:**
 - **Bootstrap:** Based on resampling the original residuals to create new samples.
 - **Simulation:** Based on setting initial conditions and generating random errors to recursively produce data.
- **Application:**
 - **Bootstrap:** Suitable for estimating parameter precision and hypothesis testing.
 - **Simulation:** Suitable for performance evaluation and scenario analysis.

5.5.4 EGARCH Model

The EGARCH(p, q) model is defined by the following equation:

$$\ln(\sigma_t^2) = \omega + \sum_{i=1}^q \beta_i \ln(\sigma_{t-i}^2) + \sum_{j=1}^p \alpha_j \left(\frac{|\epsilon_{t-j}|}{\sigma_{t-j}} \right) + \sum_{j=1}^p \gamma_j \left(\frac{\epsilon_{t-j}}{\sigma_{t-j}} \right)$$

Parameters and Variables

- σ_t^2 : Conditional variance (volatility) at time t .
- ω : Constant term, setting the baseline level of volatility.
- β_i : Coefficients for lagged conditional variances, capturing the persistence of volatility.
- α_j : Coefficients for lagged standardized absolute residuals, measuring the impact of past shocks on current volatility.
- γ_j : Coefficients for lagged standardized residuals, capturing asymmetries (e.g., leverage effects).
- ϵ_t : Residual or error term at time t , typically modeled as $\epsilon_t = \sigma_t z_t$ where z_t is a standardized residual with $E(z_t) = 0$ and $\text{Var}(z_t) = 1$.

Special Case: EGARCH(1,1)

In practice, the most commonly used specification is the EGARCH(1, 1) model, which simplifies the general form as follows:

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \alpha \left(\frac{|\epsilon_{t-1}|}{\sigma_{t-1}} \right) + \gamma \left(\frac{\epsilon_{t-1}}{\sigma_{t-1}} \right)$$

Or equivalently,

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \alpha |z_{t-1}| + \gamma z_{t-1}$$

Parameter Interpretation

1. **Log Conditional Variance** ($\ln(\sigma_t^2)$): Taking the logarithm ensures that the conditional variance remains positive without imposing additional constraints on the parameters.
2. **Constant Term** (ω): Adjusts the baseline level of volatility in the model.
3. **Lagged Conditional Variance Coefficient** (β): Reflects the persistence of volatility from one period to the next. Typically, $0 < \beta < 1$.
4. **Lagged Absolute Residual Coefficient** (α): Measures the impact of past shocks (regardless of sign) on current volatility. A larger α indicates greater sensitivity to new information.
5. **Lagged Residual Coefficient** (γ): Captures the asymmetry in the response of volatility to positive and negative shocks. If $\gamma < 0$, it implies that negative shocks have a larger impact on volatility than positive shocks, embodying the leverage effect.

5.6 Model Selection

In time series modeling, model selection is a crucial step to find the best-fitting model for the data. Common criteria for model selection include Log Likelihood (LL), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC). Here's a detailed explanation of these criteria and their uses:

5.6.1 Log Likelihood (LL)

Log Likelihood measures the goodness of fit of a model. It is calculated as the logarithm of the likelihood function. For time series models, **a higher log likelihood value indicates a better fit to the data.**

5.6.2 Akaike Information Criterion (AIC)

AIC is a criterion that balances the goodness of fit and the complexity of the model. Its formula is:

$$\text{AIC} = 2k - 2\ln(L) \quad (8)$$

where k is the number of parameters in the model, and L is the maximum likelihood estimate.

The goal of AIC is to find a model that achieves a balance between complexity and fit. **A lower AIC value indicates a better model that fits the data well** while accounting for its complexity.

5.6.3 Bayesian Information Criterion (BIC)

BIC also balances the goodness of fit and model complexity, but it does so differently from AIC. Its formula is:

$$\text{BIC} = \ln(n)k - 2\ln(L) \quad (9)$$

where n is the sample size, k is the number of parameters in the model, and L is the maximum likelihood estimate.

BIC tends to favor simpler models, especially when the sample size is large. **A lower BIC value indicates a better model that fits the data well** while considering its complexity.

5.6.4 Comparison and Selection

When selecting models, you can compare the Log Likelihood, AIC, and BIC values of different models:

- **Log Likelihood (LL):** Mainly used to measure the goodness of fit. A higher log likelihood value indicates a better fit but does not consider model complexity.
- **AIC:** Balances goodness of fit and model complexity, suitable for smaller sample sizes. A lower AIC value indicates a better model.
- **BIC:** Puts more emphasis on model simplicity, suitable for larger sample sizes. A lower BIC value indicates a better model.

Overall, AIC and BIC are essential criteria for model selection, helping us choose a model that fits the data well without being overly complex. The choice of criterion depends on the specific application scenario and sample size. In practice, it is often necessary to consider these criteria comprehensively and combine them with the research objectives and data characteristics to make the best model selection.

6 Regression Analysis

6.1 Motivations and assumptions

A linear regression model can be represented as follows:

$$y = \beta_0 + \beta_1 x + \epsilon$$

where y is the dependent variable, x is the independent variable, β_0 is the intercept, β_1 is the slope, and ϵ is the error term.

1. Linearity

The relationship between the independent variable x and the dependent variable y is linear. This implies that the expected value of y given x is a linear function of x :

$$E(y|x) = \beta_0 + \beta_1 x$$

2. Independence

The residuals (errors) are independent. This means there is no correlation between different errors. Additionally, the residuals are independent of the explanatory variable x_i .

$$\text{Cov}(\epsilon_i, \epsilon_j) = 0 \quad \text{for } i \neq j$$

$$\text{Cov}(x_i, \epsilon_i) = 0$$

3. Homoskedasticity

The residuals (errors) have constant variance at every level of x .

$$\text{Var}(\epsilon_i) = \sigma^2 \quad \forall i$$

4. Normality

The residuals (errors) of the model are normally distributed.

$$\epsilon_i \sim N(0, \sigma^2)$$

5. No Multicollinearity

In the case of multiple regression, **the independent variables are not correlated with each other.** No multicollinearity requires that all the columns of X should be linearly independent: otherwise, some k th column can be expressed as a linear combination of the other columns; k th regressor is redundant.

The variance inflation factor (VIF) is used to detect the presence of multicollinearity:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination of the regression of x_j on all other independent variables. A VIF value greater than 10 indicates high multicollinearity.

6.2 Simple linear regression

6.2.1 Ordinary least squares estimation

For a Simple linear regression:

$$Y_i = \beta_1 + \beta_2 X_i + \epsilon_i, \quad 1 \leq i \leq n$$

Distance between Y_i and $\hat{\beta}_1 + \hat{\beta}_2 X_i$ on the estimated line:

$$Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i$$

Ordinary least squares estimation: find $\hat{\beta}_1, \hat{\beta}_2$ that minimize:

$$L(\hat{\beta}_1, \hat{\beta}_2) = \sum_{i=1}^n (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2$$

Solve **normal equations**: $\frac{\partial L}{\partial \hat{\beta}_1} = 0, \frac{\partial L}{\partial \hat{\beta}_2} = 0$

$$\sum_{i=1}^n (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) = 0 \Rightarrow \bar{Y}_n - \hat{\beta}_1 - \hat{\beta}_2 \bar{X}_n = 0$$

$$\sum_{i=1}^n X_i (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) = 0 \Rightarrow \sum_{i=1}^n X_i Y_i - \hat{\beta}_1 \sum_{i=1}^n X_i - \hat{\beta}_2 \sum_{i=1}^n X_i^2 = 0$$

where $\bar{X}_n$ and $\bar{Y}_n$ are sample means.

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^n (X_i - \bar{X}_n)^2}, \quad \hat{\beta}_1 = \bar{Y}_n - \hat{\beta}_2 \bar{X}_n$$

Denote:

$$S_{XX} = \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

$$S_{XY} = \sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)$$

Then

$$\hat{\beta}_2 = \frac{S_{XY}}{S_{XX}}, \quad \hat{\beta}_1 = \bar{Y}_n - \hat{\beta}_2 \bar{X}_n$$

Fitted values and residuals

- The i th **fitted value**

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$$

Part of the dependent variable explained by the regressors

- The i th **OLS residual**

$$\hat{\epsilon}_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i$$

Part of the dependent variable not explained by the regressors

Estimating σ^2

Residuals can be used to estimate σ^2

- Note that sample mean of $\hat{\epsilon}_i$'s is zero (1st normal equation)

$$\frac{1}{n} \sum_{i=1}^n \hat{\epsilon}_i = \bar{Y}_n - \hat{\beta}_1 - \hat{\beta}_2 \bar{X}_n = 0$$

- OLS estimator for σ^2

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- $\hat{\sigma}^2$ is an **unbiased** estimator of σ^2 ; $\hat{\sigma}$ known as **residual standard error**; $n-2$ is the **degrees of freedom**

6.2.2 Analysis of Simple Regression Model**Unbiasedness**

- OLS estimator $\hat{\beta}_2$ is unbiased with $\mathbb{E}[\hat{\beta}_2] = \beta_2$ since

$$\begin{aligned} \mathbb{E}[\hat{\beta}_2] &= \mathbb{E} \left[\frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{S_{XX}} \right] \\ &= \frac{1}{S_{XX}} \sum_{i=1}^n (X_i - \bar{X}_n) \mathbb{E}[(Y_i - \bar{Y}_n)] \\ &= \frac{1}{S_{XX}} \sum_{i=1}^n (X_i - \bar{X}_n) (\beta_1 + \beta_2 X_i - (\beta_1 + \beta_2 \bar{X}_n)) \\ &= \beta_2 \end{aligned}$$

- $\hat{\beta}_1$ is unbiased with $\mathbb{E}[\hat{\beta}_1] = \beta_1$ since

$$\mathbb{E}[\hat{\beta}_1] = \mathbb{E}[\bar{Y}_n - \hat{\beta}_2 \bar{X}_n] = \beta_1 + \beta_2 \bar{X}_n - \beta_2 \bar{X}_n = \beta_1$$

Standard error for $\hat{\beta}_2$

- Note that

$$\begin{aligned} \hat{\beta}_2 &= \sum_{i=1}^n \frac{X_i - \bar{X}_n}{S_{XX}} Y_i \\ &= \sum_{i=1}^n \frac{X_i - \bar{X}_n}{S_{XX}} (\beta_1 + \beta_2 X_i + \epsilon_i) \end{aligned}$$

- Recall the zero conditional correlation assumption:

$$\text{var}(\hat{\beta}_2) = \frac{1}{S_{XX}^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \text{var}(\epsilon_i) = \frac{\sigma^2}{S_{XX}}$$

- Estimated standard error: $\text{se}(\hat{\beta}_2) = \frac{\hat{\sigma}}{\sqrt{S_{XX}}}$
- When X_i 's are more spread out: better estimation with smaller $\text{se}(\hat{\beta}_2)$

Standard error for $\hat{\beta}_1$

- Recall that $\hat{\beta}_1 = \bar{Y}_n - \hat{\beta}_2 \bar{X}_n$

$$\begin{aligned}\text{cov}(\hat{\beta}_2, \bar{Y}_n) &= \sum_{i=1}^n \frac{X_i - \bar{X}_n}{S_{XX}} \text{cov}(Y_i, \bar{Y}_n) \\ &= \frac{\sigma^2}{nS_{XX}} \sum_{i=1}^n (X_i - \bar{X}_n) = 0\end{aligned}$$

- Variance and estimated standard error of $\hat{\beta}_1$

$$\begin{aligned}\text{var}(\hat{\beta}_1) &= \text{var}(\bar{Y}_n) + \bar{X}_n^2 \text{var}(\hat{\beta}_2) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}_n^2}{S_{XX}} \right) \\ \text{se}(\hat{\beta}_1) &= \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{S_{XX}}}\end{aligned}$$

6.2.3 Analysis of variance (ANOVA)

- Variation in the dependent variable: total sum of squares

$$SST = \sum_{i=1}^n (Y_i - \bar{Y}_n)^2$$

- Variation explained by the regression model: regression sum of squares

$$SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y}_n)^2 \quad \left(\text{note that } \frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}_n \right)$$

- Variation due to the noise: error sum of squares (or residual sum of squares)

$$SSE = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad \left(\text{note that } \frac{1}{n} \sum_{i=1}^n \hat{\epsilon}_i = 0 \right)$$

Decomposition of variation: $SST = SSR + SSE$

$$\begin{aligned}SST &= \sum_{i=1}^n (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y}_n)^2 \\ &= SSE + SSR + 2 \sum_{i=1}^n (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y}_n)\end{aligned}$$

$$\text{Recall } \hat{\beta}_1 = \bar{Y}_n - \hat{\beta}_2 \bar{X}_n \Rightarrow \hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i = \bar{Y}_n + \hat{\beta}_2 (X_i - \bar{X}_n)$$

$$\begin{aligned}\sum_{i=1}^n (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y}_n) &= \hat{\beta}_2 \sum_{i=1}^n (Y_i - \bar{Y}_n - \hat{\beta}_2 (X_i - \bar{X}_n))(X_i - \bar{X}_n) \\ &= \hat{\beta}_2 (S_{XY} - \hat{\beta}_2 S_{XX}) = 0\end{aligned}$$

6.2.4 R squared

R^2 (coefficient of determination) measures the proportion of variation in Y that is explained by the regression

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- R^2 close to 1 indicates a good fitting
- R = sample correlation in the simple regression model

$$\begin{aligned}\rho^2 &= \frac{(\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n))^2}{(\sum_{i=1}^n (X_i - \bar{X}_n)^2)(\sum_{i=1}^n (Y_i - \bar{Y}_n)^2)} = \frac{S_{XY}^2}{S_{XX}S_{YY}} \\ &= \frac{\hat{\beta}_2^2 S_{XX}}{SST} = \frac{SSR}{SST} = R^2 \quad \left(\text{Recall } \hat{Y}_i = \bar{Y}_n + \hat{\beta}_2(X_i - \bar{X}_n) \right)\end{aligned}$$

6.3 Multiple linear regression

General expression of multiple linear regression:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \varepsilon$$

where Y is the dependent variable, $X_1, X_2, \dots, X_K$ are the independent variables, $\beta_0, \beta_1, \dots, \beta_K$ are the coefficients, and ε is the error term.

6.3.1 Normal equation

$$\mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{Y}$$

where $\mathbf{X}$ is the matrix of independent variables, β is the vector of coefficients, and $\mathbf{Y}$ is the vector of the dependent variable.

Derivation of the Normal Equation

- The objective is to minimize the sum of squared residuals:

$$\text{Minimize } S(\beta) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_p X_{ip})^2$$

- Rewrite in matrix form:

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta)$$

- Take the derivative with respect to β and set it to zero:

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\beta) = 0$$

- Simplify to obtain the normal equation:

$$\mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{Y}$$

- Solving the normal equation for β gives the estimate $\hat{\beta}$:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

6.3.2 Fitted values

- Fitted values using matrix notation

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y = PY$$

- Recall the first normal equation in $X^T(Y - X\hat{\beta}) = X^T(Y - \hat{Y}) = 0$, where $X_{i1} = 1$:

$$\sum_{i=1}^n X_{i1}(Y_i - \hat{Y}_i) = \sum_{i=1}^n (Y_i - \hat{Y}_i) = 0 \Rightarrow \bar{Y}_n := \frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}_n$$

- $P = X(X^T X)^{-1} X^T$ is called the projection matrix, $M = I_n - P = I_n - X(X^T X)^{-1} X^T$ is called the annihilator matrix. They are symmetric and idempotent

$$\begin{aligned} P^T &= P, & P^2 &= P, & PX &= X \\ M^T &= M, & M^2 &= M, & MX &= 0 \end{aligned}$$

6.3.3 Estimating σ^2

- Residuals using matrix notation

$$\hat{\epsilon} = Y - \hat{Y} = (I_n - P)Y = MY$$

- Error sum of squares

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \hat{\epsilon}^T \hat{\epsilon} = Y^T M Y$$

The following OLS estimator for σ^2 is unbiased:

$$\hat{\sigma}^2 = \frac{SSE}{n - K}$$

6.3.4 Analysis of variance (ANOVA)

- $SST = SSR + SSE$

$$\begin{aligned} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2 &= \sum_{i=1}^n (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y}_n)^2 \\ &= SSE + SSR + 2 \sum_{i=1}^n (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y}_n) \\ &= SSE + SSR + 2 \sum_{i=1}^n (Y_i - \hat{Y}_i) \hat{Y}_i \\ &= SSE + SSR + 2(Y - \hat{Y})^T \hat{Y} \\ &= SSE + SSR + 2Y^T P Y - 2Y^T P^T P Y \\ &= SSE + SSR \end{aligned}$$

- SST has $n - 1$ degrees of freedom

$$SST = \sum_{i=1}^n (Y_i - \bar{Y}_n)^2$$

- **SSE has $n - K - 1$ degrees of freedom**

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Recall the normal equations $X^T(Y - \hat{Y}) = 0$

$$\sum_{i=1}^n X_{ik}(Y_i - \hat{Y}_i) = 0, \quad k = 1, \dots, K$$

- **SSR has K degrees of freedom**

$$SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y}_n)^2$$

Recall that

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_{i2} + \dots + \hat{\beta}_K X_{iK}$$

$$\bar{Y}_n = \hat{\bar{Y}}_n = \hat{\beta}_1 + \hat{\beta}_2 \bar{X}_{.2} + \dots + \hat{\beta}_K \bar{X}_{.K}, \quad \bar{X}_{.k} = \frac{1}{n} \sum_{i=1}^n X_{ik}$$

$$\hat{Y}_i - \bar{Y}_n = \sum_{k=2}^K \hat{\beta}_k (X_{ik} - \bar{X}_{.k})$$

6.3.5 R-squared and Adjusted R-squared

R-squared (R^2)

R-squared is a statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model. The formula for R^2 is:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

where:

- SSE is the Sum of Squares for Error (residual sum of squares).
- SST is the Total Sum of Squares.

Adjusted R-squared (R^2_{adj})

Adjusted R-squared adjusts the R^2 value for the number of predictors in the model. It is defined as:

$$R^2_{\text{adj}} = 1 - \left(\frac{SSE/(n-k-1)}{SST/(n-1)} \right)$$

where:

- n is the sample size.
- k is the number of independent variables.

Explanation

- R^2 indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. Its value ranges from 0 to 1, with higher values indicating better model fit.
- However, R^2 tends to increase with the addition of more predictors, even if they are not significant.
- Adjusted R^2 provides a more accurate measure by penalizing the addition of non-significant predictors. It can decrease if the added variable does not improve the model significantly.

6.4 Regression diagnostics

6.4.1 t-Test in Linear Regression

Hypotheses:

$$H_0: \beta_k = 0$$

$$H_1: \beta_k \neq 0$$

Method 1: Computing Confidence Interval

1. Calculate the Standard Error of $\hat{\beta}_k$:

$$SE(\hat{\beta}_k) = \hat{\sigma} \sqrt{(\mathbf{X}^\top \mathbf{X})_{kk}^{-1}}$$

where $\hat{\sigma}$ is the standard error of the residuals, and $(\mathbf{X}^\top \mathbf{X})_{kk}^{-1}$ is the k th diagonal element of the matrix $(\mathbf{X}^\top \mathbf{X})^{-1}$.

2. Compute the Confidence Interval:

$$\hat{\beta}_k \pm t_{\alpha/2, n-k} \cdot SE(\hat{\beta}_k)$$

where $t_{\alpha/2, n-k}$ is the critical value from the t-distribution with $n - k$ degrees of freedom.

3. Check if the Confidence Interval Contains 0:

- If the confidence interval contains 0, fail to reject the null hypothesis.
- If the confidence interval does not contain 0, reject the null hypothesis.

Method 2: Computing t-Statistic

1. Calculate the t-Statistic:

$$t_{0k} = \frac{\hat{\beta}_k}{SE(\hat{\beta}_k)}$$

where $\hat{\beta}_k$ is the estimated regression coefficient, and $SE(\hat{\beta}_k)$ is its standard error.

2. Compare t-Statistic with Critical Value:

- Look up the critical value $t_{\alpha/2, n-k}$ from the t-distribution table.
- Compare $|t_{0k}|$ with $t_{\alpha/2, n-k}$:
 - If $|t_{0k}| > t_{\alpha/2, n-k}$, reject the null hypothesis.
 - If $|t_{0k}| \leq t_{\alpha/2, n-k}$, fail to reject the null hypothesis.

Method 3: Computing p-Value

1. Calculate the t-Statistic:

$$t_{0k} = \frac{\hat{\beta}_k}{SE(\hat{\beta}_k)}$$

2. Compute the p-Value:

$$p = 2 \cdot \mathbb{P}(T_{n-k} > |t_{0k}|)$$

where T_{n-k} is the t-distribution with $n - k$ degrees of freedom.

3. Compare p-Value with Significance Level α :

- If $p < \alpha$, reject the null hypothesis.
- If $p \geq \alpha$, fail to reject the null hypothesis.

6.4.2 F-Test in Linear Regression

Hypotheses:

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ (The model has no explanatory power)

$H_1 : \text{At least one } \beta_i \neq 0$ (The model has explanatory power)

Sum of Squares:

$$\text{Total Sum of Squares (SST) : } SST = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$\text{Regression Sum of Squares (SSR) : } SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

$$\text{Error Sum of Squares (SSE) : } SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Note that: $SST = SSR + SSE$

Mean Squares:

$$\text{Mean Square for Regression (MSR) : } MSR = \frac{SSR}{k}$$

$$\text{Mean Square for Error (MSE) : } MSE = \frac{SSE}{n - k - 1}$$

F-statistic:

$$F = \frac{MSR}{MSE}$$

Decision Rule:

- Find the critical value $F_{\alpha, k, n-k-1}$ from the F-distribution table.
- Compare the calculated F-statistic with the critical value:
 - If $F > F_{\alpha, k, n-k-1}$, reject the null hypothesis H_0 .
 - If $F \leq F_{\alpha, k, n-k-1}$, fail to reject the null hypothesis H_0 .

6.4.3 Residual Plots

Residual plots are essential for diagnosing issues with a regression model, such as nonlinearity, correlated error terms, and non-constant variances (heteroscedasticity).

Nonlinearity

A residual plot with a discernible pattern (e.g., a curve) indicates nonlinearity between the independent and dependent variables. For instance, if we fit a linear model but the true relationship is quadratic:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$$

A plot of residuals vs. fitted values would show a curved pattern.

Correlated Error Terms

If residuals are plotted against time or another sequential variable and show a pattern (e.g., cyclic behavior), this indicates autocorrelation. For instance, in time series data, we might have:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where

$$\epsilon_i = \delta + \phi_1 \epsilon_{i-1} + u_i$$

(first-order autoregressive process).

Non-constant Variances (Heteroscedasticity)

If the residual plot shows a funnel shape (residuals fan out or in), it suggests heteroscedasticity. For example:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where

$$\text{Var}(\epsilon_i) = \sigma_i^2$$

.

6.4.4 Studentized Residuals

Studentized residuals are residuals divided by an estimate of their standard deviation. They are useful for identifying outliers. For the regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$:

$$t_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

where e_i is the residual, s is the standard error of the regression, and h_{ii} is the leverage of the i -th observation.

Observations with $|t_i| > 2$ might be considered outliers.

6.4.5 Outliers

Outliers are data points that deviate significantly from the trend established by the rest of the data. In a regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$:

$$|t_i| > 2$$

indicates potential outliers, where t_i is the studentized residual.

6.4.6 Leverage

Leverage measures how far an independent variable's value deviates from its mean. High leverage points have a large influence on the regression model's fit. For a regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2}$$

where h_{ii} is the leverage of the i -th observation, n is the number of observations, x_i is the independent variable, and $\bar{x}$ is the mean of x .

6.4.7 High Leverage Points

High leverage points are those with leverage significantly larger than the average leverage. In a regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$:

$$h_{ii} > \frac{2(p+1)}{n}$$

indicates high leverage points, where p is the number of predictors, and n is the number of observations.

These diagnostic tools help assess the validity of a regression model and identify potential issues that need addressing.

6.5 Non-linear Regression

6.5.1 Lasso Regression

The general form of the linear regression model is:

$$y = \mathbf{x}^T \beta + \epsilon$$

Lasso regression (Least Absolute Shrinkage and Selection Operator) is a regression technique based on linear regression that adds an L1 regularization term (the sum of the absolute values of the coefficients) to the objective function. This allows Lasso to shrink some coefficients to exactly zero, which results in automatic feature selection.

The objective function for Lasso regression is:

$$\hat{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right)$$

Where:

1. $\mathbf{x}_i$ is the feature vector of the i -th sample,
2. y_i is the target value for the i -th sample,
3. β is the vector of regression coefficients,
4. n is the number of samples,
5. p is the number of features,
6. λ is the regularization strength hyperparameter that controls the weight of the L1 penalty. A larger λ will force more coefficients to be zero, effectively selecting fewer features.

Explanation:

1. **Loss function:** The Lasso regression loss function consists of two parts: the residual sum of squares and the L1 regularization term. The first part $\sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2$ is the residual sum of squares from traditional linear regression, measuring the difference between the predicted and actual values. The second part $\lambda \sum_{j=1}^p |\beta_j|$ is the L1 regularization term, penalizing the absolute values of the regression coefficients.

2. **L1 Regularization:** The L1 penalty in Lasso regression encourages sparsity by shrinking some coefficients to zero. This makes Lasso useful for feature selection, as it reduces the influence of irrelevant features by eliminating them from the model.

3. **Feature Selection:** Unlike Ridge regression, which only shrinks coefficients, Lasso can force some coefficients to be exactly zero, thus performing feature selection. Ridge regression does not eliminate any features but only reduces the magnitude of the coefficients.

Lasso regression is especially useful when the number of features exceeds the number of observations, or when feature selection is important in the model. The optimization problem in Lasso regression **does not have a closed-form solution**, meaning there is no direct formula that can be used to compute the optimal solution.

6.5.2 Ridge Regression

The general form of the linear regression model is:

$$y = \mathbf{x}^T \beta + \epsilon$$

Ridge regression is a type of linear regression that includes an L2 regularization term to prevent overfitting by penalizing large coefficients. The objective function for ridge regression is to minimize the sum of squared residuals plus the L2 penalty.

$$\hat{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

Where:

1. $\mathbf{x}_i$ is the feature vector of the i -th sample,
2. y_i is the target value for the i -th sample,
3. β is the vector of regression coefficients,
4. n is the number of samples,
5. p is the number of features,
6. λ is the regularization strength hyperparameter that controls the weight of the L2 penalty. When $\lambda = 0$, ridge regression reduces to ordinary least squares, and as λ increases, the model complexity is more strongly constrained.

Analytical Solution for Ridge regression

The loss function for Ridge regression is:

$$\ell(\beta) = (Y - X\beta)^T (Y - X\beta) + \lambda \beta^T \beta$$

Taking the derivative with respect to β , we get:

$$\frac{d\ell(\beta)}{d\beta} = -2X^T(Y - X\beta) + 2\lambda\beta$$

Setting the derivative to zero, we obtain:

$$(X^T X + \lambda I)\beta = X^T Y$$

Solving for β , the analytical solution for Ridge regression is:

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T Y$$

6.5.3 Logistics Regression

Logistic regression is a type of regression used when the dependent variable is categorical, typically binary (0 or 1). It models the probability that a given input belongs to a certain class using a logistic function (also called a sigmoid function) to map any real-valued number into a value between 0 and 1.

Sigmoid Function

Logistic regression uses the sigmoid function, defined as:

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

The output of the sigmoid function is always between 0 and 1, which makes it suitable for modeling probabilities.

The expression for logistic regression is:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{x}^T \beta)}$$

Where:

1. $P(y = 1|\mathbf{x})$ is the probability that the dependent variable y equals 1 (the positive class) given the feature vector $\mathbf{x}$,
2. $\mathbf{x}$ is the vector of independent variables (features),
3. β is the vector of coefficients (parameters),
4. $\exp(-\mathbf{x}^T \beta)$ is the exponent of the negative linear combination of the features and their coefficients.

6.6 Classic Questions

6.6.1 Linear Regression Assumptions

Linear regression relies on several assumptions to produce reliable results:

1. **Linearity:** The relationship between the independent variables and the dependent variable should be linear.
2. **Independence:** The residuals (errors) should be independent of each other.
3. **Homoscedasticity:** The variance of the residuals should be constant across all levels of the independent variables.
4. **Normality of Residuals:** The residuals should follow a normal distribution.
5. **No Multicollinearity:** The independent variables should not be highly correlated with each other.

6.6.2 Statistical Tests for Assumptions

1. Linearity:

- *Graphical Test:* Use a residual plot (plot residuals vs. predicted values). No clear pattern indicates linearity.
- *Ramsey's RESET Test:* Can detect non-linearity in the functional form.

2. Independence:

- *Durbin-Watson Test:* Assesses autocorrelation in residuals. A value around 2 suggests no autocorrelation.

3. Homoscedasticity:

- *Breusch-Pagan Test:* Tests for constant variance of residuals.
- *White's Test:* Another test for detecting heteroscedasticity.

4. Normality of Residuals:

- *Shapiro-Wilk Test:* Checks if residuals are normally distributed.
- *Q-Q Plot:* A graphical method for checking normality.
- *Kolmogorov-Smirnov Test:* Another normality test.

5. No Multicollinearity:

- *Variance Inflation Factor (VIF):* VIF greater than 10 may indicate multicollinearity.
- *Condition Index:* Values above 30 may indicate multicollinearity.

6.6.3 Why Each Assumption is Important

1. **Linearity:** Ensures that the model accurately represents the relationship between independent and dependent variables. If non-linearity exists, predictions may be inaccurate.
2. **Independence:** Violations lead to underestimation of standard errors, resulting in misleading conclusions about the significance of predictors.
3. **Homoscedasticity:** Constant variance of residuals ensures efficient and unbiased estimates. Heteroscedasticity can lead to inefficient estimates.
4. **Normality of Residuals:** Ensures valid hypothesis tests and confidence intervals.
5. **No Multicollinearity:** High multicollinearity inflates standard errors, making it difficult to assess the true effect of each independent variable.

6.6.4 Linearity Assumption

What happens if the linearity assumption in linear regression is not satisfied?

solution:

If the linearity assumption is violated in a linear regression model, it may lead to the following issues:

1. The model may not fit correctly. Linear regression is based on the assumption that a linear relationship exists between the independent variables and the dependent variable. If this assumption is not met, meaning the relationship between the variables is nonlinear, the fitting effect of the linear regression model will be affected. The model may not accurately capture the patterns in the data, resulting in poor predictive performance.
2. The normality of the error term: Linear regression models assume that the error term (residuals) is independently and normally distributed, following a normal distribution. If the linear relationship does not hold, the error term may violate this assumption. This may result in the distribution of the error term being abnormal, such as skewed or having unequal variance, which could affect the reliability of statistical inference and prediction from the model.
3. Parameter estimation errors: The parameters of the linear regression model are estimated based on the assumption of a linear relationship. If the linear relationship does not hold, the estimated regression coefficients may be biased, meaning the parameter estimates may be incorrect. These biased parameter estimates could lead to an incorrect interpretation of the influence of the independent variables on the dependent variable.

To address the problem of nonlinearity, the following methods can be considered:

- Apply appropriate transformations to the independent or dependent variables, such as logarithmic transformation, square root transformation, or polynomial transformations, to satisfy the assumption of linearity. This can be achieved by observing the characteristics of the data and using domain knowledge to determine appropriate transformation methods.
- Introduce interaction terms: Including interaction terms between the independent variables in the model can capture nonlinear relationships between the variables, which helps build a better linear model.
- Use nonlinear regression models: If the linear relationship does not hold, consider using nonlinear regression models, such as polynomial regression or spline regression, which can more flexibly model the nonlinear relationships in the data.

In summary, if the linearity assumption in a linear regression model is not satisfied, it may result in the model fitting poorly, abnormal error terms, biased parameter estimates, and unreliable inference. To address these issues, nonlinear transformations, interaction terms, or nonlinear regression models can be used to better capture the nonlinear relationships in the data.

6.6.5 Homoscedasticity Assumption

What happens if the constant variance assumption in linear regression (homoscedasticity) is violated?

solution:

If heteroscedasticity exists, it will affect linear regression analysis in the following ways:

1. **Inaccurate standard error estimates:** When the degree of heteroscedasticity increases in the model, the OLS estimates of standard errors may become inaccurate. This can lead to unreliable significance tests for coefficients (e.g., t-tests), which in turn may result in biased inference.

2. **Efficiency loss:** When heteroscedasticity is present, even though the OLS estimates remain unbiased, they are no longer the most efficient. Other methods can provide more accurate estimates, leading to results that are closer to the true values.

3. **Model misspecification:** Heteroscedasticity may indicate that the model is misspecified, such as missing necessary variables or improperly handling nonlinearity in the relationships. If this is the cause of the heteroscedasticity, the OLS estimates will be biased, and the predictive ability of the model will suffer.

Therefore, if heteroscedasticity exists, we need to detect and handle it accordingly (e.g., by transforming the data, using robust standard errors, or adding least squares adjustments).

6.6.6 Multicollinearity Assumption

What happens if the assumption of no multicollinearity in linear regression is violated?

solution:

If multicollinearity exists, it may affect linear regression analysis in the following ways:

1. **Instability of parameter estimates:** When there is a high degree of multicollinearity among explanatory variables, the parameter estimates obtained through OLS may become very unstable, and even small changes in the data can lead to large changes in the estimated coefficients.

2. **Difficulty in interpreting coefficients:** In the presence of multicollinearity, it becomes difficult to determine which variable is influencing the dependent variable due to the high correlation between variables.

3. **Ineffectiveness of significance tests:** Due to the increase in standard errors, the significance tests for variables may become ineffective. For example, even if all explanatory variables have an impact on the dependent variable, the significance tests may show non-significant results due to multicollinearity.

To address the problem of multicollinearity, the following methods can be considered:

- **Increase the sample size:** Increasing the sample size may alleviate the multicollinearity problem to some extent, although this is not always feasible due to practical constraints on data collection.

- **Remove irrelevant explanatory variables:** If two or more explanatory variables are highly correlated, consider removing some of them. Choose to keep variables based on theory or their explanatory power for the dependent variable.

- **Use principal component analysis or factor analysis:** These methods can transform the original highly correlated explanatory variables into new, independent ones.

- **Use ridge regression or similar shrinkage methods:** These methods can overcome multicollinearity to some extent, but they introduce bias in the estimates.

- **Add regularization terms:** For example, Lasso regression or Ridge regression can add a penalty term to the loss function, making the model less sensitive to multicollinearity.

6.6.7 Independence Assumption

What happens if the independence of errors assumption in linear regression is violated? How do you handle it?

solution:

In linear regression, one assumption is that the error terms (residuals) are independent of each other, meaning the error for each observation is not related to the errors of other observations. If the error terms are correlated (lack of independence), this can affect the results of the linear regression model in several ways:

1. **Inaccurate coefficient estimates:** Linear regression models estimate the coefficients using OLS. When the error terms are correlated, OLS estimates may be biased, causing the estimated coefficients to deviate from the true values. This can lead to poor model fitting and inaccurate predictions.

2. **Bias in statistical inference:** Statistical inference is usually based on the assumption of independent errors. If the error terms are correlated, it can result in biased hypothesis tests, confidence intervals, and prediction intervals. This may cause incorrect conclusions in statistical analysis.

3. **Decrease in predictive ability of the model:** Correlated error terms can decrease the predictive ability of the model on new data. Since the model does not account for the correlation among error terms, it may perform poorly when applied to future datasets.

4. **Incorrect interpretation of explanatory variables:** When the error terms are correlated, the model may incorrectly attribute the relationship between the explanatory variables and the dependent variable to the correlated error terms, leading to erroneous interpretation and explanation of the effects of the explanatory variables.

Below are some common methods to address the lack of independence in errors:

- **Weighted Least Squares (WLS):** This method assigns different weights to different samples based on the characteristics of the heteroscedastic error terms. For samples with high noise, lower weights are assigned to reduce their influence on the regression model.

- **Robust Regression:** Robust regression methods are more resistant to outliers and extreme values in the data. These methods use robust loss functions, such as the Huber loss or Tukey's bisquare loss, to minimize the impact of outliers on the model.

- **Nonparametric Regression:** Nonparametric methods do not rely on assumptions about the distribution of the data. Common methods include Locally Weighted Regression and Kernel Regression, which can fit various types of error distributions.

- **Generalized Linear Models (GLM):** GLMs extend traditional linear regression models by allowing for different distributions of the error terms and using link functions to handle non-independent error distributions. For example, when the error terms follow a normal distribution, a relaxed regression model can be used.

6.6.8 Regression Parameter

If $y = \beta_1 \cdot x$ and $x = \beta_2 \cdot y$, find the range of $\beta_1 \cdot \beta_2$.

solution:

First, we substitute $x = \beta_2 y$ into $y = \beta_1 x$, yielding $y = \beta_1(\beta_2 y)$.

Simplifying, we get $y(1 - \beta_1 \beta_2) = 0$.

This equation has two solutions: $y = 0$ or $\beta_1 \beta_2 = 1$. Therefore, if $y \neq 0$, $\beta_1 \beta_2$ must equal 1.

6.6.9 Least Squares Estimation

For the least squares estimation of the linear regression of y on x , the coefficient is 1. Find the relationship between the coefficient for the regression of x on y and 1.

solution:

$$\sqrt{\hat{\beta}_{yx} \cdot \hat{\beta}_{xy}} = \sqrt{\frac{\text{Cov}(x, y)}{\text{Var}(x)} \cdot \frac{\text{Cov}(y, x)}{\text{Var}(y)}} = \frac{|\text{Cov}(x, y)|}{\text{SD}(x) \cdot \text{SD}(y)} = |r|$$

Since $|r| \leq 1$, the regression coefficient of x on y is less than or equal to 1.

6.6.10 R-square

When adding a new independent variable X_{n+1} , how does R^2 change?

solution:

When a new independent variable is added, the value of R^2 usually increases or remains the same. This is because adding a new variable can provide more information about the variation in the dependent variable, which allows the model to better fit the data. However, this does not necessarily mean that the new

variable is always significant. Other statistical tests (such as the t-test or F-test) are needed to determine the significance of the independent variable.

6.6.11 OLS Review I

Consider the following ordered pairs (x, y) : $(-2, 3)$, $(-1, 2)$, $(0, 1)$, $(1, 1)$, $(2, 0.5)$. Compute the sum of the least-squares estimators for the parameters β_0, β_1 in the model $Y = \beta_0 + \beta_1 x + \epsilon$, where $\mathbb{E}[\epsilon] = 0$.

solution:

Recall $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$, where $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$. We find $\bar{x} = 0, \bar{y} = 1.5$. Plugging in values, we find:

$$S_{xy} = -6, \quad S_{xx} = 10, \quad \hat{\beta}_1 = -0.6, \quad \hat{\beta}_0 = 1.5$$

We conclude $\mathbb{E}[Y] = \hat{y} = 1.5 - 0.6x$, so our answer is $1.5 - 0.6 = 0.9$.

6.6.12 OLS Review II

Use the method of least squares to fit a straight line to the following dataset of ordered pairs (x, y) : $(-2, 0)$, $(-1, 0)$, $(0, 1)$, $(1, 1)$, $(2, 3)$. Find the sum of the slope and the intercept.

solution:

We assume $\mathbb{E}[Y] = \beta_0 + \beta_1 x$. The method of least squares will allow us to find $\hat{\beta}_0, \hat{\beta}_1$, the estimators for $\mathbb{E}[Y]$. We know that

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

For our data, we easily find $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 7$, and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = 10$. Hence $\hat{\beta}_1 = \frac{7}{10}$ and $\hat{\beta}_0 = 1 - 0 = 1$. The sum of the two estimators is $\frac{17}{10}$.

6.6.13 OLS Review III

Given the following data, $(x, y) : (-2, 0), (-1, 0), (0, 1), (1, 1), (2, 3)$, Bob does a simple linear regression modeling Y as

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

where $\mathbb{E}[\epsilon] = 0$ and $\text{Var}[\epsilon] = \sigma^2$. Estimate σ^2 .

solution:

It can be shown that $s^2 = \frac{\text{SSE}}{n-2}$ is an unbiased estimator for σ^2 in the case where there are 2 β parameters. First, let's determine the least-squares estimators for the β parameters. We know that

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

For our data, we easily find $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 7$, and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = 10$. Hence $\hat{\beta}_1 = \frac{7}{10}$, and $\hat{\beta}_0 = 1 - 0 = 1$. We can now compute the sum of squared estimate of errors (SSE) with the following formula:

$$\begin{aligned} \text{SSE} &= S_{yy} - \hat{\beta}_1 S_{xy} \\ &= 6 - \frac{7}{10} \cdot 7 = \frac{11}{10} \\ \Rightarrow \frac{\text{SSE}}{n-2} &= \frac{\frac{11}{10}}{5-2} = \frac{11}{30} \end{aligned}$$

6.6.14 Duplicating Data

You use OLS regression model the relationship between news sentiment, $\mathbf{X}$, and AAPL share price, $\mathbf{Y}$, and find that $\mathbf{Y} = 156.3 + 10.21\mathbf{X}$ with a correlation coefficient of 0.78. You duplicate the data and re-run the OLS regression. How much did the correlation coefficient, ρ , increase by?

solution:

Correlation is not a function of sample size, and thus correlation increases by 0.

6.6.15 Double Data Trouble I

Suppose that you run OLS regression on some dataset and obtain an R^2 value of 0.56. If you were to run linear regression again on the dataset where you duplicate each point in your original dataset, what would R^2 be? If it cannot be determined, enter -1.

solution:

Recall that $R^2 = 1 - \frac{SSE}{SST}$. With twice the data that is identical to the original, the new SSE is exactly twice what it was before, as we have duplicated every data point and the mean is unchanged. For this same reason, SST also doubles. Therefore, the ratio $\frac{SSE}{SST}$ is unchanged, meaning that R^2 is unchanged from before.

6.6.16 Double Data Trouble II

Suppose that you run linear regression on some dataset and obtain the coefficients $\hat{\beta}_{OLS}$. Recall that if X is the data and σ^2 is the variance of the IID normal errors, then $\text{Var}(\hat{\beta}_{OLS}) = \sigma^2(X^T X)^{-1}$. If you were to run linear regression again on the dataset where you duplicate each point in your original dataset and obtain new coefficients $\hat{\beta}'_{OLS}$, find the constant c such that $\text{Var}(\hat{\beta}'_{OLS}) = c \text{Var}(\hat{\beta}_{OLS})$. If no such constant exists, enter -1.

solution:

When we duplicate the data, we are just replacing the matrix $\mathbf{X}$ with the matrix $\mathbf{X}' = \begin{bmatrix} \mathbf{X} \\ \mathbf{X} \end{bmatrix}$, so $(\mathbf{X}')^T \mathbf{X}' = 2\mathbf{X}^T \mathbf{X}$. Therefore, $\sigma^2(2\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{2} \cdot \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$, which proves our statement.

6.6.17 Double Data Trouble III

Suppose that you run linear regression on some dataset and obtain the coefficients $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Let t_0 be the t -statistic obtained when testing the null hypothesis $\mathbf{H}_0 : \beta_1 = 0$ against $\mathbf{H}_A : \beta_1 \neq 0$. Suppose that we duplicate our data and then run linear regression again on this new data set. Let $\hat{y}_i = \hat{\beta}'_0 + \hat{\beta}'_1 x_i$ be the new coefficients that are obtained. If t_1 is the corresponding t -statistic obtained when testing the null hypothesis $\mathbf{H}_1 : \beta'_1 = 0$ against $\mathbf{H}_A : \beta'_1 \neq 0$, find $\frac{t_1^2}{t_0^2}$. If it can't be determined, enter -1.

solution:

From Double Data Trouble II, we know that the variance of our estimate is reduced by $\frac{1}{2}$, so the standard error is reduced by $\frac{1}{\sqrt{2}}$. As $t_0 = \frac{\hat{\beta}_1}{s_{\{\hat{\beta}_1\}}}$, we know that the estimate $\hat{\beta}'_1$ is the same from Double Data Trouble I, and that the standard error in the denominator is $\frac{1}{\sqrt{2}}$ as large for $\hat{\beta}'_1$, we get that $t_1 = \sqrt{2}t_0$, meaning that $t_1^2 = 2t_0^2$.

6.6.18 Double Data Trouble V

Suppose that you run linear regression on some dataset and obtain the coefficients $\hat{\beta}_{OLS}$. Recall that if X is the data and σ^2 is the variance of the IID normal errors, then $\text{Var}(\hat{\beta}_{OLS}) = \sigma^2(X^T X)^{-1}$. If you were to run linear regression again on the dataset where you double the values of each point in your original dataset

and obtain new coefficients $\hat{\beta}'_{OLS}$, find the constant c such that $\text{Var}(\hat{\beta}'_{OLS}) = c\text{Var}(\hat{\beta}_{OLS})$. If no such constant exists, enter -1 .

solution:

When we double the data, we are just replacing the matrix $\mathbf{X}$ with the matrix $\mathbf{X}' = 2\mathbf{X}$. However, we also have to replace σ with 2σ . Accordingly, $(\mathbf{X}')^T \mathbf{X}' = 4\mathbf{X}^T \mathbf{X}$ by simple matrix multiplication. Then

$$(2\sigma)^2(4\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{4} \cdot 4 \cdot \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1},$$

which proves our statement.

6.6.19 Double Data Trouble IV

Suppose that you run linear regression on some dataset and obtain an R^2 value of 0.48. If you were to run linear regression again on the dataset where you double all of the values of each point in the dataset (for example, if $(2, 5)$ were in the original dataset, it would now be $(4, 10)$), what would R^2 be? If it can't be determined, enter -1 .

solution:

Recall that $R^2 = 1 - \frac{SSE}{SST}$. With each of the data points doubled, the new SSE is 4 times as large as it was before. This is because we are summing squared errors and we have doubled each of the points. Therefore, the errors (in magnitude) are doubled. Once we square these, the SSE would then be 4 times as large. For this same reason, SST also is quadrupled. Accordingly, the ratio $\frac{SSE}{SST}$ is unchanged, meaning that R^2 is unchanged from before.

6.6.20 Double Data Trouble VI

Suppose that you run linear regression on some dataset and obtain the coefficients $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Let t_0 be the t -statistic obtained when testing the null hypothesis $\mathbf{H}_0 : \beta_1 = 0$ against $\mathbf{H}_A : \beta_1 \neq 0$. Suppose that we double the values of each of our data points and then run linear regression again on this new data set. Let $\hat{y}_i = \hat{\beta}'_0 + \hat{\beta}'_1 x_i$ be the new coefficients that are obtained. If t_1 is the corresponding t -statistic obtained when testing the null hypothesis $\mathbf{H}_1 : \beta'_1 = 0$ against $\mathbf{H}_A : \beta'_1 \neq 0$, find $\frac{t_1^2}{t_0^2}$. If it cannot be determined, enter -1 .

solution:

From Double Data Trouble V, we know that the variance of our estimate is the same, so the standard error is also the same. As

$$t_0 = \frac{\hat{\beta}_1}{s\{\hat{\beta}_1\}}$$

we know that the estimate $\hat{\beta}'_1$ is the same from Double Data Trouble I, and that the standard error in the denominator is the same for $\hat{\beta}'_1$. Therefore, we get that $t_1 = t_0$, meaning that $t_1^2 = t_0^2$.

7 Stochastic Analysis

7.1 Stochastic Process

7.1.1 Markov Chain

A Markov chain is a sequence of random variables $X_0, X_1, \dots, X_n, \dots$ with the Markov property that given the present state, the future states and the past states are independent:

$$P\{X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} = p_{ij} = P\{X_{n+1} = j | X_n = i\}, \text{ for all } n, i_0, \dots, i_{n-1}, i, \text{ and } j$$

where $i, j \in \{1, 2, \dots, M\}$ represent the state space $S = \{s_1, s_2, \dots, s_M\}$ of X .

In other words, once the current state is known, past history has no bearing on the future. For a homogenous Markov chain, the transition probability from state i to state j does not depend on n . A Markov chain with M states can be completely described by an $M \times M$ transition matrix P and the initial probabilities $P(X_0)$.

7.1.2 Transition Matrix

Transition matrix: $P = \{p_{ij}\} = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1M} \\ p_{21} & p_{22} & \cdots & p_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ p_{M1} & p_{M2} & \cdots & p_{MM} \end{pmatrix}$, where p_{ij} is the transition probability from state i to state j .

Initial probabilities:

$$P(X_0) = (P(X_0 = 1), P(X_0 = 2), \dots, P(X_0 = M)), \quad \sum_{i=1}^M P(X_0 = i) = 1$$

The probability of a path:

$$P(X_1 = i_1, X_2 = i_2, \dots, X_n = i_n | X_0 = i_0) = p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}.$$

Transition graph: A transition graph is often used to express the transition matrix graphically. The transition graph is more intuitive than the matrix, and it emphasizes the transition matrix structure possible and impossible transitions. Figure 5.1 shows the transition graph and the transition matrix of a Markov chain with four states:

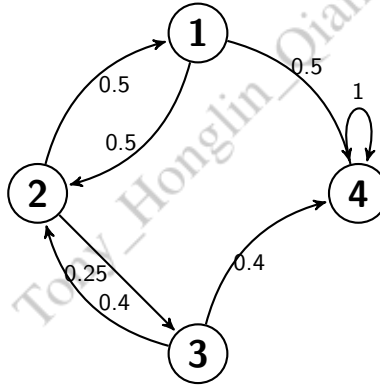


Figure 1: Transition graph and transition matrix of the Play

$$P = \begin{pmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.25 & 0.25 \\ 0 & 0.4 & 0.4 & 0.2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

7.1.3 Classification of states

- **Accessible (Reachable) State:** A state j is said to be accessible from state i if there is a positive probability of transitioning from state i to state j in some finite number of steps. This is denoted as $i \rightarrow j$.
- **Communicating States:** Two states i and j are said to communicate with each other if each is accessible from the other. This means there exists a sequence of transitions from i to j and from j to i . If states i and j communicate, it is denoted as $i \leftrightarrow j$.
- **Recurrent State:** A state i is recurrent (also known as persistent) if, starting from i , the probability of returning to i at some future time is 1. In other words, once the process enters a recurrent state, it is guaranteed to return to it eventually.
- **Transient State:** A state i is transient if, starting from i , there is a positive probability that the process will never return to i . This means that, over the long run, the process may leave the transient state and never come back.
- **Absorbing Markov chains:** A state i is called **absorbing** if it is impossible to leave this state ($p_{ii} = 1, p_{ij} = 0, \forall j \neq i$). A Markov chain is absorbing if it has at least one absorbing state and if from every state it is possible to go to an absorbing state. In Figure 1, state 4 is an absorbing state. The corresponding Markov chain is an absorbing Markov chain.
- **Equations for absorption probability:** The probability to reach a specific absorbing state s , $a_1, a_2, \dots, a_M$, are unique solutions to equations $a_s = 1, a_i = 0$ for all absorbing state(s) $i \neq s$, and

$$a_i = \sum_{j=1}^M a_j p_{ij}$$

for all transient states i . These equations can be easily derived using the law of total probability by conditioning the absorption probabilities on the next state.

- **Equations for the expected time to absorption:** The expected times to absorption, $\mu_1, \dots, \mu_M$, are unique solutions to the equations $\mu_i = 0$ for all absorbing state(s) i and

$$\mu_i = 1 + \sum_{j=1}^m p_{ij} \mu_j$$

for all transient states i . These equations can be easily derived using the law of total expectation by conditioning the expected times to absorption on the next state. The number 1 is added since it takes one step to reach the next state.

7.2 Martingale and Random Walk

7.2.1 Random walk

The process $\{S_n; n \geq 1\}$ is called a random walk if:

1. $\{X_i; i \geq 1\}$ are IID (identical and independently distributed) random variables

2. $S_n = X_1 + \cdots + X_n$, where $n = 1, 2, \dots$.

The term comes from the fact that we can think of S_n as the position at time n for a walker who makes successive random steps $X_1, X_2, \dots$

Properties

- If X_i takes values 1 and -1 with probabilities p and $1 - p$ respectively, S_n is called a **simple random walk** with parameter p .
- Furthermore, if $p = \frac{1}{2}$, the process S_n is a **symmetric random walk**. For symmetric random walk, it's easy to show that $E[S_n] = 0$ and $\text{var}(S_n) = E[S_n^2] - E[S_n]^2 = E[S_n^2] = n$.
- Symmetric random walk is the process that is most often tested in quantitative interviews. The interview questions on random walk often revolve around finding the first n for which S_n reaches a defined threshold α , or the probability that S_n reaches α for any given value of n .

7.2.2 Martingale

A martingale $\{Z_n; n \geq 1\}$ is a stochastic process with the properties that:

1. $E[|Z_n|] < \infty$ for all n
2. $E[Z_{n+1} | Z_n = z_n, Z_{n-1} = z_{n-1}, \dots, Z_1 = z_1] = z_n$.

The property of a martingale can be extended to $E[Z_m; m > n | Z_n = z_n, Z_{n-1} = z_{n-1}, \dots, Z_1 = z_1] = z_n$, which means the conditional expected value of future Z_m is the current value Z_n .

Properties:

- **A symmetric random walk is a martingale.** From the definition of the symmetric random walk we have:

$$S_{n+1} = \begin{cases} S_n + 1 & \text{with probability } \frac{1}{2} \\ S_n - 1 & \text{with probability } \frac{1}{2} \end{cases}$$

so $E[S_{n+1} | S_n = s_n, \dots, S_1 = s_1] = s_n$. Since $E[S_{n+1}^2 - (n+1)] = \frac{1}{2}[(S_n + 1)^2 + (S_n - 1)^2] - (n+1) = S_n^2 - n$, $S_n^2 - n$ is a martingale as well.

- **Stopping rule:** For an experiment with a set of IID random variables $X_1, X_2, \dots$, a stopping rule for $\{X_i; i \geq 1\}$ is a positive integer-valued random variable N (stopping time) such that for each $n > 1$, the event $\{N \leq n\}$ is independent of $X_{n+1}, X_{n+2}, \dots$. Basically, it says that whether to stop at n depends only on $X_1, X_2, \dots, X_n$ (i.e., no look ahead).
- **Wald's Equality:** Let N be a stopping rule for IID random variables $X_1, X_2, \dots$ and let $S_N = X_1 + X_2 + \cdots + X_N$, then $E[S_N] = E[X]E[N]$.

Since it is an important—yet relatively little known—theorem, let's briefly review its proof. Let I_n be the indicator function of the event $\{N \geq n\}$. So S_N can be written as

$$S_N = \sum_{n=1}^{\infty} X_n I_n,$$

where $I_n = 1$ if $N \geq n$ and $I_n = 0$ if $N \leq n - 1$.

From the definition of stopping rules, we know that I_n is independent of $X_n, X_{n+1}, \dots$ (it only depends on $X_1, X_2, \dots, X_{n-1}$). So $E[X_n I_n] = E[X_n]E[I_n] = E[X]E[I_n]$ and

$$E[S_N] = E\left[\sum_{n=1}^{\infty} X_n I_n\right] = \sum_{n=1}^{\infty} E[X_n I_n] = \sum_{n=1}^{\infty} E[X]E[I_n] = E[X] \sum_{n=1}^{\infty} E[I_n] = E[X]E[N].$$

A martingale stopped at a stopping time is a martingale!!!!

7.2.3 Symmetric Random Walk with Two Absorbing Barriers

We consider a random walk that stops once it reaches either a or b (where $b > 0 > a$). This random walk is called a symmetric random walk with two absorbing barriers.

Theorem 1: The probability that a symmetric random walk with two absorbing barriers starting from the origin reaches b before a (where $b > 0 > a$) is given by:

$$P(T_{a,b} = T_b) = \frac{-a}{b-a}$$

For symmetric random walks,

$$P(T_{a,b} = T_a) = \frac{b}{b-a}$$

Proof: Since we know that a random walk, even with the stopping time $\tau = T_{a,b}$, is still a martingale, we have

$$E[M_\tau] = M_0 = 0$$

Thus:

$$P(T_{a,b} = T_a) \times (-a) + (1 - P(T_{a,b} = T_a)) \times b = 0$$

Theorem 2: The expected number of steps for a symmetric random walk with two absorbing barriers to reach either a or b (where $b > 0 > a$) starting from the origin is given by:

$$E[N] = -ab$$

Proof: Using Theorem 3.2, since $M_n^2 - n$ is a martingale, and since $E[M_n^2 - n] = M_0^2 - 0 = 0$, we have:

$$E[P(T_{a,b} = T_a) \times a^2 + (1 - P(T_{a,b} = T_a)) \times b^2] - E[N] = 0$$

Thus, we get:

$$E[N] = -ab$$

7.2.4 Recurrence of Symmetric Random Walks

Theorem 1: Symmetric random walks are recurrent, meaning that a symmetric random walk returns to the origin with probability 1.

Theorem 2 The first return expectation of symmetric random walks: While symmetric random walks are recurrent, the expected number of steps to return to the origin is infinite. Similarly, starting from the origin, the expected number of steps to reach 1 is also infinite.

7.3 Brown Motion

7.3.1 Define and properties of a Brownian motion

A continuous stochastic process $W(t)$, $t \geq 0$, is a Brownian motion if

- $W(0) = 0$;
- The increments of the process $W(t_1) - W(0), W(t_2) - W(t_1), \dots, W(t_n) - W(t_{n-1})$,

$$\forall 0 \leq t_1 \leq t_2 \leq \dots \leq t_n$$

are independent;

- Each of these increments is normally distributed with distribution

$$W(t_{i+1}) - W(t_i) \sim N(0, t_{i+1} - t_i).$$

Some of the important properties of Brownian motion are the following: continuous (no jumps);

$$E[W(t)] = 0; \quad E[W(t)^2] = t; \quad W(t) \sim N(0, t);$$

martingale property

$$E[W(t+s)|W(t)] = W(t); \quad \text{cov}(W(s), W(t)) = s, \quad \forall 0 < s < t;$$

and Markov property (in continuous space).

7.3.2 Martingales in Brown Motion

There are other important martingales related to Brownian motion that are valuable tools in many applications.

- $W(t)$ itself is a martingale.
- $Y(t) = W(t)^2 - t$ is a martingale.
- $Z(t) = \exp\left\{\lambda W(t) - \frac{1}{2}\lambda^2 t\right\}$, where λ is any constant and $W(t)$ is a Brownian motion, is a martingale. (Exponential martingale).

We'll show a proof of the first martingale using Ito's lemma in the next section. A sketch for the exponential martingale is the following:

$$\begin{aligned} E[Z(t+s)] &= E\left[\exp\left\{\lambda(W(t) + W(s)) - \frac{1}{2}\lambda^2(t+s)\right\}\right] \\ &= \exp\left\{\lambda W(t) - \frac{1}{2}\lambda^2 t\right\} \exp\left\{-\frac{1}{2}\lambda^2 s\right\} E[\exp\{\lambda W(s)\}] \\ &= Z_t \exp\left\{-\frac{1}{2}\lambda^2 s\right\} \exp\left\{\frac{1}{2}\lambda^2 s\right\} = Z_t \end{aligned}$$

Example

What is the mean of the stopping time for a Brownian motion to reach either -1 or 1?

Solution: As we have discussed, $B_t^2 - t$ is a martingale. It can be proved by applying Ito's lemma:

$$d(B_t^2 - t) = \frac{\partial(B_t^2 - t)}{\partial B_t} dB_t + \frac{\partial(B_t^2 - t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2(B_t^2 - t)}{\partial B_t^2} dt = 2B_t dB_t - dt + dt = 2B_t dB_t.$$

So $d(B_t^2 - t)$ has no drift term and is a martingale. Let $T = \min\{t; B_t = 1 \text{ or } -1\}$. At continuous time and space, the following property still applies: A martingale stopped at a stopping time is a martingale! So $B_T^2 - T$ is a martingale and

$$E[B_T^2 - T] = B_0^2 - 0 = 0.$$

The probability that B_t hits 1 or -1 is 1, so $B_T^2 = 1 \Rightarrow E[T] = E[B_T^2] = 1$.

7.3.3 Quadratic Variation

Definition: Quadratic Variation of Brownian Motion

Let $f(t)$ be a function defined on $0 \leq t \leq T$. The quadratic variation of f with respect to time T is given by

$$[f, f](T) = \lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2$$

where $\Pi = \{t_0, t_1, \dots, t_n\}$ is a partition of the interval $(0, T)$; $0 = t_0 < t_1 < \dots < t_n = T$.

Theorem: Quadratic Variation of Brownian Motion

Let W_t be a Brownian motion. Then, for almost every $T > 0$, we have $[W, W](T) = T$.

Theorem: Fundamental Theorem of Stochastic Calculus !!!!!

$$(dW_t)^2 = dt$$

The notation in differential form $(dW_t)^2 = dt$ is a shorthand for the conventional integral notation used in Riemann integral

$$\int_0^T (dW_t)^2 = \int_0^T dt. \quad (3.120)$$

The intuition behind (3.120) is related to the *quadratic variation* of a Brownian path. By definition, the quadratic variation $QV_W[0, T]$ of the Brownian path W in the interval $[0, T]$ is

$$QV_W[0, T] = \lim_{n \rightarrow \infty} \sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2,$$

where $t_i = \frac{i}{n}T$, for $i = 0 : n$. Therefore, in expectation, we have that

$$E \left[\lim_{n \rightarrow \infty} \sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2 \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n E [|W_{t_i} - W_{t_{i-1}}|^2] = \lim_{n \rightarrow \infty} \sum_{i=1}^n (t_i - t_{i-1}) = T,$$

since $W_{t_i} - W_{t_{i-1}} \sim N(0, t_i - t_{i-1})$ for $i = 1 : n$. In fact, it can be shown that the convergence

$$\sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2 \rightarrow T \quad \text{as } n \rightarrow \infty$$

is in L^2 sense, i.e.,

$$\lim_{n \rightarrow \infty} E \left[\left(\sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2 - T \right)^2 \right] = 0.$$

By mimicking the notations used in the conventional Riemann integral, we write that

$$\int_0^T (dW_t)^2 = \lim_{n \rightarrow \infty} \sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2 = \int_0^T dt,$$

whose shorthand notation in differential form is

$$(dW_t)^2 = dt.$$

7.3.4 Geometric Brownian Motion

Definition: The process $\{X(t), t \geq 0\}$ defined by $X(t) = e^{B(t)}$ ($t \geq 0$) is called a Geometric Brownian Motion.

Since the moment generating function of Brownian motion is $E[e^{sB(t)}] = e^{\frac{ts^2}{2}}$,

Theorem: The mean and variance functions of Geometric Brownian Motion are given by:

$$E[X(t)] = E[e^{B(t)}] = e^{\frac{t}{2}}$$

$$\text{Var}[X(t)] = E[X^2(t)] - \{E[X(t)]\}^2$$

$$= E[e^{2B(t)}] - e^t = e^{2t} - e^t$$

Stock Geometric Brown Motion:

Now consider adding a drift term μt related to time t to a standard Brownian motion with diffusion term σ . This results in a process with a mean value μt and variance proportional to time. This process is often called **Brownian motion with drift**, denoted as:

$$X(t) = \mu t + \sigma B(t)$$

where $B(t)$ is standard Brownian motion. The differential form can be written as:

$$dX(t) = \mu dt + \sigma dB(t)$$

Here, μ represents the drift term, which is the expected rate of return for the stock price. In finance, this drift term helps us evaluate the long-term growth expectation of the stock. It does not reflect short-term volatility but instead captures the stock's overall expected price trend.

Given that stock prices fluctuate randomly, the stock price process $S(t)$ follows a stochastic differential equation (SDE):

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dB(t)$$

Thus, we have:

$$dS(t) = \mu S(t)dt + \sigma S(t)dB(t)$$

Definition The standard geometric Brownian motion is a process $S(t)$ governed by the SDE:

$$dS(t) = \mu S(t)dt + \sigma S(t)dB(t)$$

To solve this equation, we apply Itô's Lemma to $\log S$, and we have:

$$d \log S_t = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2$$

Substituting into the equation:

$$d \log S_t = \mu dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt$$

Therefore, we have:

$$\log S_t = \log S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t$$

Exponentiating both sides, we get:

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right)$$

7.3.5 Brownian Motion with Drift

This section will be better understood after mastering Itô's Lemma in the next section, but we can still learn some properties of Brownian motion with drift within the current framework.

Definition: A Brownian motion with drift can be expressed as

$$dX(t) = m dt + dW(t)$$

Theorem: The stochastic process $X(t)$ satisfies $W(t) = X(t) - mt$, which is a standard Brownian motion.

Theorem: The probability that a Brownian motion with drift $m dt$ starting from 0 reaches b (where $b > 0 > a$) before a is given by

$$P_b = \frac{e^{-2ma} - 1}{e^{-2ma} - e^{-2mb}}.$$

Proof: We use the exponential martingale. Let

$$Z(t) = \exp\left(\lambda W(t) - \frac{1}{2}\lambda^2 t\right)$$

be a martingale. Since $W(t) = X(t) - mt$, $X(t) - mt$ is a Brownian motion. Applying the martingale property, for any λ , we have

$$E\left[\exp\left(\lambda(X(t) - mt) - \frac{1}{2}\lambda^2 t\right)\right] = 1.$$

We can choose $\lambda = -2m$, then the expectation becomes $E[\exp(-2mX)] = 1$. Because the martingale at stopping time is still a martingale, we have

$$P_b \exp(-2mb) + (1 - P_b) \exp(-2m \times a) = 1 \implies P_b = \frac{e^{-2ma} - 1}{e^{-2ma} - e^{-2mb}}.$$

Theorem: The probability that a Brownian motion with drift mdt (with $m > 0$) has ever passed a ($a < 0$) is e^{2ma} .

Note: This definition implies that a Brownian motion with drift is not necessarily recurrent because it tends to drift. However, we still have $E[\exp(-2mX)] = 1$. When we have boundary conditions at a and ∞ , applying the stopping time, this equation becomes:

$$P_a \exp(-2m \times a) + (1 - P_a) \exp(-2m \times \infty) = P_a e^{-2ma} = 1 \implies P_a = e^{2ma}$$

7.3.6 Reflection Principle

The Reflection Principle of Brownian Motion is a significant result in probability theory and stochastic processes. This theorem describes the symmetric property of standard Brownian motion (or Wiener process), especially its behavior when it reaches its historical maximum.

Theorem

Let $B(t)$ be a standard Brownian motion, i.e., $B(0) = 0$. For any time $t \geq 0$,

$$P\left(\max_{0 \leq s \leq t} B(s) \geq a\right) = 2P(B(t) \geq a)$$

where a is a positive constant and $P(\cdot)$ denotes probability.

Intuitive Explanation of the Reflection Principle

The reflection principle states that if we observe a Brownian motion up to time t and it reaches a certain height a at some point, then the path of the Brownian motion after this point behaves as if it were reflected at this point. This means that the behavior of the Brownian motion after reaching its maximum a is symmetric to that of a Brownian motion starting from the maximum a .

7.4 Stochastic differential Equation

A Stochastic Differential Equation (SDE) describes the random component of a system. It combines deterministic differential equations and stochastic components of random processes, used to describe the random variation in systems. The general form of an SDE can be written as:

$$dX_t = b(X_t, t)dt + \sigma(X_t, t)dW_t$$

where:

X_t is the stochastic process representing the state of the system as it changes over time.

$b(X_t, t)$ represents the drift, describing the deterministic trend or drift of the system.

$\sigma(X_t, t)$ represents the diffusion, describing the stochastic volatility or spread of the system. W_t is the Brownian motion (or Wiener process), representing the stochastic term used to describe the random variation in the system;

dt represents the infinitesimal change in time, dW_t represents the infinitesimal change in Brownian motion.

The meaning of this equation is that at each time t , the change in the system dX_t , driven by the drift term $b(X_t, t)$ and the diffusion term $\sigma(X_t, t)$, is simultaneously influenced by the stochastic term dW_t . The drift term describes the deterministic trend of the system, while the diffusion term describes the stochastic volatility.

As can be seen from the SDE, as a function of X and t , f itself is also an Itô process. More importantly, Itô's lemma explains that the differential form on the right-hand side, involving dB , exactly matches the Brownian motion term dX in the expression. In other words, the randomness in f and X is determined by a single Brownian motion, not two independent Brownian motions. This point is crucial in the Black-Scholes-Merton (BSM) equation.

7.5 Itô process

An Itô process is a generic term referring to a stochastic process X_t determined by the solution of a stochastic differential equation (SDE) of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t,$$

where W_t is a Wiener process. The coefficient $a(x, t)$ of the dt term is the drift of X_t ; the coefficient $b(x, t)$ of the dW_t term is the diffusion of X_t . The SDE is, by definition, the shorthand notation for the stochastic integral equation

$$X_t = X_0 + \int_0^t a(X_s, s)ds + \int_0^t b(X_s, s)dW_s.$$

We note that a sufficient condition for the existence and uniqueness of the (strong) solution to an SDE is for the drift and the diffusion coefficients $a(x, t)$ and $b(x, t)$ to be locally Lipschitz functions of at most linear growth in x .

7.6 Ito's Lemma

Definition: Brownian motion SDE

A Brownian motion with drift and diffusion can be described by the following stochastic differential equation:

$$dX(t) = \mu dt + \sigma dB(t)$$

More generally, the drift and diffusion parameters can be functions of the stochastic process $X(t)$ and time t . Suppose we define $a(X(t), t)$ and $b(X(t), t)$ as the drift and diffusion parameters (in the above example, $a(X(t), t) = \mu$ and $b(X(t), t) = \sigma$). The drift represents the long-term trend of a stochastic process, while diffusion measures the short-term volatility of this stochastic process.

Itô's lemma, also known as Itô's formula, states that, if X_t is an Itô process satisfying the SDE

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t,$$

then for any function $f(x, t)$ with continuous second order partial derivative in x and continuous first order partial derivative in t , the process $f(X_t, t)$ is also an Itô process, driven by the same Wiener process W_t , and the drift and diffusion parts of $f(X_t, t)$ are determined according to the Taylor expansion of $f(x, t)$ to first order for the t part and up to second order for the x part:

$$\begin{aligned} df(X_t, t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2 \\ &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}[a(X_t, t)dt + b(X_t, t)dW_t] + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}[a(X_t, t)dt + b(X_t, t)dW_t]^2 \\ &= \left[\frac{\partial f}{\partial t} + a(X_t, t)\frac{\partial f}{\partial x} + \frac{b^2(X_t, t)}{2}\frac{\partial^2 f}{\partial x^2} \right]dt + b(X_t, t)\frac{\partial f}{\partial x}dW_t. \quad (3.157) \end{aligned}$$

Note that for (3.157) we used the fact that

$$(dX_t)^2 = [a(X_t, t)dt + b(X_t, t)dW_t]^2 = b^2(X_t, t)dt$$

since $(dW_t)^2 = dt$, $(dt)^2 = 0$, and $dW_t dt = 0$.

General Form of Itô's Lemma

Itô's lemma is the stochastic counterpart of the chain rule in ordinary calculus. Let $X(t)$ be an Ito process satisfying

$$dX(t) = \beta(t, X)dt + \gamma(t, X)dW(t)$$

and $f(X(t), t)$ be a twice-differentiable function of $X(t)$ and t . Then $f(X(t), t)$ is an Ito process satisfying

$$df = \left(\frac{\partial f}{\partial t} + \beta(t, X) \frac{\partial f}{\partial x} + \frac{1}{2} \gamma^2(t, X) \frac{\partial^2 f}{\partial x^2} \right) dt + \gamma(t, X) \frac{\partial f}{\partial x} dW(t)$$

Explanation of Terms in the Equation

- $\frac{\partial f}{\partial t}$: Partial derivative of function f with respect to time t .
- $\frac{\partial f}{\partial x}$: Partial derivative of function f with respect to $X(t)$.
- $\frac{\partial^2 f}{\partial x^2}$: Second partial derivative of function f with respect to $X(t)$.
- $\beta(t, X)$: Drift term of the Ito process.
- $\gamma(t, X)$: Diffusion term of the Ito process.
- $dW(t)$: Increment of the Wiener process (Brownian motion).

Drift Rate

The drift rate represents the expected change of the process in the time interval dt . The drift rate is given by:

$$\frac{\partial f}{\partial t} + \beta(t, X) \frac{\partial f}{\partial x} + \frac{1}{2} \gamma^2(t, X) \frac{\partial^2 f}{\partial x^2}$$

7.7 Itô Integral

7.7.1 Definition of Itô Integral

Let $W(t)$ be a standard Brownian motion and $f(t, W(t))$ be an appropriate function. The Itô integral $\int_0^T f(t, W(t)) dW(t)$ is defined as the following limit:

$$\int_0^T f(t, W(t)) dW(t) = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(t_i, W(t_i)) (W(t_{i+1}) - W(t_i))$$

where $0 = t_0 < t_1 < \dots < t_n = T$ is a partition of the interval $[0, T]$, and as $n \rightarrow \infty$, the maximum length of the partition intervals approaches zero.

7.7.2 Example

Example 1

Let W_t be a Wiener process. Find the mean and the variance of

$$\int_0^t W_s^2 dW_s.$$

Answer: Let $\mathcal{B}_t$ be the Borel σ -algebra over the time interval $[0, t]$ and let $\mathcal{F}_t$ be the filtration for the probability space in which the Wiener process W_t resides. We will use the following results:

Theorem 3.1. (Martingality)

Let f_s be progressively measurable and square integrable in $[0, T]$, i.e., f_t is $\mathcal{B}_t \otimes \mathcal{F}_t$ -measurable for every $t \in [0, T]$ and $E \left[\int_0^T |f_t|^2 dt \right] < \infty$. Then, the stochastic integral

$$\int_0^t f_s dW_s$$

defines a zero mean, square integrable martingale for $t \in [0, T]$. In particular,

$$E \left[\int_0^t f_s dW_s \right] = 0, \quad \forall t \in [0, T]. \quad (3.136)$$

Theorem 3.2. (Itô's isometry)

Let f_t and g_t be progressively measurable and square integrable processes. Then,

$$E \left[\int_0^t f_s dW_s \int_0^t g_s dW_s \right] = \int_0^t E[f_s g_s] ds.$$

In particular, if $f_s = g_s$, it follows that

$$E \left[\left(\int_0^t f_s dW_s \right)^2 \right] = \int_0^t E[f_s^2] ds. \quad (3.137)$$

For our problem, we need to compute $E[X]$ and $\text{var}(X)$, where

$$X = \int_0^t W_s^2 dW_s.$$

We first check that the integrand W_s^2 is progressively measurable and square integrable in $[0, t]$. Note that W_s^2 is progressively measurable because it is adapted and continuous.

Furthermore, since $W_s \sim N(0, s)$, it follows that $W_s = \sqrt{s}Z$, where Z is a standard normal random variable, and therefore

$$E[W_s^4] = E[(\sqrt{s}Z)^4] = s^2 E[Z^4] = 3s^2, \quad (3.138)$$

where we used the fact that the fourth moment of the standard normal distribution is 3, i.e., $E[Z^4] = 3$. From (3.138), it follows that

$$\int_0^t E[W_s^4] ds = \int_0^t 3s^2 ds = t^3 < \infty. \quad (3.139)$$

In other words, W_s^2 is square integrable in $[0, t]$. We can therefore apply both Theorems 3.1 and 3.2 with $f_s = W_s^2$ and $g_s = W_s^2$.

From (3.136) for $f_s = W_s^2$, we find that

$$E[X] = E \left[\int_0^t W_s^2 dW_s \right] = 0. \quad (3.140)$$

From (3.137) for $f_s = g_s = W_s^2$, and using (3.139), it follows that

$$E[X^2] = E \left[\left(\int_0^t W_s^2 dW_s \right)^2 \right] = \int_0^t E[W_s^4] ds = t^3.$$

Since $E[X] = 0$, see (3.140), we find that

$$\text{var}(X) = E[X^2] - (E[X])^2 = t^3.$$

Example 2

Calculate $\int_0^t B_s ds$

$$\begin{aligned} \int_0^t B_s ds &= sB_s \Big|_0^t - \int_0^t s dB_s \\ &= \int_0^t t dB_s - \int_0^t s dB_s = \int_0^t (t-s) dB_s \\ &\sim \mathcal{N} \left(0, \int_0^t (t-s)^2 ds \right) \quad (\text{Ito isometry}) \\ &\sim \mathcal{N} \left(0, \frac{1}{3} t^3 \right) \end{aligned}$$

Example 3

If W_t is a Wiener process, find the variance of

$$\int_0^t s dW_s.$$

Answer: Recall that, for any deterministic square integrable function $f : [a, b] \rightarrow \mathbb{R}$, the stochastic integral $\int_a^b f(s) dW_s$ is normally distributed with mean 0 and variance equal to the square of the L^2 norm of f , i.e.,

$$\int_a^b f(s) dW_s \sim N \left(0, \int_a^b f^2(s) ds \right).$$

Then,

$$\int_0^t s dW_s \sim N \left(0, \int_0^t s^2 ds \right) = N \left(0, \frac{t^3}{3} \right),$$

and we conclude that

$$\text{var} \left(\int_0^t s dW_s \right) = \frac{t^3}{3}.$$

7.8 Classic Questions

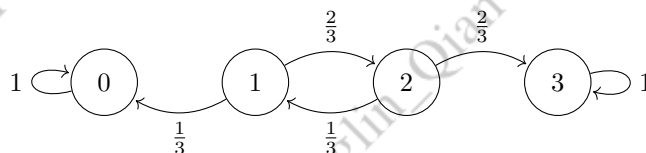
7.8.1 Gambler's ruin problem

Player M has 1 dollar and player N has 2 dollars. Each game gives the winner 1 dollar from the other. As a better player, M wins $2/3$ of the games. They play until one of them is bankrupt. What is the probability that M wins?

solution:

The transition graph and the corresponding transition matrix are shown in Figure.

$$P = \{p_{ij}\} = \begin{pmatrix} p_{0,0} & p_{0,1} & p_{0,2} & p_{0,3} \\ p_{1,0} & p_{1,1} & p_{1,2} & p_{1,3} \\ p_{2,0} & p_{2,1} & p_{2,2} & p_{2,3} \\ p_{3,0} & p_{3,1} & p_{3,2} & p_{3,3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



The initial state is $X_0 = 1$ (M has \$1 at the beginning). At state 1, the next state is 0 (M loses a game) with probability $\frac{1}{3}$ and 2 (M wins a game) with probability $\frac{2}{3}$. So $p_{1,0} = \frac{1}{3}$ and $p_{1,2} = \frac{2}{3}$. Similarly we can get $p_{2,1} = \frac{1}{3}$ and $p_{2,3} = \frac{2}{3}$. Both state 3 (M wins the whole game) and state 0 (M loses the whole game) are absorbing states.

To calculate the probability that M reaches absorbing state 3, we can apply absorption probability equations:

$$a_3 = 1, a_0 = 0, \text{ and } a_1 = \sum_{j=0}^3 p_{1,j}a_j, a_2 = \sum_{j=0}^3 p_{2,j}a_j$$

Plugging in the transition probabilities using either the transition graph or transition matrix, we have

$$\begin{cases} a_1 = \frac{1}{3} \times 0 + \frac{2}{3} \times a_2 \\ a_2 = \frac{1}{3} \times a_1 + \frac{2}{3} \times 1 \end{cases} \Rightarrow \begin{cases} a_1 = \frac{4}{7} \\ a_2 = \frac{6}{7} \end{cases}$$

So, starting from \$1, player M has $\frac{4}{7}$ probability of winning.

7.8.2 Dice question

Two players bet on roll of the total of two standard six-face dice. Player A bets that a sum of 12 will occur first. Player B bets that two consecutive 7s will occur first. The players keep rolling the dice and record the sums until one player wins. What is the probability that A will win?

solution:

let's try a Markov chain approach. Again the key part is to choose the right state space and define the transition probabilities. It is apparent that we have two absorbing states, 12 (A wins) and 7-7 (B wins), at least two transient states, S (starting state) and 7 (one 7 occurs, yet no 12 or 7-7 occurred). Do we need any other states? Theoretically, you can have other states. In fact, you can use all combination of the outcomes of one roll and two consecutive rolls as states to construct a transition matrix and you will get the same final result. Nevertheless, we want to consolidate as many equivalent states as possible. As we just discussed in the conditional probability approach, if no 12 has occurred and the most recent roll did not yield 7, we essentially go back to the initial starting state S . So all we need are states S , 7, 7-7 and 12. The transition graph and probability to reach state 12 are shown in Figure.

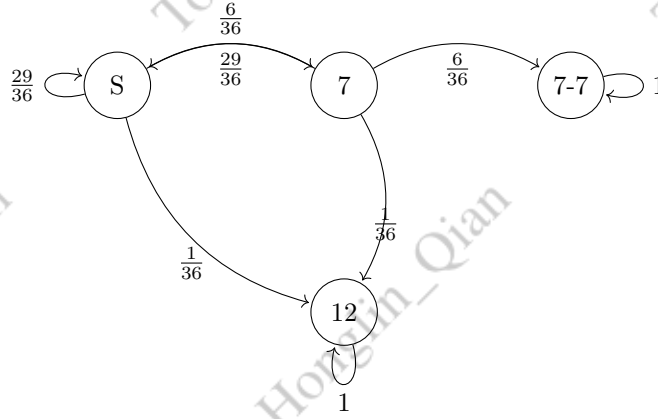
Probability to absorption state 12:

$$a_{12} = 1, a_{7-7} = 0$$

$$a_S = \frac{1}{36} \times 1 + \frac{6}{36} \times a_7 + \frac{29}{36} \times a_S \Rightarrow a_S = \frac{7}{13}$$

$$a_7 = \frac{1}{36} \times 1 + \frac{6}{36} \times 0 + \frac{29}{36} \times a_S$$

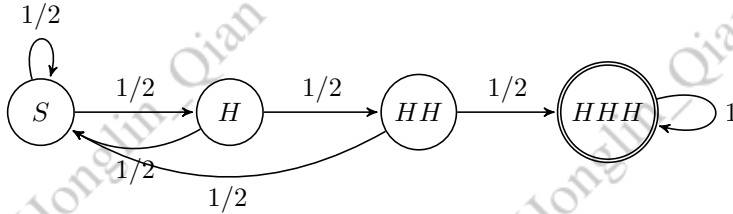
Hence, $a_S = \frac{7}{13}$



7.8.3 Coin triplets 1

If you keep on tossing a fair coin, what is the expected number of tosses such that you can have HHH (heads heads heads) in a row? What is the expected number of tosses to have THH (tails heads heads) in a row?

solution:



At state S , after a coin toss, the state will stay at S when the toss gives an H , the state becomes H . At state H , it has $\frac{1}{2}$ probability the next toss is T ; otherwise, it goes to state HH . At state HH , it also has $\frac{1}{2}$ probability goes back to state S if the next toss is T ; otherwise, it reaches the absorbing state HHH .

So we have the following transition probabilities: $P_{S,S} = \frac{1}{2}$, $P_{S,H} = \frac{1}{2}$, $P_{H,S} = \frac{1}{2}$, $P_{H,HH} = \frac{1}{2}$, $P_{HH,S} = \frac{1}{2}$, $P_{HH,HHH} = \frac{1}{2}$, and $P_{HHH,HHH} = 1$.

We are interested in the expected number of tosses to get HHH , which is the expected time to absorption starting from state S . Applying the standard equations for the expected time to absorption, we have

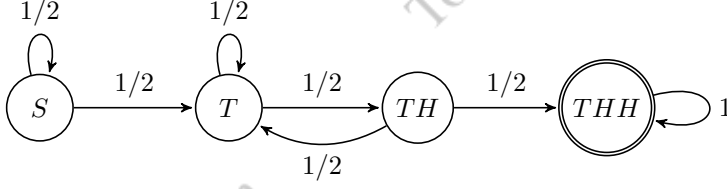
$$\begin{cases} \mu_S = 1 + \frac{1}{2}\mu_S + \frac{1}{2}\mu_H \\ \mu_H = 1 + \frac{1}{2}\mu_S + \frac{1}{2}\mu_{HH} \\ \mu_{HH} = 1 + \frac{1}{2}\mu_S + \frac{1}{2}\mu_{HHH} \\ \mu_{HHH} = 0 \end{cases} \Rightarrow \begin{cases} \mu_S = 14 \\ \mu_H = 12 \\ \mu_{HH} = 8 \\ \mu_{HHH} = 0 \end{cases}$$

So from the starting state, the expected number of tosses to get HHH is 14.

Similarly for the expected time to reach THH , we can construct the following transition graph and estimate the corresponding expected time to absorption:

$$\begin{cases} \mu_S = 1 + \frac{1}{2}\mu_S + \frac{1}{2}\mu_T \\ \mu_T = 1 + \frac{1}{2}\mu_T + \frac{1}{2}\mu_{TH} \\ \mu_{TH} = 1 + \frac{1}{2}\mu_T + \frac{1}{2}\mu_{THH} \\ \mu_{THH} = 0 \end{cases} \Rightarrow \begin{cases} \mu_S = 8 \\ \mu_T = 4 \\ \mu_{TH} = 2 \\ \mu_{THH} = 0 \end{cases}$$

So from the starting state S , the expected number of tosses to get THH is 8.

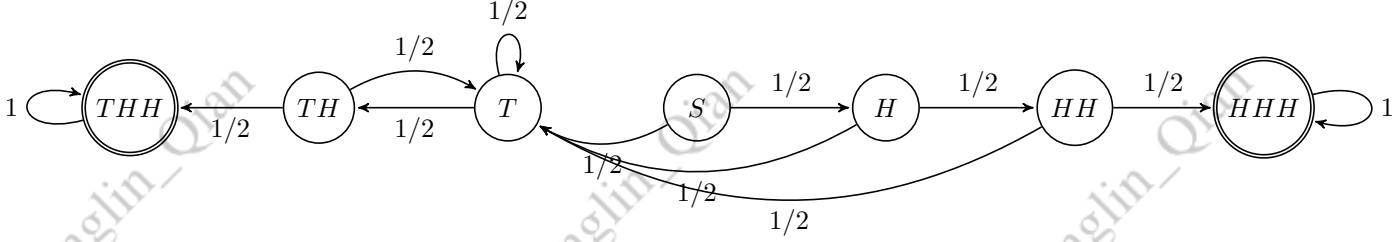


7.8.4 Coin triplets 2

Keep flipping a fair coin until either HHH or THH occurs in the sequence. What is the probability that you get an HHH subsequence before THH ?

solution:

Solution: Let's try a standard Markov chain approach. Again the focus is on choosing the right state space. In this case, we begin with starting state S . We only need ordered subsequences of either HHH or THH . After one coin is flipped, we have either state T or H . After two flips, we have states TH and HH . We do not need TT (which is equivalent to T for this problem) or HT (which is also equivalent to T as well). For three coin sequences, we only need THH and HHH states, which are both absorbing states. Using these states, we can build the following transition graph:



We want to get the probability to reach absorbing state HHH from the starting state S . Applying the equations for absorption probability, we have

$$\begin{cases} a_{HHH} = 1, a_{THH} = 0 \\ a_S = \frac{1}{2}a_T + \frac{1}{2}a_H \\ a_T = \frac{1}{2}a_T + \frac{1}{2}a_{THH}, a_H = \frac{1}{2}a_T + \frac{1}{2}a_{HH} \\ a_{TH} = \frac{1}{2}a_T + \frac{1}{2}a_{THH}, a_{HH} = \frac{1}{2}a_T + \frac{1}{2}a_{HHH} \end{cases} \Rightarrow \begin{cases} a_T = 0, a_{THH} = 0 \\ a_S = \frac{1}{8} \\ a_H = \frac{1}{4} \\ a_{HH} = \frac{1}{2} \end{cases}$$

So the probability that we end up with the HHH pattern is $1/8$.

This problem actually has a special feature that renders the calculation unnecessary. You may have noticed that $a_T = 0$. Once a tail occurs, we will always get THH before HHH .

The reason is that the last two coins in THH is HH , which is the first two coins in sequence HHH . In fact, the only way that the sequence reaches state HHH before THH is that we get three consecutive H s in the beginning. Otherwise, we always have a T before the first HH sequence and always end in THH first. So if we don't start the coin flipping sequence with HHH , which has a probability of $1/8$, we will always have THH before HHH .

7.8.5 Drunk man

A drunk man is at the 17th meter of a 100-meter-long bridge. He has a 0.5 probability of staggering forward or backward one meter each step. What is the probability that he will make it to the end of the bridge (the 100th meter) before the beginning (the 0th meter)? What is the expected number of steps he takes to reach either the beginning or the end of the bridge?

solution:

The probability part of the problem—often appearing in different disguises—is among the most popular martingale problems asked by quantitative interviewers. Interestingly, few people use a clear-cut martingale argument. Most candidates either use Markov chain with two absorbing states or treat it as a special version of the gambler's ruin problem with $p = 1/2$. These approaches yield the correct results in the end, yet a martingale argument is not only simpler but also illustrates the insight behind the problem.

Let's set the current position (the 17th meter) to 0; then the problem becomes a symmetric random walk that stops at either 83 or -17. We also know that both S_n and $S_n^2 - n$ are martingales. Since a martingale stopped at a stopping time is a martingale, S_N and $S_N^2 - N$ (where $S_N = X_1 + X_2 + \dots + X_N$ with N being the stopping time) are martingales as well. Let p_α be the probability that it stops at $\alpha = 83$, p_β be the probability it stops at $-\beta = -17$ ($p_\beta = 1 - p_\alpha$), and N be the stopping time. Then we have

$$\begin{aligned} E[S_N] &= p_\alpha \times 83 - (1 - p_\alpha) \times 17 = S_0 = 0 \\ E[S_N^2 - N] &= E[p_\alpha \times 83^2 + (1 - p_\alpha) \times 17^2] - E[N] = S_0^2 - 0 = 0 \end{aligned} \Rightarrow \begin{cases} p_\alpha = 0.17 \\ E[N] = 1411 \end{cases}$$

Hence, the probability that he will make it to the end of the bridge (the 100th meter) before reaching the beginning is 0.17, and the expected number of steps he takes to reach either the beginning or the end of the bridge is 1411.

7.8.6 Dice game

Suppose that you roll a dice. For each roll, you are paid the face value. If a roll gives 4, 5 or 6, you can roll the dice again. If you get 1, 2 or 3, the game stops. What is the expected payoff of this game?

solution:

A simpler approach—requiring more knowledge—is to apply Wald's Equality since the problem has clear stopping rules. For each roll, the process has $1/2$ probability of stopping. So the stopping time N follows a geometric distribution with $p = 1/2$ and we have $E[N] = 1/p = 2$. For each roll, the expected face value is $E[X] = 7/2$. The total expected payoff is $E[S_N] = E[X]E[N] = 7/2 \times 2 = 7$.

7.8.7 Coin sequence

Assume that you have a fair coin. What is the expected number of coin tosses to get n heads in a row?

solution:

method 1:

Let $E[f(n)]$ be the expected number of coin tosses to get n heads in a row. In the Markov chain section, we discussed the case where $n = 3$ (to get the pattern HHH). For any integer n , we can consider an induction approach. Using the Markov chain approach, we can easily get that $E[f(1)] = 2$, $E[f(2)] = 6$ and $E[f(3)] = 14$. A natural guess for the general formula is that $E[f(n)] = 2^{n+1} - 2$. As always, let's prove the formula using induction. We have shown the formula is true for $n = 1, 2, 3$. So we only need to prove that if $E[f(n)] = 2^{n+1} - 2$, $E[f(n+1)] = 2^{n+2} - 2$. The following diagram shows how to prove that the equation holds for $E[f(n+1)]$:

The state before $(n+1)$ heads in a row (denoted as $(n+1)H$) must be n heads in a row (denoted as nH). It takes an expected $E[f(n)] = 2^{n+1} - 2$ tosses to reach nH . Conditioned on state nH , there is a $1/2$ probability it will go to $(n+1)H$ (the new toss yields H) and the process stops. There is also a $1/2$ probability that it will go to the starting state 0 (the new toss yields T) and we need another expected $E[f(n+1)]$ tosses to reach $(n+1)H$. So we have

$$E[f(n+1)] = E[f(n)] + \frac{1}{2} \times 1 + \frac{1}{2} \times E[f(n+1)]$$

$$\Rightarrow E[f(n+1)] = 2 \times E[f(n)] + 2 = 2^{n+2} - 2.$$

method 2:

General Martingale approach: Let's use $HH \dots H_n$ to explain a general approach for the expected time to get any coin sequence by exploring the stopping times of martingales. Imagine a gambler has 1 dollar to

bet on a sequence of n heads ($HH \cdots H_n$) in a fair game with the following rule: Bets are placed on up to n consecutive games (tosses) and each time the gambler bets all his money (unless he goes bankrupt). For example, if H appears at the first game, he will have 2 and he will put all 2 into the second game. He stops playing either when he loses a game or when he wins n games in a row, in which case he collects 2^n (with probability $1/2^n$). Now let's imagine, instead of one gambler, before each toss a new gambler joins the game and bets on the same sequence of n heads with a bankroll of 1 as well. After the i -th game, i gamblers have participated in the game and the total amount of money they have put in the game should be i . Since each game is fair, the expected value of their total bankroll is i as well. In other words, if we denote x_i as the amount of money all the participating gamblers have after the i -th game, then $(x_i - i)$ is a martingale.

Now, let's add a stopping rule: the whole game will stop if one of the gamblers becomes the first to get n heads in a row. A martingale stopped at a stopping time is a martingale. So we still have $E[(x_i - i)] = 0$. If the sequence stops after the i -th toss ($i \geq n$), the $(i - n + 1)$ -th player is the (first) player who gets n heads in a row with payoff 2^n . So all the $(i - n)$ players before him went bankrupt; the $(i - n + 2)$ -th player gets $(n - 1)$ heads in a row with payoff 2^{n-1} ; ...; the i -th player gets one head with payoff 2. So the total payoff is fixed and $x_i = 2^n + 2^{n-1} + \cdots + 2^1 = 2^{n+1} - 2$.

Hence, $E[(x_i - i)] = 2^{n+1} - 2 - E[i] = 0 \implies E[i] = 2^{n+1} - 2$.

This approach can be applied to any coin sequences—as well as dice sequences or any sequences with arbitrary number of elements. For example, let's consider the sequence $HHTTHH$. We can again use a stopped martingale process for sequence $HHTTHH$. The gamblers join the game one by one before each toss to bet on the same sequence $HHTTHH$ until one gambler becomes the first to get the sequence $HHTTHH$. If the sequence stops after the i -th toss, the $(i - 5)$ th gambler gets the $HHTTHH$ with payoff 2^6 . All the $(i - 6)$ players before him went bankrupt; the $(i - 4)$ th player loses in the second toss (HT); the $(i - 3)$ th player and the $(i - 2)$ th player lose in the first toss (T); the $(i - 1)$ th player gets sequence HH with payoff 2^2 and the i -th player gets H with payoff 2.

Hence, $E[(x_i - i)] = 2^6 + 2^2 + 2^1 - E[i] = 0 \implies E[i] = 70$.

7.8.8 Dynamic Programming 1 - Dice game

You can roll a 6-side dice up to 3 times. After the first or the second roll, if you get a number x , you can decide either to get x dollars or to choose to continue rolling. But once you decide to continue, you forgo the number you just rolled. If you get to the third roll, you'll just get x dollars if the third number is x and the game stops. What is the game worth and what is your strategy?

solution:

This is a simple dynamic programming strategy game. As all dynamic programming questions, the key is to start with the final stage and work backwards. For this question, it is the stage where you have forgone the first two rolls. It becomes a simple dice game with one roll. Face values 1, 2, 3, 4, 5, and 6 each have a $1/6$ probability and your expected payoff is 3.5.

Now let's go back one step. Imagine that you are at the point after the second roll, for which you can choose either to have a third roll with an expected payoff of 3.5 or keep the current face value. Surely you will keep the face value if it is larger than 3.5; in other words, when you get 4, 5 or 6, you stop rolling. When you get 1, 2 or 3, you keep rolling. So your expected payoff before the second roll is:

$$\frac{3}{6} \times 3.5 + \frac{1}{6} \times (4 + 5 + 6) = 4.25.$$

Now let's go back one step further. Imagine that you are at the point after the first roll, for which you can choose either to have a second roll with expected payoff 4.25 (when face value is 1, 2, 3 or 4) or keep the current face value. Surely you will keep the face value if it is larger than 4.25; in other words, when you get 5 or 6, you stop rolling. So your expected payoff before the first roll is:

$$\frac{4}{6} \times 4.25 + \frac{1}{6} \times (5 + 6) = \frac{14}{3}.$$

This backward approach—called **tail policy** in dynamic programming—gives us the strategy and also the expected value of the game at the initial stage, $\frac{14}{3}$.

7.8.9 Dynamic Programming 2 - World series

The Boston Red Sox and the Colorado Rockies are playing in the World Series finals. In case you are not familiar with the World Series, there are a maximum of 7 games and the first team that wins 4 games claims the championship. You have 100 dollars to place a double-or-nothing bet on the Red Sox.

Unfortunately, you can only bet on each individual game, not the series as a whole. How much should you bet on each game so that if the Red Sox wins the whole series, you win exactly 100 dollars, and if Red Sox loses, you lose exactly 100 dollars.

solution:

Let (i, j) represent the state that the Red Sox has won i games and the Rockies has won j games, and let $f(i, j)$ be our net payoff, which can be negative when we lose money, at state (i, j) . From the rules of the game, we know that there may be between 4 and 7 games in total. We need to decide on a strategy so that whenever the series is over, our final net payoff is either +100—when Red Sox wins the championship—or -100—when Red Sox loses. In other words, the state space of the final stage includes $\{(4, 0), (4, 1), (4, 2), (4, 3)\}$ with payoff $f(i, j) = 100$ and $\{(0, 4), (1, 4), (2, 4), (3, 4)\}$ with payoff $f(i, j) = -100$. As all dynamic programming questions, the key is to start with the final stage and work backwards—even though in this case the number of stages is not fixed. For each state (i, j) , if we bet y on the Red Sox for the next game, we will have $(f(i, j) + y)$ if the Red Sox wins and the state goes to $(i + 1, j)$, or $(f(i, j) - y)$ if the Red Sox loses and the state goes to $(i, j + 1)$. So clearly we have

$$\begin{aligned} f(i+1, j) &= f(i, j) + y \\ f(i, j+1) &= f(i, j) - y \end{aligned} \implies f(i, j) = \frac{f(i+1, j) + f(i, j+1)}{2} \implies y = \frac{f(i+1, j) - f(i, j+1)}{2}.$$

For example, we have

$$f(3, 3) = \frac{f(4, 3) + f(3, 4)}{2} = \frac{100 + (-100)}{2} = 0.$$

Let's set up a table with the columns representing i and the rows representing j . Now we have all the information to fill in $f(4, 0), f(4, 1), f(4, 3), f(4, 2), f(0, 4), f(1, 4), f(2, 4), f(3, 4)$, as well as $f(3, 3)$. Similarly we can also fill in all $f(i, j)$ for the states where $i = 3$ or $j = 3$.

Going further backward, we can fill in the net payoffs at every possible state. Using equation $y = \frac{f(i+1, j) - f(i, j+1)}{2}$, we can also calculate the bet we need to place at each state, which is essentially our strategy.

7.8.10 Dynamic Programming 3 - Dynamic dice game

A casino comes up with a fancy dice game. It allows you to roll a dice as many times as you want unless a 6 appears. After each roll, if 1 appears, you will win \$1; if 2 appears, you will win \$2; $\dots$; if 5 appears, you win \$5; but if 6 appears all the moneys you have won in the game is lost and the game stops. After each roll, if the dice number is 1-5, you can decide whether to keep the money or keep on rolling. How much are you willing to pay to play the game (if you are risk neutral)?

solution:

Assuming that we have accumulated n dollars, the decision to have another roll or not depends on the expected profit versus expected loss. If we decide to have an extra roll, our expected payoff will become

$$\frac{1}{6}(n+1) + \frac{1}{6}(n+2) + \frac{1}{6}(n+3) + \frac{1}{6}(n+4) + \frac{1}{6}(n+5) + \frac{1}{6} \cdot 0 = \frac{5}{6}n + 2.5.$$

We have another roll if the expected payoff $\frac{5}{6}n + 2.5 > n$, which means that we should keep rolling if the money is no more than \$14. Considering that we will stop rolling when $n \geq 15$, the maximum payoff of the game is \$19 (the dice rolls a 5 after reaching the state $n = 14$). We then have the following: $f(19) = 19$, $f(18) = 18$, $f(17) = 17$, $f(16) = 16$, and $f(15) = 15$. When $n \leq 14$, we will keep on rolling, so

$$E[f(n) \mid n \leq 14] = \frac{1}{6} \sum_{i=1}^5 E[f(n+i)].$$

Using this equation, we can calculate the value for $E[f(n)]$ recursively for all $n = 14, 13, \dots, 0$. The results are summarized in Table. Since $E[f(0)] = 6.15$, we are willing to pay at most \$6.15 for the game.

n	19	18	17	16	15	14	13	12	11	10
$E[f(n)]$	19.00	18.00	17.00	16.00	15.00	14.17	13.36	12.59	11.85	11.16
n	9	8	7	6	5	4	3	2	1	0
$E[f(n)]$	10.52	9.91	9.34	8.80	8.29	7.81	7.36	6.93	6.53	6.15

Table 2: Expected payoff of the game when the player has accumulated n dollars

7.8.11 Dynamic Programming 4 - Dynamic card game

A casino offers yet another card game with the standard 52 cards (26 red, 26 black). The cards are thoroughly shuffled and the dealer draws cards one by one. (Drawn cards are not returned to the deck.) You can ask the dealer to stop at any time you like. For each red card drawn, you win \$1; for each black card drawn, you lose \$1. What is the optimal stopping rule in terms of maximizing expected payoff and how much are you willing to pay for this game?

solution:

It is another problem perceived to be difficult by many interviewees. Yet it is a simple dynamic programming problem. Let (b, r) represent the number of black and red cards left in the deck, respectively. By symmetry, we have

$$\text{red cards drawn} - \text{black cards drawn} = \text{black cards left} - \text{red cards left} = b - r.$$

At each (b, r) , we face the decision whether to stop or keep on playing. If we ask the dealer to stop at (b, r) , the payoff is $b - r$. If we keep on going, there is a $\frac{b}{b+r}$ probability that the next card will be black—in which case the state changes to $(b-1, r)$ —and $\frac{r}{b+r}$ probability that the next card will be red—in which case the state changes to $(b, r-1)$. We will stop if and only if the expected payoff of drawing more cards is less than $b - r$. That also gives us the system equation:

$$E[f(b, r)] = \max \left(b - r, \frac{b}{b+r} E[f(b-1, r)] + \frac{r}{b+r} E[f(b, r-1)] \right).$$

Using the boundary conditions $f(0, r) = 0$, $f(b, 0) = b$, $\forall b, r = 0, 1, \dots, 26$, and the system equation for $E[f(b, r)]$, we can recursively calculate $E[f(b, r)]$ for all pairs of b and r .

The expected payoff at the beginning of the game is $E[f(26, 26)] = 2.62$.

7.8.12 Brownian Motion 1

What is the correlation of a Brownian motion and its square

solution:

The solution to this problem is surprisingly simple. At time t , $B_t \sim N(0, t)$, by symmetry, $E[B_t] = 0$ and $E[B_t^3] = 0$. Applying the equation for covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y],$$

we have

$$\text{Cov}(B_t, B_t^2) = E[B_t^3] - E[B_t]E[B_t^2] = 0 - 0 = 0.$$

So the correlation of a Brownian motion and its square is 0, too.

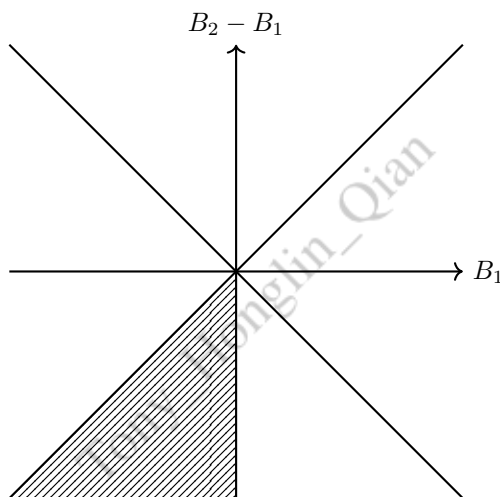
7.8.13 Brownian Motion 2

Let B_t be a Brownian motion. What is the probability that $B_1 > 0$ and $B_2 < 0$?

solution:

If we fully take advantage of the facts that B_1 and $B_2 - B_1$ are two IID $N(0, 1)$. Using conditional probability and independence, we can reformulate the equation as

$$\begin{aligned}
 & P(B_1 > 0, B_2 < 0) \\
 &= P(B_1 > 0, B_2 - B_1 < -B_1) \\
 &= P(B_1 > 0) \cdot P(B_2 - B_1 < -B_1 \mid B_1 > 0) \\
 &= \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}
 \end{aligned}$$

**7.8.14 Stopping time/ first passage time 1**

What is the mean of the stopping time for a Brownian motion to reach either -1 or 1?

Solution:

As we have discussed, $B_t^2 - t$ is martingale. It can be proved by applying Itô's lemma:

$$d(B_t^2 - t) = \frac{\partial(B_t^2 - t)}{\partial B_t} dB_t + \frac{\partial(B_t^2 - t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2(B_t^2 - t)}{\partial B_t^2} dt = 2B_t dB_t - dt + dt = 2B_t dB_t.$$

So $d(B_t^2 - t)$ has no drift term and is a martingale. Let $T = \min\{t; B_t = 1 \text{ or } -1\}$. At continuous time and space, the following property still applies: A martingale stopped at a stopping time is a martingale! So $B_T^2 - T$ is a martingale and $\mathbb{E}[B_T^2 - T] = B_0^2 - 0 = 0$.

The probability that B_t hits 1 or -1 is 1, so $B_T^2 = 1$ implies $\mathbb{E}[T] = \mathbb{E}[B_T^2] = 1$.

7.8.15 Stopping time/ first passage time 2

Let $W(t)$ be a standard Wiener process and τ_x ($x > 0$) be the first passage time to level x ($\tau_x = \min\{t; W(t) = x\}$). What is the probability density function of τ_x and the expected value of τ_x ?

Solution:

This is a textbook problem that is elegantly solved using the reflection principle, so we will simply summarize the explanation. For any Wiener process paths that reach x before t ($\tau_x \leq t$), they have equal probability of ending above x or below x at time t , $P(\tau_x \leq t, W(t) \geq x) = P(\tau_x \leq t, W(t) \leq x)$. The explanation lies in the reflection principle. For each path that reaches x before t and is at a level y above x at time t , we can switch the sign of any move starting from τ_x , and the reflected path will end at $2x - y$ that is below x at time t . For a standard Wiener process (Brownian motion), both paths have equal probability.

$$\begin{aligned} P(\tau_x \leq t) &= P(\tau_x \leq t, W(t) \geq x) + P(\tau_x \leq t, W(t) \leq x) = 2P(\tau_x \leq t, W(t) \geq x) \\ &= 2P(W(t) \geq x) = 2 \int_x^\infty \frac{1}{\sqrt{2\pi t}} e^{-w^2/2t} dw \end{aligned}$$

Let $v = \frac{w}{\sqrt{t}}$, we have $e^{-w^2/2t} = e^{-v^2/2}$ and $dw = \sqrt{t}dv$.

$$\therefore P(\tau_x \leq t) = 2 \int_{x/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi}} e^{-v^2/2} dv = 2 \int_{x/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi}} e^{-v^2/2} dv = 2 - 2N(x/\sqrt{t}).$$

Take the derivative with respect to t , we have

$$f_{\tau_x}(t) = \frac{dP(\tau_x \leq t)}{dt} = \frac{dP(\tau_x \leq t)}{d(x/\sqrt{t})} \frac{d(x/\sqrt{t})}{dt} = 2N' \left(\frac{x}{\sqrt{t}} \right) \times \frac{x}{2} t^{-3/2} = \frac{xe^{-x^2/2t}}{t\sqrt{2\pi t}}, \quad \forall x > 0.$$

it's easy to show that the expected stopping time to reach either α ($\alpha > 0$) or $-\beta$ ($\beta > 0$) is again $E[N] = \alpha\beta$. The expected first passage time to level x is essentially the expected stopping time to reach either x or $-\infty$ and $E[\tau_x] = x \times \infty = \infty$.

Although we have $P(\tau_x \leq \infty) = 2 - 2N(x/\sqrt{\infty}) = 1$, the expected value of τ_x is ∞ !

7.8.16 Stopping time/ first passage time 3

Suppose that X is a Brownian motion with no drift, i.e. $dX(t) = dW(t)$. If X starts at 0, what is the probability that X hits 3 before hitting -5? What if X has drift m , i.e. $dX(t) = mdt + dW(t)$?

Solution:

A Brownian motion is a martingale. Let p_3 be the probability that the Brownian motion hits 3 before -5. Since a martingale stopped at a stopping time is a martingale, we have

$$3p_3 + (-5)(1 - p_3) = 0 \implies p_3 = \frac{5}{8}.$$

Similar to random walk, if we have stopping boundaries $\alpha > 0$ and $-\beta$ ($\beta > 0$), the probability that it stops at α instead of $-\beta$ is $p_\alpha = \beta/(\alpha + \beta)$. The expected stopping time to reach either α or $-\beta$ is again $E[N] = \alpha\beta$.

A different approach for question 2 takes advantage of the exponential martingale: $Z(t) = \exp\{\lambda W(t) - \frac{1}{2}\lambda^2 t\}$. Since $W(t) = X(t) - mt$, $X(t) - mt$ is a Brownian motion as well. Applying the exponential martingale, we have

$$E \left[\exp \left(\lambda(X - mt) - \frac{1}{2}\lambda^2 t \right) \right] = 1$$

for any constant λ . To remove the terms including time t , we can set $\lambda = -2m$ and the equation becomes

$$E[\exp(-2mX)] = 1.$$

Since a martingale stopped at a stopping time is a martingale, we have

$$p_3 \exp(-2m \times 3) + (1 - p_3) \exp(-2m \times (-5)) = 1 \implies p_3 = \frac{e^{10m} - 1}{e^{10m} - e^{-6m}}.$$

7.8.17 Martingale

What is a martingale? How is it related to option pricing?

Solution:

Let $(\Omega, \mathcal{F}_t, P)$ be a filtered probability space, where Ω is the sample space, $\mathcal{F}_t$ is a filtration, and P is a probability measure on Ω . A stochastic process X_t is called a martingale with respect to the filtration $\{\mathcal{F}_t\}$ if and only if

- (i) X_t is adapted, i.e., X_t is $\mathcal{F}_t$ -measurable for all t ;
- (ii) X_t is integrable for all t , i.e., $E[|X_t|] < \infty$ for all t ;
- (iii) $E[X_t | \mathcal{F}_s] = X_s$ almost surely for all $s < t$.

In other words, a martingale is a stochastic process in which, given the available information $\mathcal{F}_s$ up to the current time s , the optimal estimation (in the least square sense) of the process in the future time t , i.e., $E[X_t | \mathcal{F}_s]$, is the current value X_s almost surely.

The martingale concept is one of the cornerstones of option pricing theory. The fundamental theorem of asset pricing states that, if a market model is arbitrage-free, then there exists a risk-neutral probability so that each discounted asset price process under the risk-neutral probability is a martingale. Thus, a way to price a derivative security is to figure out a partial differential equation, called the pricing equation, usually deduced by applying Itô's formula, so that the discounted price process of the derivative is a martingale under risk-neutral probability.

Example: Pricing a European Call Option using Risk-Neutral Pricing

Let's say we want to price a European call option on a stock. A European call option gives the holder the right (but not the obligation) to buy the stock at a fixed strike price K at a future expiration date T .

Step 1: Define the Market Dynamics

Assume that the stock price S_t follows a geometric Brownian motion under the real-world probability measure $\mathbb{P}$:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where:

- μ is the drift of the stock price.
- σ is the volatility.
- dW_t is the increment of a Wiener process under $\mathbb{P}$.

Step 2: Construct the Risk-Neutral Measure

We switch to a risk-neutral measure $\mathbb{Q}$ such that the drift of the stock price changes to the risk-free rate r . Under $\mathbb{Q}$, the stock price follows:

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}},$$

where $W_t^{\mathbb{Q}}$ is a Wiener process under $\mathbb{Q}$.

Step 3: Define the Discounted Stock Price Process

We then define the discounted stock price process as:

$$\tilde{S}_t = e^{-rt} S_t.$$

Under the risk-neutral measure $\mathbb{Q}$, this process $\tilde{S}_t$ is a martingale, which means:

$$\mathbb{E}^{\mathbb{Q}} [\tilde{S}_T | \mathcal{F}_t] = \tilde{S}_t.$$

Step 4: Price the Call Option using the Martingale Property

Let $C(t, S_t)$ denote the price of the European call option at time t when the underlying stock price is S_t . The payoff at maturity T is $(S_T - K)^+$. Using the risk-neutral pricing formula, the price of the option at time t is given by the expected discounted payoff under $\mathbb{Q}$:

$$C(t, S_t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [(S_T - K)^+ | \mathcal{F}_t].$$

By solving the Black-Scholes partial differential equation (PDE), which is derived using Itô's lemma to ensure that the discounted price process of the option is a martingale under $\mathbb{Q}$, we get the Black-Scholes formula for the price of a European call option:

$$C(t, S_t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2),$$

where:

$$d_1 = \frac{\ln(S_t/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, d_2 = d_1 - \sigma\sqrt{T-t}.$$

Here, $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

7.8.18 Brown Motion Itô's lemma

A: Let B_t be a Brownian motion and $Z_t = \sqrt{t}B_t$. What is the mean and variance of Z_t ? Is Z_t a martingale process?

Solution:

As a Brownian motion, $B_t \sim N(0, t)$, which is symmetric about 0. Since $\sqrt{t}$ is a constant at t , $Z_t = \sqrt{t}B_t$ is symmetric about 0 and has mean 0 and variance $t \times \text{var}(B_t) = t^2$. More exactly, $Z_t \sim N(0, t^2)$.

Although Z_t has an unconditional expected value 0, it is not a martingale. Applying Itô's lemma to $Z_t = \sqrt{t}B_t$, we have

$$dZ_t = \frac{\partial Z_t}{\partial B_t} dB_t + \frac{\partial Z_t}{\partial t} dt + \frac{1}{2} \times \frac{\partial^2 Z_t}{\partial B_t^2} dt = \frac{1}{2} t^{-1/2} B_t dt + \sqrt{t} dB_t.$$

For all the cases that $B_t \neq 0$, which has probability 1, the drift term $\frac{1}{2} t^{-1/2} B_t dt$ is not zero. Hence, the process $Z_t = \sqrt{t}B_t$ is not a martingale process.

B: Let $W(t)$ be a Brownian motion. Is $W(t)^3$ a martingale process?

Solution:

Applying Itô's lemma to $f(W(t), t) = W(t)^3$, we have

$$\frac{\partial f}{\partial W(t)} = 3W(t)^2, \quad \frac{\partial f}{\partial t} = 0, \quad \text{and} \quad \frac{\partial^2 f}{\partial W(t)^2} = 6W(t).$$

Thus,

$$df(W(t), t) = 3W(t)dt + 3W(t)^2 dW(t).$$

So again, for the cases $W(t) \neq 0$, which has probability 1, the drift term is not zero. Hence, $W(t)^3$ is not a martingale process.

7.8.19 Silly SDE

Let W_t be a standard Brownian Motion. Let X_t be a process satisfying the SDE

$$dX_t = \kappa(\theta - X_t)dt - \sigma\sqrt{X_t}dW_t$$

with $X_0 = x > 0$. It can be shown (you do not need to do this) that if $\frac{2\kappa}{\theta} > \sigma^2$, $X_t > 0$ with probability 1, so $\sqrt{X_t}$ is defined almost surely. For $T > 0$, $\mathbb{E}[X_T]$ can be written as a function of $x, \kappa, \theta, \sigma$, and T .

Evaluate this function when $T = 10$, $\kappa = 0.2$, $\theta = 2$, $x = 5$, and $\sigma = 0.1$. The answer will be in the form $A + Be^C$ for integers A , B , and C . Find ABC .

solution:

First, we are going to calculate $d(e^{\kappa t} X_t)$. By the product rule, we have that

$$d(e^{\kappa t} X_t) = \kappa e^{\kappa t} X_t dt + e^{\kappa t} dX_t = \kappa e^{\kappa t} X_t dt + e^{\kappa t} (\kappa(\theta - X_t)dt - \sigma\sqrt{X_t}dW_t)$$

Distributing everything, we get the first term cancels with the $-\kappa X_t dt$ term inside the parentheses, so we are left with

$$d(e^{\kappa t} X_t) = \kappa\theta e^{\kappa t} dt - \sigma e^{\kappa t} \sqrt{X_t} dW_t$$

We can use this to get $e^{\kappa t} X_t$ very simply by integrating both sides. Namely,

$$e^{\kappa t} X_t - X_0 = \int_0^t \kappa\theta e^{\kappa s} ds - \int_0^t \sigma e^{\kappa s} \sqrt{X_s} dW_s$$

Rearranging to isolate X_t , we have that

$$X_t = X_0 e^{-\kappa t} + \int_0^t \kappa\theta e^{\kappa(s-t)} ds - \int_0^t \sigma e^{\kappa(s-t)} \sqrt{X_s} dW_s$$

Now, we know that $X_0 = x$, so we can substitute that in. In addition, the second integral is Ito and the function inside is "sufficiently nice" to conclude that the mean of it is 0. Thus,

$$\mathbb{E}[X_T] = x e^{-\kappa T} + \int_0^T \kappa\theta e^{\kappa(s-T)} ds = x e^{-\kappa T} + \theta \kappa \int_0^T e^{\kappa s} ds \Big|_0^T = x e^{-\kappa T} + \theta (1 - e^{-\kappa T})$$

Evaluating this at the specific values, our expectation evaluates to $5e^{-2} + 2(1 - e^{-2}) = 2 + 3e^{-2}$. The answer is thus $3 \cdot 2 \cdot (-2) = -12$.

7.8.20 Squared GBM

Let S_t be a geometric Brownian Motion with drift and volatility parameters both 1. Furthermore, let $S_0 = 1$. Find $\mathbb{E}[S_2^2]$. The answer is in the form e^k for a constant k . Find k .

solution:

The SDE that S_t satisfies is $dS_t = S_t(dt + dW_t)$ by plugging in $\mu = \sigma = 1$. Use Ito's Lemma to compute dS_t^2 by applying it to $f(x) = x^2$. This yields that

$$df(S_t) = dS_t^2 = 2S_t dS_t + \frac{1}{2} \cdot 2d[S, S]_t$$

The quadratic variation of S_t is simply just

$$d[S, S]_t = S_t^2(dt + dW_t)(dt + dW_t) = S_t^2 dt$$

Plugging dS_t and $d[S, S]_t$ in, we get that

$$dS_t^2 = 2S_t \cdot S_t(dt + dW_t) + S_t^2 dt = S_t^2(3dt + 2dW_t)$$

Therefore, we see that S_t^2 satisfies the same SDE as S_t but with $\mu' = 3$ and $\sigma' = 2$.

For a GBM S_t with drift μ and volatility σ , $\mathbb{E}[S_t] = S_0 e^{\mu t}$. We have that $S_0^2 = 1$ as well. Furthermore, as S_t^2 is also a GBM with $\mu' = 3$ and $\sigma' = 2$, plugging these in, we quickly see that

$$\mathbb{E}[S_2^2] = 1 \cdot e^{3 \cdot 2} = e^6$$

Therefore, $k = 6$.

7.8.21 Hitting MGF

Let W_t be a standard Brownian Motion and define $T_a = \inf\{t > 0 : |W_t| > a\}$ for $a > 0$. Find $\mathbb{E}[e^{-\lambda T_a}]$ as a function of λ and a . Evaluate this function with $\lambda = 4$ and $a = \ln(2)$.

solution:

Consider the exponential martingale $Z_t = e^{\theta W_t - \theta^2 t/2}$. If you have not shown this is a martingale before, it is a good exercise to do so. We can apply the Optional Stopping Theorem to Z_t with T_a and obtain

$$\mathbb{E}[Z_{T_a}] = \mathbb{E}[Z_0] = e^0 = 1$$

Note that when T_a occurs, W_{T_a} is equally likely to be either a or $-a$, since they are equidistant from the origin. Therefore, we need to take a conditional expectation based on what W_{T_a} is. In particular, we obtain

$$\frac{1}{2}\mathbb{E}\left[e^{a\theta - \theta^2 T_a/2}\right] + \frac{1}{2}\mathbb{E}\left[e^{-a\theta - \theta^2 T_a/2}\right] = 1$$

by conditioning on $W_{T_a} = \pm a$. We can multiply by 2 on both sides and extract the leading term of each expectation to get that

$$(e^{a\theta} + e^{-a\theta})\mathbb{E}\left[e^{-\theta^2 T_a/2}\right] = 2 \iff \mathbb{E}\left[e^{-\theta^2 T_a/2}\right] = \frac{1}{\cosh(a\theta)}$$

We are looking for $\mathbb{E}[e^{-\lambda T_a}]$. This can be remedied by re-parameterizing and letting $\theta = \sqrt{2\lambda}$ so that we get our desired quantity on the LHS. In particular, we get that

$$\mathbb{E}\left[e^{-\lambda T_a}\right] = \frac{1}{\cosh(a\sqrt{2\lambda})}$$

Plugging in $\lambda = 4$ and $a = \ln(2)$, our answer is

$$\frac{2}{e^{4\ln(2)} + e^{-4\ln(2)}} = \frac{2}{16 + \frac{1}{16}} = \frac{32}{257}$$

7.8.22 Reciprocal SDE

Let W_t be a standard Brownian Motion. Suppose that X_t is some process such that $\frac{1}{X_t}$ satisfies the SDE $d\left(\frac{1}{X_t}\right) = \frac{1}{X_t}(2dt - dW_t)$. The SDE that X_t satisfies is in the form $dX_t = X_t(ad t + b dW_t)$ for some integers a and b . Find ab .

solution:

Let $Y_t = \frac{1}{X_t}$. We can write the SDE in the question as $dY_t = Y_t(2dt - dW_t)$. Note that $X_t = \frac{1}{Y_t}$ from definition. Therefore, let $f(y) = \frac{1}{y}$. We have that $f'(y) = -\frac{1}{y^2}$ and $f''(y) = \frac{2}{y^3}$.

From Ito's Formula, we get that

$$dX_t = df(Y_t) = f'(Y_t)dY_t + \frac{1}{2}f''(Y_t)d[Y, Y]_t$$

where $d[Y, Y]_t$ is the quadratic variation of Y_t . We already know dY_t from the SDE above. We get that

$$d[Y, Y]_t = (Y_t(2dt - dW_t))^2 = Y_t^2(4dt \cdot dt + d[t, W]_t) = Y_t^2 dt$$

by the fact that $d[t, t]_t = d[t, W]_t = 0$.

Plugging everything in,

$$dX_t = -\frac{1}{Y_t^2} \cdot Y_t(2dt - dW_t) + \frac{1}{Y_t^3} \cdot Y_t^2 dt = \frac{1}{Y_t}(-dt + dW_t) = X_t(-dt + dW_t)$$

This yields that our answer is $-1 \cdot 1 = -1$.

7.8.23 Make Your Martingale I

Let $X_1, X_2, \dots$ be IID random variables that take the values ± 1 with equal probability. Fix a constant c . Let $M_0 = 1$ and $M_n = M_{n-1}e^{X_n+c}$. There is a unique value of c such that M_n is a martingale with respect to the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Find c . The answer will be in the form $-\log(ae^b + ce^d)$ for integers a, b, c, d . $\log$ here refers to the natural logarithm. Find $a + b + c + d$.

solution:

Our goal is to compute $\mathbb{E}[M_{n+1} | M_n]$ as a function of c . Afterwards, we must find the value of c so that the conditional expectation is M_n . Doing this, $\mathbb{E}[M_{n+1} | M_n] = \mathbb{E}[M_n e^{X_{n+1}+c} | M_n] = M_n e^c \mathbb{E}[e^{X_{n+1}} | M_n]$ by taking out constants and what is known. Note that X_{n+1} is independent of M_n , so $\mathbb{E}[e^{X_{n+1}} | M_n] = \mathbb{E}[e^{X_{n+1}}]$. The computation of this is quite simple as X_{n+1} only takes two values, so

$$\mathbb{E}[e^{X_{n+1}}] = e \cdot \frac{1}{2} + e^{-1} \cdot \frac{1}{2}$$

Therefore, $\mathbb{E}[M_{n+1} | M_n] = M_n e^c \cdot \frac{e+e^{-1}}{2}$.

To make M_n a martingale, we need all the constants to simplify to 1. Namely, this means $e^c = \frac{2}{e+e^{-1}}$. Therefore, $c = \log\left(\frac{2}{e+e^{-1}}\right) = -\log\left(\frac{1}{2}e + \frac{1}{2}e^{-1}\right)$. For our question, the answer is $a+b+c+d = \frac{1}{2} + \frac{1}{2} + 1 - 1 = 1$.

7.8.24 Make Your Martingale II

Let W_t be a standard Brownian Motion and let $X_t = W_t^3 - ctW_t$ for some constant c . Find the value of c that makes X_t a martingale.

solution:

Using Ito's Formula on $f(t, w) = w^3 - ctw$, we have that

$$dX_t = f_t(t, w)dt + f_w(t, w)dW_t + \frac{1}{2}f_{ww}(t, w)dt$$

The last term comes from the quadratic variation of W_t . This is a martingale exactly when there is no dt term, i.e., when $f_t(t, w) = -\frac{1}{2}f_{ww}(t, w)$. We can quickly calculate that $f_t(t, w) = -cw$ and that $f_{ww}(t, w) = 6w$. Therefore, $-cw = -\frac{1}{2} \cdot 6w = -3w$, which means $c = 3$.

7.8.25 Make Your Martingale III

Let $X_1, X_2, \dots \sim \text{Exp}(\lambda)$ IID. Define the process $\{M_n\}_{n \geq 0}$ by $M_0 = 1$ and $M_n = M_{n-1} \cdot \frac{1}{2}e^{\frac{\lambda}{2}X_n}$ with the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. If $0 < a < p < b$ for some real values a and b , then $\{M_n^p\}_{n \geq 0}$ is a sub-martingale. Find $a + b$.

solution:

By direct computation, we have that

$$\mathbb{E}[M_{n+1}^p | M_n^p] = \mathbb{E}\left[M_n^p \cdot \frac{1}{2^p} e^{\frac{p\lambda}{2}X_{n+1}} | M_n^p\right] = M_n^p \cdot \frac{1}{2^p} \mathbb{E}\left[e^{\frac{p\lambda}{2}X_{n+1}} | M_n^p\right] = M_n^p \cdot \frac{1}{2^p} \mathbb{E}\left[e^{\frac{p\lambda}{2}X_{n+1}}\right]$$

We know that X_{n+1} is independent of M_n^p , as M_n is only a function of $X_1, \dots, X_n$. Therefore,

$$\mathbb{E}\left[e^{\frac{p\lambda}{2}X_{n+1}} | M_n^p\right] = \mathbb{E}\left[e^{\frac{p\lambda}{2}X_{n+1}}\right]$$

is the MGF of an $\text{Exp}(\lambda)$ random variable. Substituting in,

$$\mathbb{E}\left[e^{\frac{p\lambda}{2}X_{n+1}}\right] = \frac{\lambda}{\lambda - \frac{p\lambda}{2}} = \frac{2}{2-p}$$

This means that

$$\mathbb{E}[M_{n+1}^p | M_n^p] = M_n^p \cdot \frac{1}{2^p} \cdot \frac{2}{2-p} = M \left(\frac{p\lambda}{2} \right) \cdot 2^{1-p}$$

We have a sub-martingale whenever

$$\frac{2^{1-p}}{2-p} > 1$$

which means $2-p < 2^{1-p}$.

Note that $0 < p < 2$ because of the assumption $p > 0$ and that our step where we plug into the MGF of the exponential distribution is otherwise invalid if $p > 2$. This is since the MGF is only defined for $\theta < \lambda$, so as $\theta = \frac{p\lambda}{2}$ here, for $p \geq 2$, this is at least λ , which is not in our domain of definition. Thus, we restrict ourselves to $(0, 2)$. In particular, we have equality at $p = 0, 1$. As $2^{1-p} > 2-p$ for $p < 0$, we must have that $2^{1-p} < 2-p$ on $(0, 1)$ and $2^{1-p} > 2-p$ for $1 < p < 2$. This means that $1 < p < 2$ yields a sub-martingale, so $a + b = 3$.

7.8.26 Make Your Martingale V

Let W_t be a standard Brownian Motion. Define a process $Y_t = \int_0^t s dW_s$. Furthermore, define a process $Z_t = tY_t - a_t$ for some process a_t . Let a_t be specifically selected so that Z_t is a martingale. The form of a_t is $a_t = k \int_0^t Y_s ds$ for a constant k . Find k .

solution:

We can quickly compute $dY_t = t dW_t$ quite simply. First, let $f(t, y) = ty$. We can see that $f_t(t, y) = y$, $f_y(t, y) = t$, and $f_{yy}(t, y) = 0$. Therefore, by Ito's Formula,

$$dZ_t = Y_t dt + t^2 dW_t$$

Therefore, if $a_t = \int_0^t Y_s ds$, Z_t would be a martingale. This is because $da_t = Y_t dt$, so it would cancel out the $Y_t dt$ that currently exists in dZ_t .

Therefore, this means $k = 1$.

7.8.27 Make Your Martingale IV

Let W_t be a standard Brownian Motion. Define $X_t = W_t^{10} - \int_0^t a_s ds$ for some process $\{a_t\}_{t \geq 0}$ adapted to the natural filtration of the probability space. The process a_t that makes X_t a martingale can be written as $a_t = AW_t^B$ for some real constants A and B . Find AB .

solution:

The objective here is to calculate dX_t via Ito's Formula. The process a_t will cancel out the dt portion. Namely,

$$dX_t = 10W_t^9 dW_t - a_t dt + \frac{1}{2} \cdot 90W_t^8 dt$$

as the integral portion has 0 second derivative. In particular, we can write this as

$$dX_t = 10W_t^9 dW_t + (45W_t^8 - a_t) dt$$

so $a_t = 45W_t^8$. This means our answer is $45 \cdot 8 = 360$.

7.8.28 Deriving Log-Price Dynamics

Let's say S_t follows the dynamics:

$$dS_t = 3S_t dt + 4S_t dW_t$$

where W_t is a standard Brownian motion. Find the dynamics of $X_t = \ln(S_t)$, where we can write the dynamics as:

$$dX_t = a dt + b dW_t$$

Find $a^2 + b^2$

solution:

We can use Ito's Lemma on $f(x) = \ln(x)$. We have $f'(x) = \frac{1}{x}$ and $f''(x) = -\frac{1}{x^2}$. Ito's Lemma says that $df = f'(S_t)dS_t + \frac{1}{2}f''(S_t)(dS_t)^2$. Plugging in the dynamics from above into dS_t , the derivatives, and using the fact that $(dS_t)^2 = 16S_t^2 dt$, we get the following.

$$\begin{aligned} df &= \frac{1}{S_t} (3S_t dt + 4S_t dW_t) + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) 16S_t^2 dt \\ &= 3dt + 4dW_t - 8dt \\ &= -5dt + 4dW_t \end{aligned}$$

So, we have $a = -5$ and $b = 4$. Plugging the values in, we get $(-5)^2 + 4^2 = 41$.

7.8.29 Integral Variance I

Let W_t be a standard Brownian Motion. Find $\text{Var} \left(\int_0^t W_s ds \right)$ as a function of t . The answer is in the form kt^3 for a constant k . Find k .

solution:

Using integration by parts, we see that

$$\int_0^t W_s ds = sW_s \Big|_0^t - \int_0^t s dW_s = t \int_0^t dW_s - \int_0^t s dW_s = \int_0^t (t-s) dW_s$$

The second equality comes from writing $W_t = \int_0^t dW_s$. Now, if $f(t)$ is a deterministic square-integrable function, then the stochastic integral $\int_0^t f(s) dW_s$ is known to be normal with mean 0 and variance $\int_0^t |f(s)|^2 ds$. This is the Ito Isometry. This theorem applies here to the function $f(s) = t-s$, so the variance of the integral is just

$$\int_0^t (t-s)^2 ds = \frac{t^3}{3}$$

This means that $k = \frac{1}{3}$.

7.8.30 Integral Variance II

Let W_t be a standard Brownian Motion. Compute $\text{Var} \left(\int_0^t W_s dW_s \right)$. The answer is in the form kt^2 for a constant k . Find k .

solution:

We first attempt to evaluate $\int_0^t W_s dW_s$. From basic calculus, we know that

$$\int x dx = \frac{x^2}{2},$$

so we first attempt to evaluate $d\left(\frac{W_t^2}{2}\right)$ as a starting point.

Let $f(x) = \frac{x^2}{2}$. We know that $f'(x) = x$ and $f''(x) = 1$. By Ito's Formula, we have that

$$d\left(\frac{W_t^2}{2}\right) = df(W_t) = W_t dW_t + \frac{1}{2} d[W, W]_t = W_t dW_t + \frac{1}{2} dt.$$

This implies to us that $\frac{W_t^2 - t}{2}$ is the function we are looking to take the derivative of, as that would produce $W_t dW_t$ on the RHS when we take the derivative. Integrating both sides, this yields that

$$\int_0^t W_s dW_s = \frac{W_t^2 - t}{2} + C.$$

As $W_0 = 0$, we get that $C = 0$ as well. Therefore, we really want to evaluate

$$\text{Var}\left(\frac{W_t^2 - t}{2}\right) = \frac{1}{4} \text{Var}(W_t^2) \text{ by properties of variance.}$$

We use the fact here that for $X \sim N(0, 1)$, $\mathbb{E}[X^4] = 3$. This can be proven with MGFs or even Brownian Motion itself! Since $W_t \sim N(0, t)$, $W_t = \sqrt{t}X$. Using this,

$$\text{Var}(W_t^2) = \mathbb{E}[W_t^4] - (\mathbb{E}[W_t^2])^2 = t^2 \mathbb{E}[X^4] - t^2 = 2t^2.$$

Since we have the constant of $1/4$ out front, this means that $k = 2 \cdot 1/4 = 1/2$.

7.8.31 Integral Variance III

Let W_t be a standard Brownian Motion. Compute $\text{Var}\left(\int_0^t W_s^2 dW_s\right)$. The answer will be in the form kt^3 for a constant k . Find k .

solution:

As W_s^2 is square-integrable, adapted, and continuous almost surely, $\int_0^t W_s^2 dW_s$ is a martingale with mean 0. By Ito Isometry, we have that

$$\mathbb{E}\left[\left(\int_0^t W_s^2 dW_s\right)^2\right] = \int_0^t \mathbb{E}[(W_s^2)^2] ds = \int_0^t \mathbb{E}[W_s^4] ds$$

As $W_s \sim N(0, s)$, $W_s = \sqrt{s}Z$, where $Z \sim N(0, 1)$. In particular, since $\mathbb{E}[Z^4] = 3$, $\mathbb{E}[W_s^4] = \mathbb{E}[(\sqrt{s}Z)^4] = 3s^2$. Now, we evaluate that integral to simply be

$$\int_0^t 3s^2 ds = t^3$$

As we already know the integral is mean 0, t^3 is also the variance, so $k = 1$.

7.8.32 Integral Variance V

Let W_t be a standard Brownian Motion. Compute the variance of $X_t = \int_0^t s dW_s$ as a function of t . The answer will be in the form kt^3 for a constant k . Find k .

solution:

As this integrand is a deterministic function $f(s)$, the integral is normally distributed with mean 0 and variance $\int_0^t (f(s))^2 ds$. In this case, $f(s) = s$, so the variance is

$$\int_0^t s^2 ds = \frac{t^3}{3}.$$

This implies $k = \frac{1}{3}$.

7.8.33 Integral Variance IV

Let W_t be a standard Brownian Motion. Define $X_t = W_t^{10} - \int_0^t a_s ds$ for some process $\{a_t\}_{t \geq 0}$ adapted to the natural filtration of the probability space. The process a_t that makes X_t a martingale can be written as $a_t = AW_t^B$ for some real constants A and B . Find AB .

solution:

The objective here is to calculate dX_t via Ito's Formula. The process a_t will cancel out the dt portion. Namely, $dX_t = 10W_t^9 dW_t - a_t dt + \frac{1}{2} \cdot 90W_t^8 dt$, as the integral portion has 0 second derivative. In particular, we can write this as $dX_t = 10W_t^9 dW_t + (45W_t^8 - a_t)dt$, so $a_t = 45W_t^8$. This means our answer is $45 \cdot 8 = 360$.

7.8.34 Absolutely Brownian

Let W_t be a standard Brownian Motion and let $T_4 = \inf\{t > 0 : |W_t| > 4\}$. Find $\text{Var}(T_4)$.

solution:

We are going to solve this for general $T_a = \inf\{t > 0 : |W_t| > a\}$ for any $a > 0$. It is a well-known fact that $Z_t = f(t, W_t)$ defines a martingale, where $f(t, w) = w^4 - 6w^2t + 3t^2$. By using the Optional Stopping Theorem, we have that

$$\mathbb{E}[Z_{T_a}] = \mathbb{E}[W_{T_a}^4 - 6W_{T_a}^2 T_a + 3T_a^2] = \mathbb{E}[Z_0] = 0$$

Since $W_{T_a} = \pm a$ with equal probability, we know that $W_{T_a}^4 = a^4$ with probability 1. Similarly, $W_{T_a}^2 = a^2$ with probability 1. Therefore, by linearity, we have that $a^4 - 6a^2 \mathbb{E}[T_a] + 3\mathbb{E}[T_a^2] = 0$. From a more basic martingale argument on $W_t^2 - t$, one can easily prove that $\mathbb{E}[T_a] = a^2$. Therefore, we have that

$$\mathbb{E}[T_a^2] = \frac{5}{3}a^4 \text{ by rearrangement.}$$

To find the variance, we just use our classic relationship $\text{Var}(T_a) = \mathbb{E}[T_a^2] - (\mathbb{E}[T_a])^2 = \frac{5}{3}a^4 - a^4 = \frac{2}{3}a^4$. In our case, $a = 4$, so the answer is $\frac{512}{3}$.

7.8.35 Shifty Cylinder

The radius of a cylinder is growing at a rate of 2 meters per hour, and the height of the cylinder is decreasing at a rate of 5 meters per hour. At what rate is the volume of the cylinder changing (in cubic meters per hour) at the instant where the base radius is 3 meters and the height is 6 meters? The answer is in the form $k\pi$ for a constant k . Find k .

solution:

$V(r, h) = \pi r^2 h$ is the volume of a cylinder as a function of r and h . Furthermore, we are given that $r' = 2$ and $h' = -5$ from the question. Therefore, we have that $V' = 2\pi r h r' + \pi r^2 h'$ by applying the product rule to the above. Plugging in all of our values, we get that the volume changes at a rate of

$$2\pi(3)(6)(2) + \pi(3)^2(-5) = 27\pi$$

7.8.36 Vasicek Equation

Let W_t be a standard Brownian Motion. Let R_t satisfy the SDE

$$dR_t = (a - bR_t)dt + \sigma dW_t$$

for $a, b, \sigma > 0$ and $R_0 = r > 0$. $\mathbb{E}[R_T]$ can be written as a function of a, b, r , and T . Evaluate this function with parameters $T = 10, a = 0.2, b = 0.1, \sigma = 0.1$, and $r = 1$. The answer will be in the form $A + Be^C$ for integers A, B , and C . Find $A + B + C$.

solution:

We can calculate $d(e^{bt}R_t)$ via Ito's Formula. This can be done via the product rule. In particular,

$$d(e^{bt}R_t) = be^{bt}R_t dt + e^{bt}dR_t = be^{bt}R_t dt + e^{bt}((a - bR_t)dt + \sigma dW_t) = ae^{bt}dt + \sigma e^{bt}dW_t$$

Now, we integrate both sides to get

$$e^{bT}R_T - R_0 = \int_0^T ae^{bs}ds + \int_0^T \sigma e^{bs}dW_s$$

We can substitute a couple of things. Namely, $R_0 = r$ by the question. In addition, the Ito integral on the RHS is "sufficiently nice" so that it has 0 mean. Therefore,

$$e^{bT}\mathbb{E}[R_T] - R_0 = \frac{a}{b}(e^{bT} - 1)$$

After adding r and multiplying by e^{-bT} on both sides, we get that $\mathbb{E}[R_T] = re^{-bT} + \frac{a}{b}(1 - e^{-bT})$.

Evaluating this with the parameters in question, our expectation evaluates to $e^{-1} + 2(1 - e^{-1}) = 2 - e^{-1}$. In this case, we have that $2 - 1 - 1 = 0$.

7.8.37 Positive Brownian I

Let W_t be a standard Brownian Motion. Find $\mathbb{P}[W_1 > 0, W_2 > 0]$.

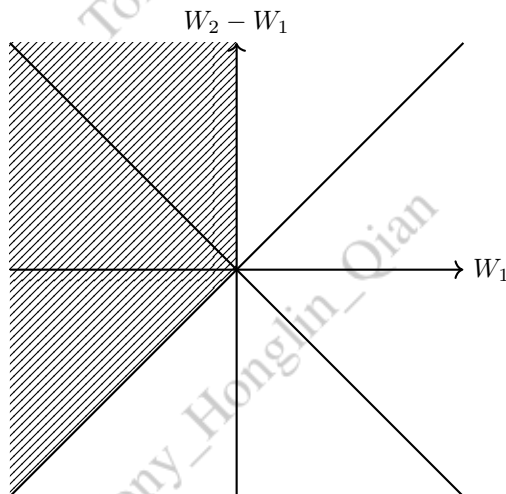
solution:

The trick here is to write $W_2 > 0$ as $W_2 - W_1 > -W_1$. This is because now $W_2 - W_1$ is independent of W_1 . Doing this, we have that

$$\mathbb{P}[W_1 > 0, W_2 - W_1 > -W_1] = \mathbb{P}[W_2 - W_1 > -W_1 \mid W_1 > 0]\mathbb{P}[W_1 > 0]$$

The latter term is just $\frac{1}{2}$ as $W_1 \sim N(0, 1)$ which is symmetric about 0. For the former term, we can note that in the plane, this region covers the entire right half of the plane except below $y = -x$. As normal random variables have radial symmetry, the region below $y = -x$ is $\frac{\pi}{4}$ radians of an entire π radians in the right half of the plane. Therefore, the probability of being below that is $\frac{1}{4}$, meaning the conditional probability in question is $1 - \frac{1}{4} = \frac{3}{4}$. Therefore, our answer is

$$\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$$



7.8.38 Brownian Supremum

Let W_t be a standard Brownian Motion. Define $M_t = \sup_{s \in [0, t]} W_s$. For any $t > 0$, find $\mathbb{P}[M_t > 0]$.

solution:

Define the events $A_n = \left\{ W_{1/n} > \frac{1}{\sqrt{n}} \right\}$ and $B = \limsup A_n = \{A_n \text{ i.o.}\}$. Then we have that

$$\mathbb{P}[B] = \mathbb{P}[\limsup A_n] \geq \limsup \mathbb{P}[A_n] \quad (\text{Fatou's Lemma})$$

$$= \limsup \mathbb{P} \left[W_{1/n} > \frac{1}{\sqrt{n}} \right]$$

$$= \limsup \mathbb{P}[Z > 1] \quad (\cdot \sqrt{n} \text{ to get } Z \sim N(0, 1))$$

$$= \mathbb{P}[Z > 1] = 1 - \Phi(1) > 0$$

By Blumenthal's 0-1 Law, we know that $\mathbb{P}[B] = 0$ or 1 . Since we already know $\mathbb{P}[B] > 0$, we know that $\mathbb{P}[B] = 1$.

7.8.39 Log Square

Consider solving the SDE $dM = \sigma M dW_t$. The two methods below yield different answers in calculating $d \ln M^2$, so one of them is incorrect. Let $a = 1$ if the first method is incorrect and $a = 2$ if the second is incorrect. Let l be the first line number in the solution that is incorrect. Find $100a + l$.

Method 1

$$dM^2 = (M^2)' dM + \frac{1}{2} (M^2)'' d[M, M]$$

$$dM^2 = 2M dM + d[M, M]$$

$$dM^2 = 2\sigma M^2 dW_t + \sigma^2 M^2 dt$$

$$\frac{dM^2}{M^2} = 2\sigma dW_t + \sigma^2 dt$$

$$d \ln M^2 = 2\sigma dW_t + \sigma^2 dt$$

Method 2

$$d \ln M^2 = (\ln M^2)' dM + \frac{1}{2} (\ln M^2)'' d[M, M]$$

$$d \ln M^2 = \frac{2M}{M^2} dM + \frac{1}{2} \left(\frac{2M}{M^2} \right)' d[M, M]$$

$$d \ln M^2 = \frac{2}{M} dM + \frac{1}{2} \left(\frac{2}{M} \right)' d[M, M]$$

$$d \ln M^2 = \frac{2}{M} dM - \frac{1}{2} \frac{2}{M^2} d[M, M]$$

$$d \ln M^2 = \frac{2}{M} dM - \frac{1}{M^2} d[M, M]$$

$$d \ln M^2 = 2\sigma dW_t - \sigma^2 dt$$

solution:

Method 1 is incorrect. The line that is incorrect is Line 5, as $\frac{dM^2}{M^2} \neq d \ln M^2$. Namely, it is missing the Ito term out back involving the quadratic variation. One can also see that Method 1 is incorrect by considering $\ln M^2 = 2 \ln M$, so calculating the dynamics of $\ln M$ is even easier.

8 The Quant Interview Cheat Sheet

8.1 Preface

This part is meant to assist you in your preparation for interviews. It will refresh you on fundamental ideas that universally show up in Quantitative Trading (QT) and Research (QR) in Buy side or Sell side math interviews. Sections labelled QT and QR are to represent the type of interview that the content is relevant for. Note that this document only consists of important results without derivation/proof. It's imperative that you properly practice questions and review the concepts listed here, as this is not a substitute for studying.

8.2 Calculus

8.2.1 Differentiation (QT/QR)

At all points x where the functions and the derivatives are defined,

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

$$\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$$

$$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$$

$$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

$$\frac{d}{dx}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\operatorname{arcsec}(x)) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + g'(x)f(x)$$

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$\frac{d}{dx}(f(x)^{g(x)}) = f(x)^{g(x)} \left[g'(x) \ln(f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \right]$$

$$\frac{d}{dx}(x^x) = x^x(\ln(x) + 1)$$

8.2.2 Integration (QT/QR)

Disregarding the $+C$ on all the integrals,

$$\int x^n dx = \frac{x^{n+1}}{n+1}, n \neq -1$$

$$\int \sin(x) dx = -\cos(x)$$

$$\int \cos(x) dx = \sin(x)$$

$$\int \tan(x) dx = -\ln |\cos(x)|$$

$$\int \sec(x) dx = \ln |\sec(x) + \tan(x)|$$

$$\int \csc(x) dx = \ln |\csc(x) - \cot(x)|$$

$$\int \cot(x) dx = \ln |\sin(x)|$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x)$$

$$\int \frac{1}{1+x^2} dx = \arctan(x)$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} dx = \operatorname{arcsec}(x)$$

$$\int e^x dx = e^x$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int u(x)v'(x) dx = u(x)v(x) - \int v(x)u'(x) dx$$

$$\int f'(g(x))g'(x) dx = f(g(x))$$

8.2.3 Taylor Series (QT/QR)

Select some point $x = x_0$. If $x_0 = 0$, we have the Maclaurin series. Generally, $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$.

Common Maclaurin series expansions:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

8.2.4 Common Summation Formulae (QT/QR)

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=0}^{\infty} a \cdot r^k = \frac{a}{1-r}, \quad |r| < 1$$

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

8.3 Linear Algebra**8.3.1 Fundamental Knowledge (QT/QR)**

Let A and B be square $n \times n$ matrices. Then all of the following hold:

$$\cos(\theta) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

$$(AB)^T = B^T A^T$$

$$(AB)^{-1} = B^{-1} A^{-1}$$

$$A^{-1} A = A A^{-1} = I$$

$$\text{rank}(A) + \text{null}(A) = n$$

$$Av = \lambda v$$

$$(A - \lambda I)v = 0 \Rightarrow \det(A - \lambda I) = 0$$

$$\det(A) = \frac{1}{\det(A^{-1})}$$

$$\det(A) = \det(A^T)$$

$$\det(AB) = \det(A) \det(B)$$

$$\det(cA) = c^n \det(A)$$

$$\det(A) = \prod_{i=1}^n \lambda_i$$

$$\text{trace}(A) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i$$

Nonsingular Matrices (QR)

A nonsingular matrix is invertible. A $(n \times n)$ is nonsingular if and only if any (and therefore all) of the following hold:

1. Columns of A span $\mathbb{R}^n$, or equivalently, $\text{rank}(A) = \dim(\text{range}(A)) = n$
2. A^T is nonsingular
3. $\det(A) \neq 0$
4. $Ax = 0$ has only the trivial solution $x = 0$; $\dim(\text{null}(A)) = 0$

Note that if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Larger inverses may be found via Gauss-Jordan Elimination:

$$[A \mid I] \xrightarrow{\text{elementary row operations}} [I \mid A^{-1}]$$

2D Rotation Matrices (QR)

2D Rotation matrices by θ radians counter-clockwise about the origin are matrices in the form

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Orthogonal Matrices (QR)

Orthogonal matrices (unitary matrices in the reals) are square with orthonormal row and column vectors. They are nonsingular and satisfy $Q^T = Q^{-1}$. Orthogonal matrices can be interpreted as rotation matrices.

Idempotent Matrices (QR)

Idempotent matrices are square matrices satisfying $A^2 = A$. In other words, the effect of applying the linear transformation A twice is the same as applying it once. Projection matrices are Idempotent.

Positive Semi-definite Matrices (QR)

Covariance and correlation matrices are always positive semi-definite and positive definite if there is no perfect linear dependence among random variables. Each of the following conditions is a necessary and sufficient condition for A to be positive semi-definite/definite:

Positive Semi-Definite	Positive Definite
$\mathbf{z}^T A \mathbf{z} \geq 0$ for all column vectors $\mathbf{z}$	$\mathbf{z}^T A \mathbf{z} > 0$ for all nonzero column vectors $\mathbf{z}$
All eigenvalues are nonnegative	All eigenvalues are positive
All upper left/lower right submatrices have nonnegative determinants	All upper left/lower right submatrices have positive determinants

Note that if A has negative diagonal elements, then A cannot be positive semi-definite.

8.3.2 Matrix Decompositions

Diagonalizable Matrices (QR)

A is diagonalizable if and only if it has linearly independent eigenvectors, or equivalently, if the geometric multiplicity and the algebraic multiplicity of all the eigenvalues agree. A special case of this is if A has n distinct eigenvalues. Suppose we have eigenvalues $\lambda_1, \dots, \lambda_n$ and corresponding eigenvectors $v_1, \dots, v_n$. Then

$$A = XDX^{-1}, \quad X = [v_1 \ \cdots \ v_n], \quad D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & & \\ 0 & \cdots & \lambda_n & \end{bmatrix}$$

Intuitively, this says that we can find a basis consisting of the eigenvectors of A . Useful for computing large powers of A , as $A^n = XD^nX^{-1}$. An important example is A being real and symmetric implies A is diagonalizable.

Singular Value Decomposition (QR)

SVD is powerful in low-rank approximations of matrices. Unlike eigenvalue decomposition, SVD uses two unique bases (left/right singular vectors). For orthogonal matrices $U(m \times m)$, $V(n \times n)$ and diagonal matrix $\Sigma(m \times n)$ with nonnegative diagonal entries in nonincreasing order, we can write any $m \times n$ matrix A as:

$$A = U\Sigma V^T$$

Intuitively, this says that we can express A as a diagonal matrix with suitable choices of (orthogonal) bases.

QR Decomposition (QR)

For nonsingular A , we can write $A = QR$, where Q is orthogonal and R is an upper triangular matrix with positive diagonal elements. QR decomposition assists in increasing the efficiency of solving $Ax = b$ for nonsingular A :

$$Ax = b \Rightarrow QRx = b \Rightarrow Rx = Q^T b \Rightarrow x = Q^T b$$

QR decomposition is very useful in efficiently solving large numerical systems and inversion of matrices. Furthermore, it is also used in least-squares when our data is not full rank.

LU and Cholesky Decompositions (QR)

For nonsingular A , we can write $A = LU$, where L is a lower and U is an upper triangular matrix. This decomposition assists in solving $Ax = b$ as well as computing the determinant:

$$\det(A) = \det(L) \det(U) = \prod_{i=1}^n L_{ii} \prod_{j=1}^n U_{jj}$$

If A is symmetric positive definite, then A can be expressed as $A = R^T R$ via Cholesky decomposition, where R is an upper triangular matrix with positive diagonal entries. Cholesky decomposition is essentially LU decomposition with $L = U^T$. These decompositions are both useful for solving large linear systems.

8.3.3 Projections (QR)

Fix a vector $v \in \mathbb{R}^n$. The projection of $x \in \mathbb{R}^n$ onto v is given by

$$\text{proj}_v(x) = P_v x = \frac{vv^T}{\|v\|^2} x = \frac{x \cdot v}{\|v\|^2} v$$

More generally, if $S = \text{Span}\{v_1, \dots, v_k\} \subseteq \mathbb{R}^n$ has orthogonal basis $\{v_1, \dots, v_k\}$, then the projection of $x \in \mathbb{R}^n$ onto S is given by

$$\text{proj}_S(x) = \sum_{i=1}^k \frac{x \cdot v_i}{\|v_i\|^2} v_i$$

The main property is that $\text{proj}_S(x) \in S$ and $x - \text{proj}_S(x)$ is orthogonal to any $s \in S$. Linear Regression can be viewed as a projection of our observed data onto the subspace formed by the span of the collected data.

8.4 Probability

8.4.1 Key Distributions

Name	Modeling Intuition	PMF/PDF	MGF	μ	σ^2
Bernoulli	Toss a coin, 1 if heads, else 0, coin lands heads with probability p	$f(t; p) = \begin{cases} p & \text{if } t = 1 \\ 1 - p & \text{if } t = 0 \end{cases}$	$pe^\theta + (1 - p)$	p	$p(1 - p)$
Binomial	Toss a coin n times, probability of t heads, coin lands heads with probability p	$f(t; n, p) = \binom{n}{t} p^t (1 - p)^{n-t}$	$[pe^\theta + (1 - p)]^n$	np	$np(1 - p)$
Geometric	Probability of tossing coin t times until heads, coin lands heads with probability p	$f(t; p) = p(1 - p)^{t-1}$	$\frac{pe^\theta}{1 - (1 - p)e^\theta}$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
Poisson	Probability of t occurrences within a fixed time interval or space; parameter λ represents average number of occurrences	$f(t; \lambda) = \frac{\lambda^t e^{-\lambda}}{t!}$	$e^{\lambda(e^\theta - 1)}$	λ	λ
Exponential	Probability distribution of time between events in a Poisson process occurring with rate λ	$f(t; \lambda) = \lambda e^{-\lambda t}, t \geq 0$	$\frac{\lambda}{\lambda - \theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Uniform	Uniform on a line	$f(t; a, b) = \frac{1}{b - a} 1_{[a, b]}$	$\frac{e^{\theta b} - e^{\theta a}}{\theta(b - a)}$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
Normal	Standard univariate normal, z-score transform	$f(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$	$e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$	μ	σ^2

8.4.2 Formulas and Laws

Common Relationships Between Distributions (QT/QR)

- $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ IID $\implies \sum_{i=1}^n X_i \sim \text{Binom}(n, p)$
- $X_1, \dots, X_r \sim \text{Geom}(p)$ IID $\implies \sum_{i=1}^r X_i \sim \text{NegBinom}(r, p)$
- $X_i \sim \text{Poisson}(\lambda_i), 1 \leq i \leq n$ independent $\implies \sum_{i=1}^n X_i \sim \text{Poisson}(\sum_{i=1}^n \lambda_i)$
- $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ IID $\implies \sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda^{-1})$ (Shape-Scale parameterization)
- $X_i \sim N(\mu, \sigma^2), 1 \leq i \leq n$ independent $\implies \sum_{i=1}^n X_i \sim N(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)$
- If $X \sim \text{Unif}(0, 1)$ and $F(x)$ is an invertible CDF, then $Y = F^{-1}(X)$ has CDF $F(x)$

Conditional Probability, Bayes, and Law of Total Probability (QT/QR)

Consider events $A_1, \dots, A_n$, which form a partition of the sample space as well as event B . Then,

$$\mathbb{P}(A_i | B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B | A_i) \mathbb{P}(A_i)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B | A_i) \mathbb{P}(A_i)}{\sum_{i=1}^n \mathbb{P}(B \cap A_i)} = \frac{\mathbb{P}(B | A_i) \mathbb{P}(A_i)}{\sum_{i=1}^n \mathbb{P}(B | A_i) \mathbb{P}(A_i)}$$

Law of Total Expectation and Variance (QT/QR)

For two random variables X, Y defined on the same sample space,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]$$

- Discrete Y :** $\mathbb{E}[X] = \sum_{i=1}^{\infty} \mathbb{P}(Y = y_i) \mathbb{E}[X | Y = y_i]$
- Continuous Y :** $\mathbb{E}[X] = \int_{\mathbb{R}} \mathbb{E}[X | Y = y] f_Y(y) dy$

$$\text{Var}(X) = \text{Var}(\mathbb{E}[X | Y]) + \mathbb{E}[\text{Var}(X | Y)], \quad \text{where } \text{Var}(X | Y) = \mathbb{E}[(X - \mathbb{E}[X | Y])^2 | Y]$$

Intuitively, the Law of Total Expectation says that if we "average over all averages" of X obtained by some information about Y , we obtain the true average. Similarly, the Law of Total Variance says that the true variance comes from two sources: between samples (the first term) and within samples (the second term).

Covariance and Correlation (QT/QR)

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Covariance and correlation are measurements of linear association of X and Y . An example of uncorrelated but not independent random variables are Z and Z^2 , where $Z \sim N(0, 1)$.

Properties of Expectation, Variance, and Covariance (QT/QR)

Let a, b, c , and d be real constants and X and Y be random variables with finite mean and variance. Then all of the following hold:

1. $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$
2. $\text{Var}(aX + b) = a^2\text{Var}(X)$
3. $\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$
4. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
5. If X and Y are independent and have finite mean, then X and Y are uncorrelated.
6. $\text{Corr}(aX + b, cY + d) = \text{sign}(ac)\text{Corr}(X, Y)$
7. $\text{Cov}(X, X) = \text{Var}(X)$
8. The correlation and covariance matrices are both positive semidefinite.
9. $|\text{Corr}(X, Y)| \leq 1$

Tail Sum and Integral (QT/QR)

If X is a non-negative random variable, then

$$\mathbb{E}[X] \xrightarrow{\text{continuous } X} = \int_0^\infty \mathbb{P}(X > t) dt \xrightarrow{\text{integer-valued } X} = \sum_{t=0}^\infty \mathbb{P}(X > t)$$

Law of the Unconscious Statistician (LOTUS) (QT/QR)

Let X be a random variable and $g(x)$ be a function. Then we have

$$\mathbb{E}[g(X)] \xrightarrow{\text{continuous } X} = \int_{\mathbb{R}} g(x) f_X(x) dx \xrightarrow{\text{discrete } X} = \sum_{k \in X} g(k) \mathbb{P}(X = k)$$

Central Limit Theorem (QT/QR)

Suppose that $X_1, X_2, \dots, X_n$ are IID random variables with finite mean μ and variance σ^2 . Furthermore, suppose that n is large. Define $S_n = X_1 + \dots + X_n$ and $\bar{X}_n = \frac{S_n}{n}$. Then we have that

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

This intuitively says that sums of large amounts of IID random variables approach a standard normal distribution when re-scaled to have mean 0 and variance 1. This is useful to approximate sums whose exact distribution is not known.

Law of Large Numbers (QT/QR)

Suppose that $X_1, X_2, \dots$ are IID random variables with finite mean μ . Define $\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$. With probability 1, we have that

$$\lim_{n \rightarrow \infty} \bar{X}_n = \mu$$

This intuitively says that the sample average approaches the true average as the sample size grows large. This theorem is the basis of Monte Carlo Sampling, as we are attempting to find the true average by averaging the results of simulations.

8.4.3 Markov Chains

An $n \times n$ transition matrix P is defined as follows for a discrete state space with n states $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$, where P_{ij} is the probability of transitioning from state x_i to state x_j . Each entry in P is within $[0, 1]$, and the sum of entries for each row must total 1. Note that not all transition matrices are stationary.

Stationary Distribution (QR)

Solve for row vector $\pi = (\pi_1, \dots, \pi_n)$, where π_i is the stationary probability of the i -th state.

$$\pi = \pi P, \quad \pi \mathbf{1} = 1$$

Expected Time to State (Example)

Suppose we have a 3×3 transition matrix P , and we wish to find the expected time to reach state x_3 from x_1 . We can solve the following system of equations for μ_1 , where μ_i denotes the expected time to reach state x_3 starting from x_i :

$$\mu_1 = 1 + P_{11}\mu_1 + P_{12}\mu_2 + P_{13}$$

$$\mu_2 = 1 + P_{21}\mu_1 + P_{22}\mu_2 + P_{23}$$

Gambler's Ruin/Random Walks (QT/QR)

Suppose that you and a friend start with a dollar and b dollar, respectively, where $a, b > 0$ are integers. Each round, a fair coin is flipped. You receive 1 dollar from your friend if it appears heads, while you must pay 1 dollar to your friend if it appears tails. The game ends once one of the players has no money left. The probability that you are ruined i.e., have no money left is

$$\frac{a}{a+b}$$

If the coin has probability p of appearing heads, define $\rho = \frac{p}{1-p}$. The probability that you are ruined is

$$\frac{1 - \rho^b}{1 - \rho^{a+b}}$$

8.5 Regression Analysis

8.5.1 OLS Assumptions:

Linearity, Independent observations, constant variance (homoscedasticity), Normality and no multicollinearity.

8.5.2 Simple Linear Regression (QR)

Suppose we have a variable X and another variable Y which we assume has some sort of linear relationship with X , and suppose we have m samples in a data set. Specifically, we assume

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \mathbb{E}[\epsilon] = 0, \quad \text{Var}(\epsilon) = \sigma^2, \quad \epsilon \text{ and } X \text{ independent}$$

The best estimates for β_0, β_1 that minimize the residual sum of squares ($RSS = \sum_{i=1}^m (y_i - \hat{y}_i)^2$) are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^m (x_i - \bar{x})^2} \approx \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^m (x_i - \bar{x})^2}, \quad \text{Var}(\hat{\beta}_0) = \frac{\sigma^2 \sum_{i=1}^m x_i^2}{m \sum_{i=1}^m (x_i - \bar{x})^2}$$

$$\hat{\sigma}^2 = \frac{1}{m-2} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

8.5.3 Terms for Model Assessment and Inference (QR)

Term	Math	Intuition
RSS (Residual Sum of Squares)	$\sum_{i=1}^m (y_i - \hat{y}_i)^2$	Measures the amount of variability that is left unexplained after performing the regression
SE (Standard Error)	$\sqrt{\frac{\sum_{i=1}^m (x_i - \bar{x})^2}{m-2}} \approx \frac{\hat{\sigma}}{\sqrt{\text{Var}(X)}}$	Used in hypothesis testing on the coefficients; $t = \frac{\beta}{SE(\beta)}$ distributed as t_{m-p-1}
RSE (Residual Standard Error)	$\sqrt{\frac{RSS}{m-2}} = \sqrt{\frac{RSS}{m-p-1}}$	Absolute measure of lack of fit of model to the data, generalized for p predictors
TSS (Total Sum of Squares)	$\sum_{i=1}^m (y_i - \bar{y})^2$	Total variance in Y
R^2	$1 - \frac{RSS}{TSS}$	Proportion of variance in Y explained by X
Adjusted R^2	$1 - \frac{RSS/(m-p-1)}{TSS/(m-1)}$	Similar to R^2 , adjusted for p predictors and m examples
Corr(X , Y)	$\frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^m (x_i - \bar{x})^2 \sum_{i=1}^m (y_i - \bar{y})^2}}$	Sample correlation between X , Y

8.5.4 Multiple Linear Regression (QR)

Our model assumes a linear relationship between Y and $X_1, X_2, \dots, X_p$.

$$Y = \epsilon + \beta_0 + \sum_{i=1}^p \beta_i X_i, \quad \mathbb{E}[\epsilon] = 0, \quad \text{Var}(\epsilon) = \sigma^2$$

We can express the above in matrix form. Denote $\mathbf{X}$ as an $m \times (p+1)$ matrix with each row an input vector with 1 in the first position corresponding to β_0 . Further, let $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$, and let $\mathbf{y}$ denote the m -vector of outputs corresponding to $\mathbf{X}$. Then we have that $\mathbf{y} = \mathbf{X}\beta + \epsilon$, $\mathbb{E}[\epsilon] = 0$, and $\text{Var}(\epsilon) = \sigma^2$.

We can compute the unbiased estimator $\hat{\beta}$ as follows:

$$RSS = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta) \rightarrow \frac{\partial RSS}{\partial \beta} = -2\mathbf{X}^T (\mathbf{y} - \mathbf{X}\beta) \Rightarrow \hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\text{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2, \quad \hat{\sigma}^2 = \frac{1}{m-p-1} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

8.5.5 F Distribution for Subset Selection (QR)

1. Is a subset of $X_1, X_2, \dots, X_p$ useful in predicting Y ? We may conduct the hypothesis test by computing the F-statistic defined below, where subscripts with 1 denote relation to the larger model.

$$F = \frac{(RSS_0 - RSS_1)/(p_1 - p_0)}{RSS_1/(m - p_1 - 1)}$$

The F statistic corresponds to the $F_{p_1-p_0, m-p_1-1}$ distribution.

2. What subset of $X_1, X_2, \dots, X_p$ helps explain Y ? The F-ratio is defined as follows, where k denotes the number of predictors currently in the model, RSS_0 and RSS_1 are the measures of error for the original and new (appropriately fitted) model, respectively.

$$F = \frac{RSS_0 - RSS_1}{RSS_1/(m - k - 2)}$$

Forward selection greedily adds predictors into the model that produce the largest value of F , whereas backward selection sequentially deletes predictors producing the smallest value of F at each stage. Stopping condition is an F -ratio $\geq (100 - \alpha)\text{th}$ percentile for some α , typically 5 or 10, of the $F_{1, N-k-2}$ distribution.

8.5.6 Coefficient Shrinkage Methods (QR)

Previously, we aimed to minimize $RSS = \sum_{i=1}^m \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$. Regularization methods such as ridge and lasso shrink regression coefficients towards zero, which can significantly reduce coefficient estimates' variance.

Ridge Regression Objective function to be minimized, for hyperparameter λ :

$$\begin{aligned} \sum_{i=1}^m \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 &= RSS + \lambda \sum_{j=1}^p \beta_j^2 = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta) + \lambda \beta^T \beta \\ \Rightarrow \hat{\beta}_{\text{ridge}} &= (\mathbf{X}^T \mathbf{X} + \lambda I)^{-1} \mathbf{X}^T \mathbf{y} \end{aligned}$$

As $\lambda \rightarrow \infty$, the impact of the shrinkage penalty grows, and ridge regression coefficient estimates will approach zero. Note that we don't want to shrink the intercept, since it is, conceptually, a measure of the center of the response when predictors are at zero. Note that ridge regression will shrink all coefficients towards zero but will not set any of them exactly to zero, reducing model interpretability for large p . Lasso overcomes this disadvantage.

Lasso Regression Objective function to be minimized, for hyperparameter λ , yielding sparse models:

$$\sum_{i=1}^m \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = RSS + \lambda \sum_{j=1}^p |\beta_j|$$

Logistic Regression (QR) Suppose we now want to model our Y as a binary output for given X . We can now model the probability of X belonging in a particular category. Here, we will assign X to group 1 if $\mathbb{P}(Y = 1 | X = x) > 0.5$ and 0 otherwise. This is done via the *logistic function*:

$$f(z) = \frac{1}{1 + e^{-z}}$$

Logistic regression is also known as a linear classification model. This is due to the fact that we use a linear regression as the input into the logistic function, which then normalizes it into a probability. In other words, $z = \beta_0 + \beta_1 x$.

To find β , we can use the method of maximum likelihood, where the probabilities for each x_i is denoted by $f(\beta_0 + \beta_1 x_i)$. Unlike in linear regression, there is no closed-form solution and we must use iterative methods to determine $\hat{\beta}$. Multiple logistic regression is done in the same manner, but now we have to calculate additional values of β .